Deep Learning for Computer Vision (Generative way)

Andrii Liubonko

2025

Course structure

Module 1. Basic image processing - Oles Dobosevych

Module 2. Learning Discriminative Models - Andrii Liubonko

Module 3. Image Segmentation, Image Correspondence - Maksym Davydov

Module 4. Metric Learning, Representation Learning and Context Importance, Tracking - Igor Krashenyi

Module 5. Learning Generative Models - Andrii Liubonko

Overview of the module

Lecture I, Lecture II

- Intro to generative modeling
- Diffusion models

Lecture III, Lecture IV

- 3D Deep Learning
- 3D Generation

Logistics (this module)

4 lectures

homework:

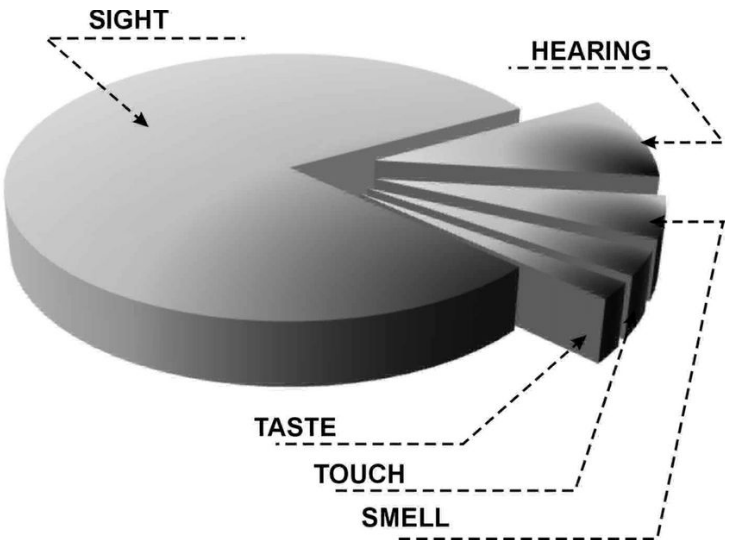
- practical assignment
- reading (paper)

deadline date: *TBD*

course repo:
https://github.com/lyubonko/ucu2025cv

Quick recap

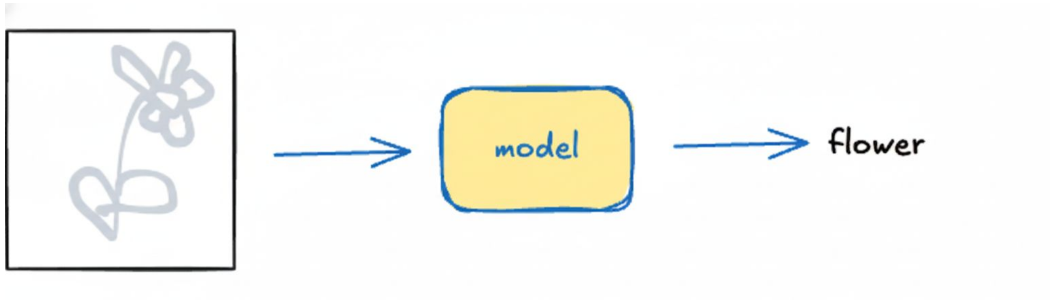
Intro



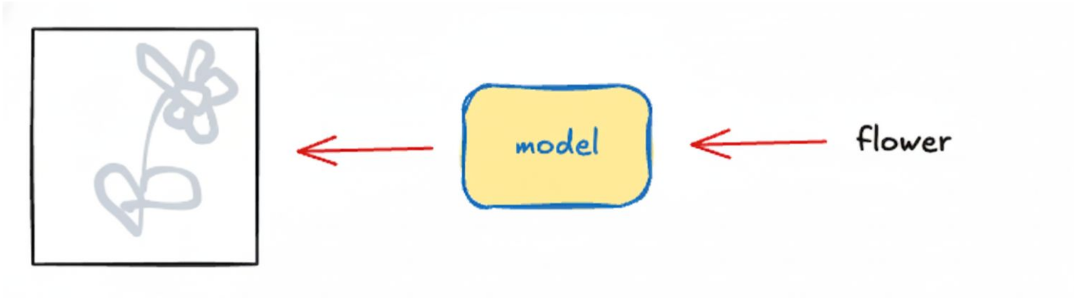
Computer Vision *aims*

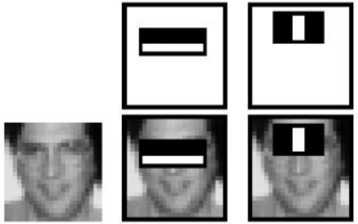
➔ *to extract useful information from visual input* **(Discriminative Models)**
➔ *to generate novel visual content* **(Generative Models)**

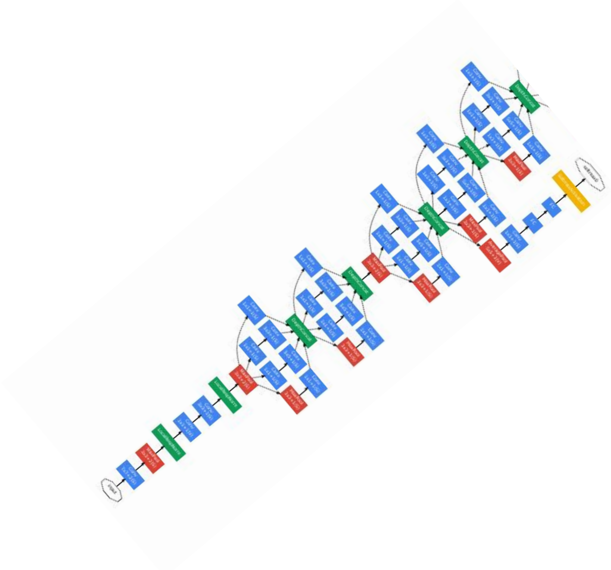
➔ *to extract useful information from visual input* **(Discriminative Models)**



➔ *to generate novel visual content* **(Generative Models)**







Era of
Human-Crafter
Features

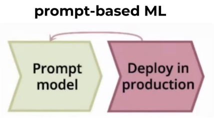
Era of
Deep Learning

Era of
LLMs

2012
AlexNet

2022
ChatGPT

APIs multimodal LLMs



Local multimodal LLMs

LLAMA, DeepSeek, GEMMA, ...

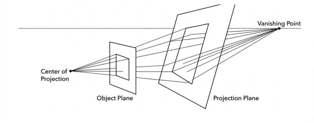
CV foundational models

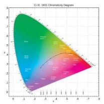
DINO, SAM, ...

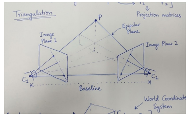
CV DL fundamentals

Image → “Deep” Neural Networks → Result

“classical” CV



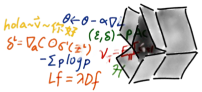




Human Crafted Features

Descriptors:
SIFT, SURF, etc.

math

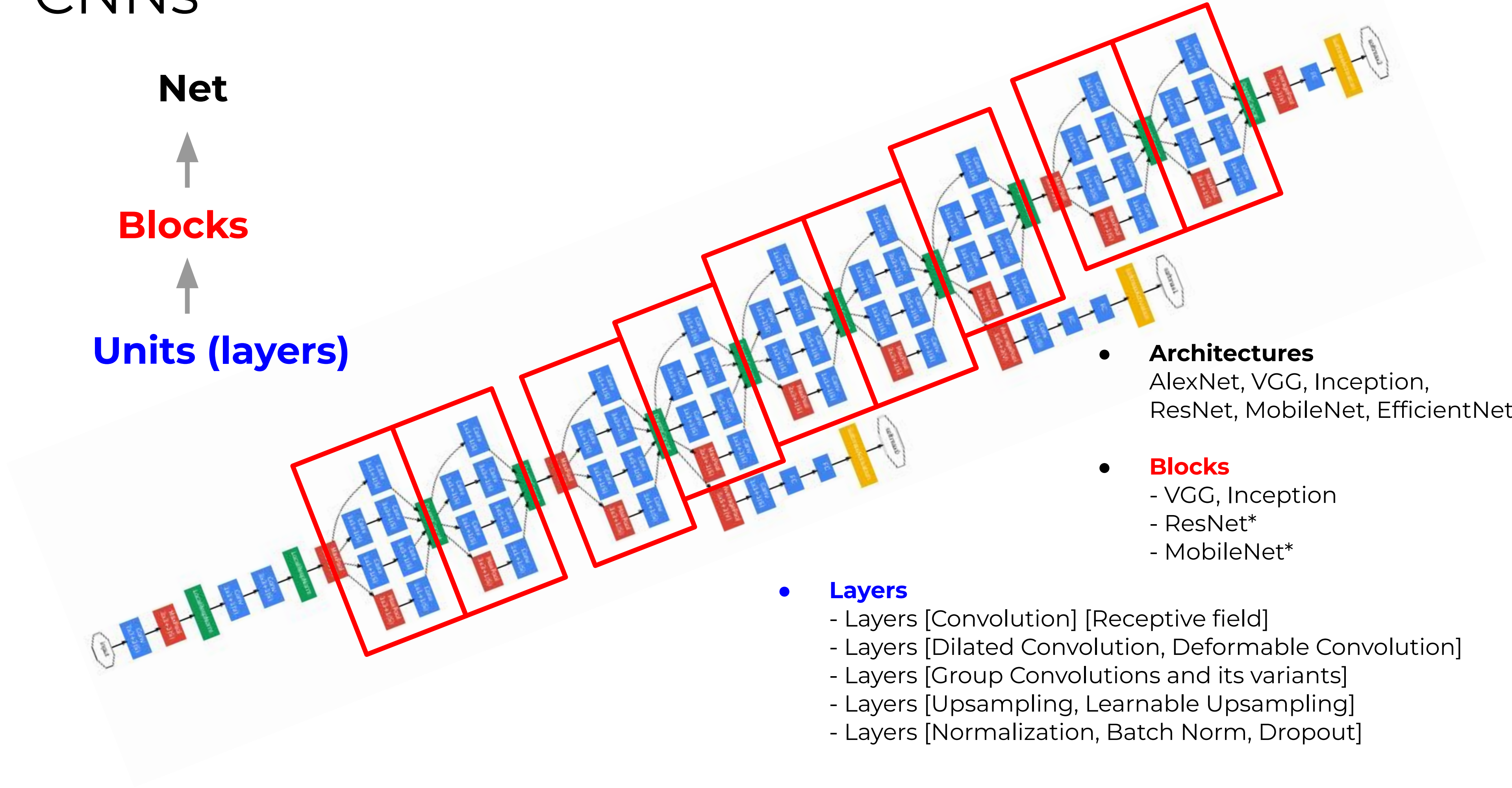


CNNs

Net

Blocks

Units (layers)

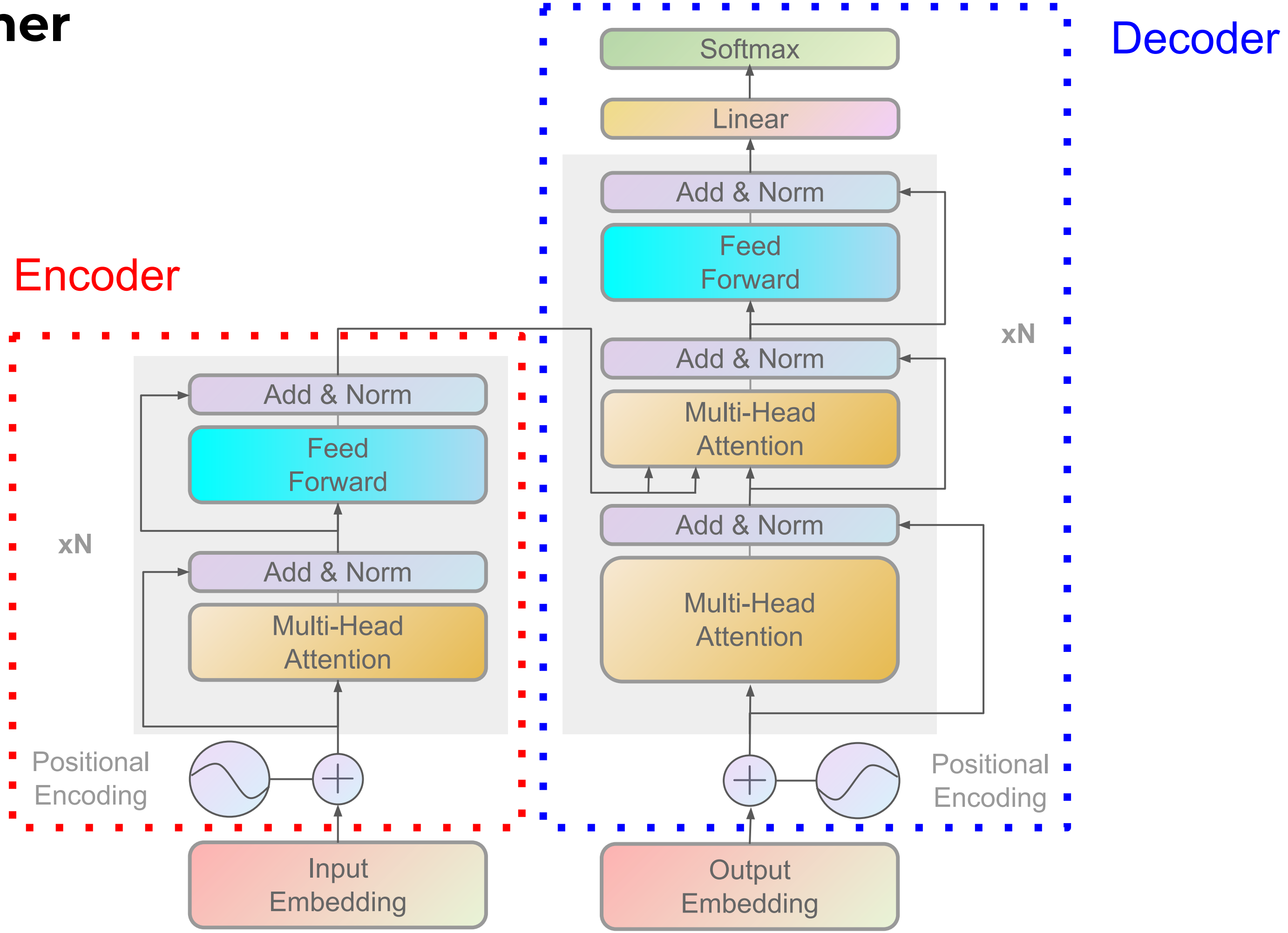


- **Architectures**
  AlexNet, VGG, Inception, ResNet, MobileNet, EfficientNet

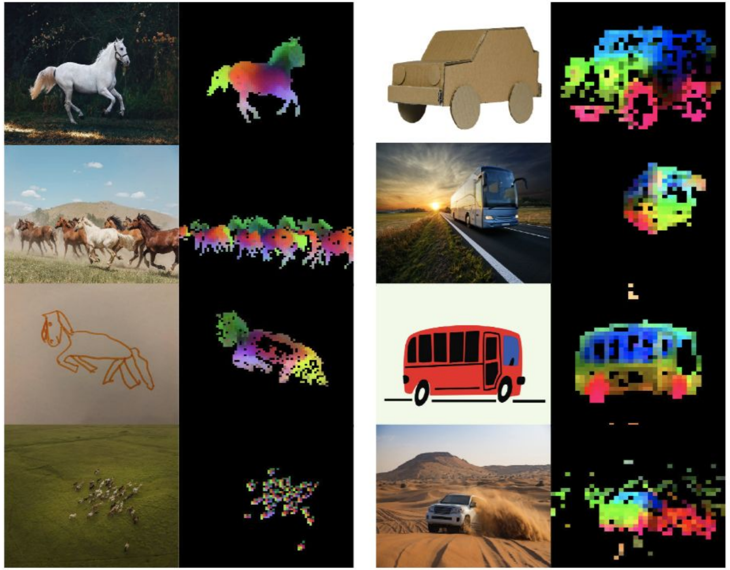
- **Blocks**
  - VGG, Inception
  - ResNet*
  - MobileNet*

- **Layers**
  - Layers [Convolution] [Receptive field]
  - Layers [Dilated Convolution, Deformable Convolution]
  - Layers [Group Convolutions and its variants]
  - Layers [Upsampling, Learnable Upsampling]
  - Layers [Normalization, Batch Norm, Dropout]

Transformer



DINOv2



Model	#Params	Inference GFLOPs Res. 256	Res. 512
CNX-Tiny	29M	5	20
CNX-Small	50M	11	46
CNX-Base	89M	20	81
CNX-Large	198M	38	152
ViT-S	21M	12	63
ViT-S+	29M	16	79
ViT-B	86M	47	216
ViT-L	300M	163	721
ViT-H+	840M	450	1903
ViT-7B	6716M	3550	14515

(a) DINOv3 family of models.

[2304.07193] DINOv2: Learning Robust Visual Features without Supervision

FEATURED

Computer Vision

Introducing Meta Segment Anything Model 3 and Segment Anything Playground

November 19, 2025 · 15 minute read

SAM3

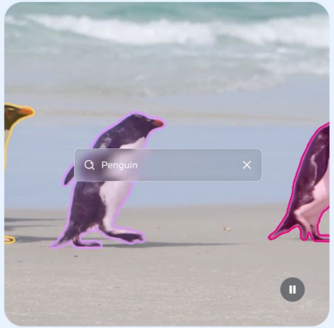
[Introducing Meta Segment Anything Model 3 and Segment Anything Playground](#)

SAM3

ENHANCED CAPABILITIES

Evolution of SAM

The Segment Anything models build on each other, offering increasingly advanced capabilities for developers and researchers to create, experiment and uplevel media workflows.

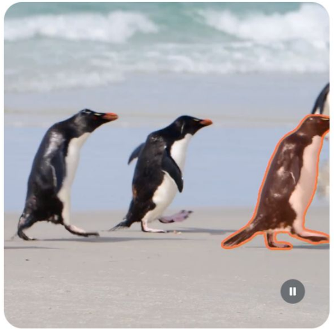


SAM 3

Detect, segment and track every example of any object category in an image or video, using text or examples

- Segment an object from a click
- Track segmented objects in videos
- Refine prediction with follow up clicks
- **Detect and segment matching instances from text**
- **Refine detection with visual examples**

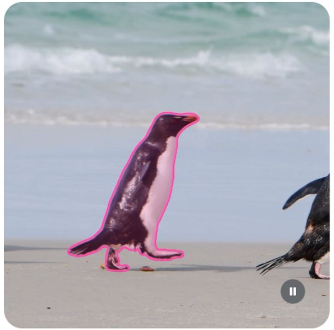
Download the model



SAM 2

Segment and track any object in any image or video using click, box or mask prompts

- Segment an object from a click
- Track segmented objects in videos
- Refine prediction with follow up clicks



SAM 1

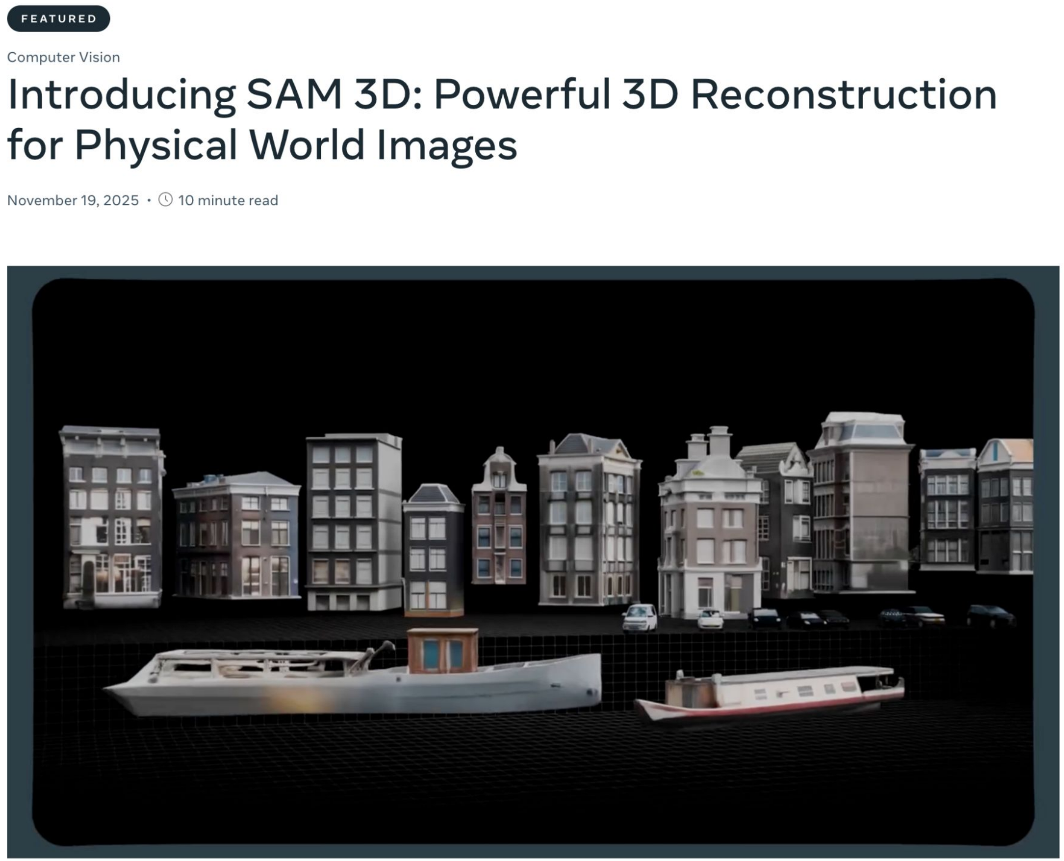
Segment any object in any image with as little as a single click

- Segment an object from a click
- Refine prediction with follow up clicks

paper: SAM 3: Segment Anything with Concepts

Introducing Meta Segment Anything Model 3 and Segment Anything Playground

SAM3 3D



paper: SAM 3D: 3Dfy Anything in Images

paper: SAM 3D Body

Introducing SAM 3D: Powerful 3D Reconstruction for Physical World Images

A new era of intelligence with Gemini 3

Nov 18, 2025
11 min read

Gemini 3 is our most intelligent model that helps you bring any idea to life.

Sundar Pichai
CEO, Google and Alphabet

Demis Hassabis
CEO, Google DeepMind

Koray Kavukcuoglu
CTO, Google DeepMind and Chief AI Architect, Google

Share

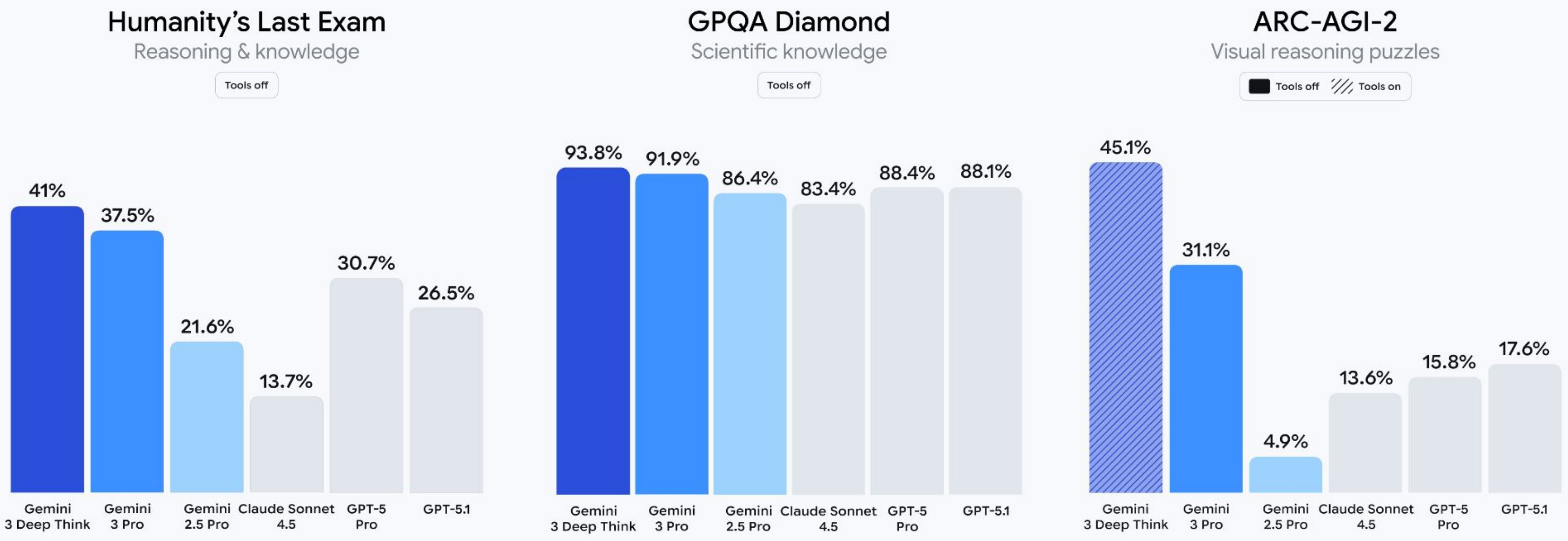
Read AI-generated summary

Benchmark	Description		Gemini 3 Pro	Gemini 2.5 Pro	Claude Sonnet 4.5	GPT-5.1
Humanity's Last Exam	Academic reasoning	No tools With search and code execution	**37.5%** **45.8%**	21.6% —	13.7% —	26.5% —
ARC-AGI-2	Visual reasoning puzzles	ARC Prize Verified	**31.1%**	4.9%	13.6%	17.6%
GPQA Diamond	Scientific knowledge	No tools	**91.9%**	86.4%	83.4%	88.1%
AIME 2025	Mathematics	No tools With code execution	**95.0%** **100%**	88.0% —	87.0% **100%**	94.0% —
MathArena Apex	Challenging Math Contest problems		**23.4%**	0.5%	1.6%	1.0%
MMMU-Pro	Multimodal understanding and reasoning		**81.0%**	68.0%	68.0%	76.0%
ScreenSpot-Pro	Screen understanding		**72.7%**	11.4%	36.2%	3.5%
CharXiv Reasoning	Information synthesis from complex charts		**81.4%**	69.6%	68.5%	69.5%
OmniDocBench 1.5	OCR	Overall Edit Distance, lower is better	**0.115**	0.145	0.145	0.147
Video-MMMU	Knowledge acquisition from videos		**87.6%**	83.6%	77.8%	80.4%
LiveCodeBench Pro	Competitive coding problems from Codeforces, ICPC, and IOI	Elo Rating, higher is better	**2,439**	1,775	1,418	2,243
Terminal-Bench 2.0	Agentic terminal coding	Terminus-2 agent	**54.2%**	32.6%	42.8%	47.6%
SWE-Bench Verified	Agentic coding	Single attempt	76.2%	59.6%	**77.2%**	76.3%
τ2-bench	Agentic tool use		**85.4%**	54.9%	84.7%	80.2%
Vending-Bench 2	Long-horizon agentic tasks	Net worth (mean), higher is better	**$5,478.16**	$573.64	$3,838.74	$1,473.43
FACTS Benchmark Suite	Held out internal grounding, parametric, MM, and search retrieval benchmarks		**70.5%**	63.4%	50.4%	50.8%
SimpleQA Verified	Parametric knowledge		**72.1%**	54.5%	29.3%	34.9%
MMMLU	Multilingual Q&A		**91.8%**	89.5%	89.1%	91.0%
Global PIQA	Commonsense reasoning across 100 Languages and Cultures		**93.4%**	91.5%	90.1%	90.9%
MRCR v2 (8-needle)	Long context performance	128k (average) 1M (pointwise)	**77.0%** **26.3%**	58.0% 16.4%	47.1% not supported	61.6% not supported

For details on our evaluation methodology please see deepmind.google/models/evals-methodology/gemini-3-pro

"Beyond text, Gemini 3 Pro redefines multimodal reasoning with 81% on MMMU-Pro and 87.6% on Video-MMMU."

Gemini 3 Deep Think



Methodology: deepmind.google/models/evals-methodology/gemini-3-pro

	Opus 4.5	Sonnet 4.5	Opus 4.1	Gemini 3 Pro	GPT-5.1
Agentic coding SWE-bench Verified	**80.9%**	77.2%	74.5%	76.2%	76.3% 77.9% Codex-Max
Agentic terminal coding Terminal-bench 2.0	**59.3%**	50.0%	46.5%	54.2%	47.6% 58.1% Codex-Max
Agentic tool use τ2-bench	Retail **88.9%** Telecom **98.2%**	Retail 86.2% Telecom 98.0%	Retail 86.8% Telecom 71.5%	Retail 85.3% Telecom 98.0%	— —
Scaled tool use MCP Atlas	**62.3%**	43.8%	40.9%	—	—
Computer use OSWorld	**66.3%**	61.4%	44.4%	—	—
Novel problem solving ARC-AGI-2 (Verified)	**37.6%**	13.6%	—	31.1%	17.6%
Graduate-level reasoning GPQA Diamond	87.0%	83.4%	81.0%	**91.9%**	88.1%
Visual reasoning MMMU (validation)	80.7%	77.8%	77.1%	—	**85.4%**
Multilingual Q&A MMMLU	90.8%	89.1%	89.5%	**91.8%**	91.0%

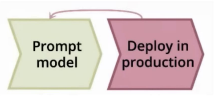
[24 Nov] Anthropic : Introducing Claude Opus 4.5

Content of today's lecture

- Generative modeling
    - appetiser
    - approaches and metrics
- GAN
- Diffusion models
    - intro
    - tale of marvelous Gaussians
    - more details
- Stable Diffusion / ControlNet

APIs multimodal LLMs

prompt-based ML



Local multimodal LLMs

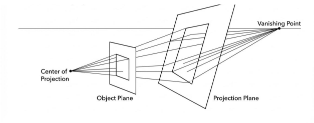
LLAMA, DeepSeek, GEMMA, ...

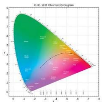
CV foundational models

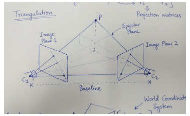
DINO, SAM, ...

CV DL fundamentals

“classical” CV



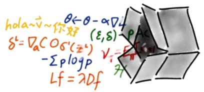




Human Crafted Features

Descriptors:
SIFT, SURF, etc.

math



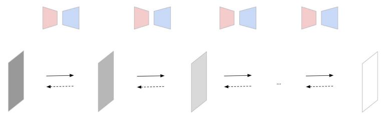
APIs visual generative model

Google Nano Banana, Grok

Local visual generative model

Stable Diffusion, Qwen-Image, ControlNet, ...

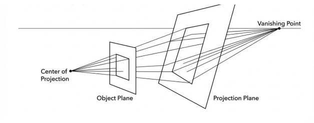
Diffusion models

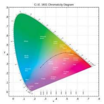


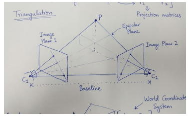
CV DL fundamentals

Image → "Deep" Neural Networks → Result

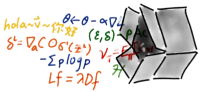
"classical" CV





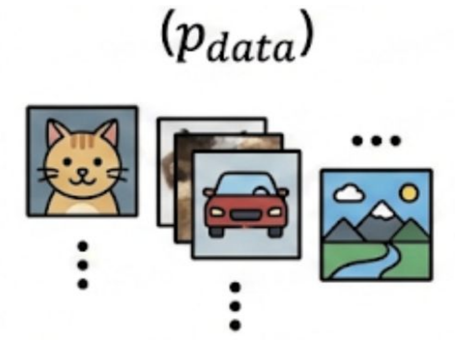


math



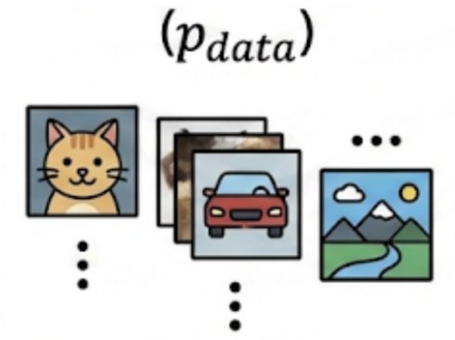
Generative modeling

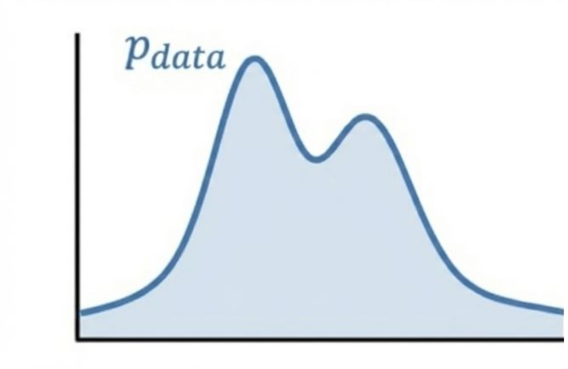
Training data



Generative modeling

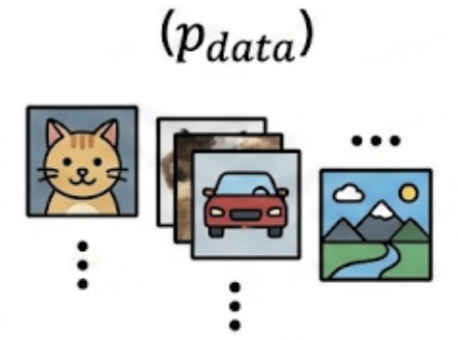
Training data

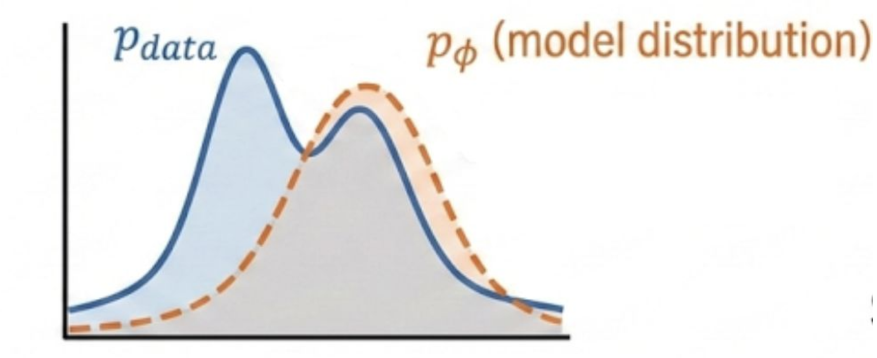




Generative modeling

Training data



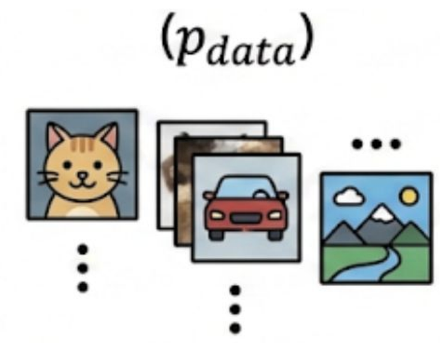


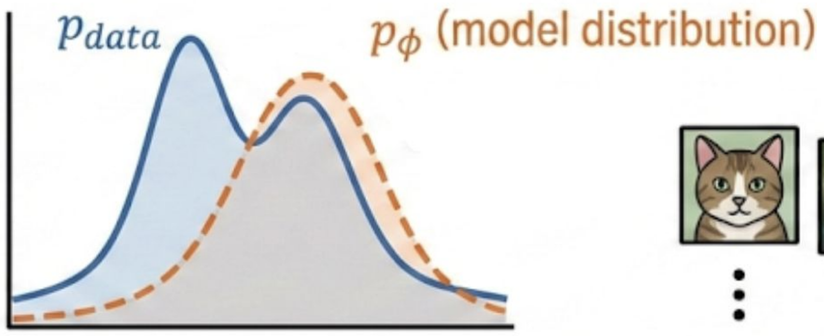
lean

Deep Generative model

Generative modeling

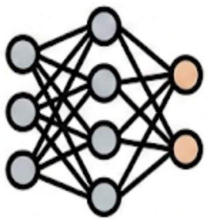
Training data





Generated samples

lean

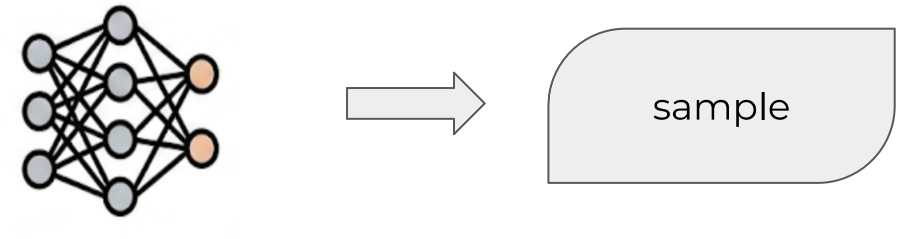


sample

Deep Generative model

Generative modeling

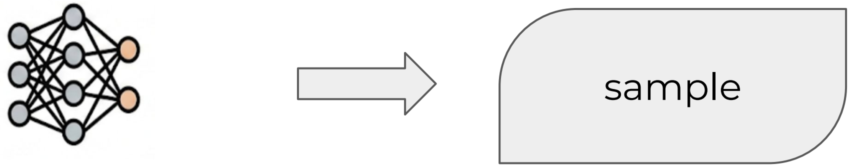
- uncontrollable (unconditional) generation:



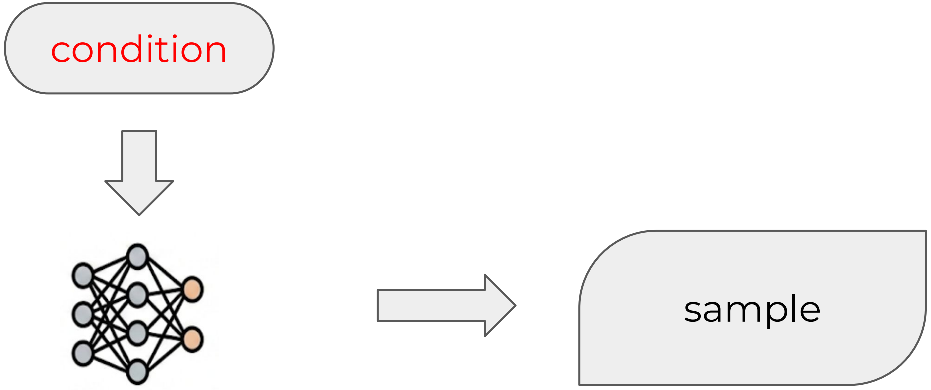
- text
- visual
- sound
- code
- …

Generative modeling

- uncontrollable (unconditional) generation:



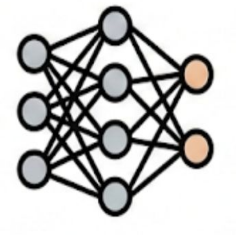
- controllable (conditional) generation:



- text
- visual
- sound
- code
- ...

Generative modeling

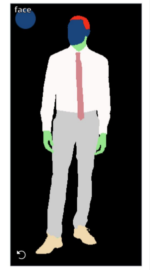
condition:
class: tiger



Generative modeling

Control over multiple conditions:

condition:



+

condition:

Texture Description

upper clothing texture | lower clothing texture | outer clothing texture

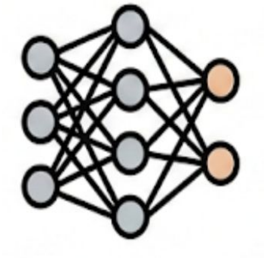
pure color | pure color | plaid/lattice

https://huggingface.co/spaces/CVPR/drawings-to-human

Generative modeling

condition:

"a cover for the cosmopolitan magazine"



Generative modeling


Andrej Karpathy
@karpathy

Fei-Fei to me after I showed her my first image captioning (image to text) network around 2015: “very cool, now do it backwards!”. Me: “haha that’s impossible” 😢. Turns out you just need a few ~B alt-text dataset scrape, transformer, diffusion, and a cluster of ~thousand A100s.

1:10 AM · Aug 31, 2022

170 Reposts 12 Quotes 2,753 Likes 102 Bookmarks

102

Generative modeling

condition:

"Christmas in Lviv"

Generative modeling

[Stable Diffusion 2-1]

[Stable Diffusion 3.5 Large]

створи картинку на якій Карлсон, Че Гевара, Шугєрмен (білий лисий чувак) думають як жити далі коли їх викрили, в стилі кумедного мультфільму

Nano Banana (previous)

Nano Banana Latest

класно, прибери будь-ласка російську літеру "ы" з текстів

все класно, але нехай персонажи сидять не на даху, а в пральні

чудово, нехай в цій пральні відмивають не речі а гроші

нехай в газеті стаття буде не "Викрито банду на Даху" а "НАБУ викрило схему в Енергоатомі"

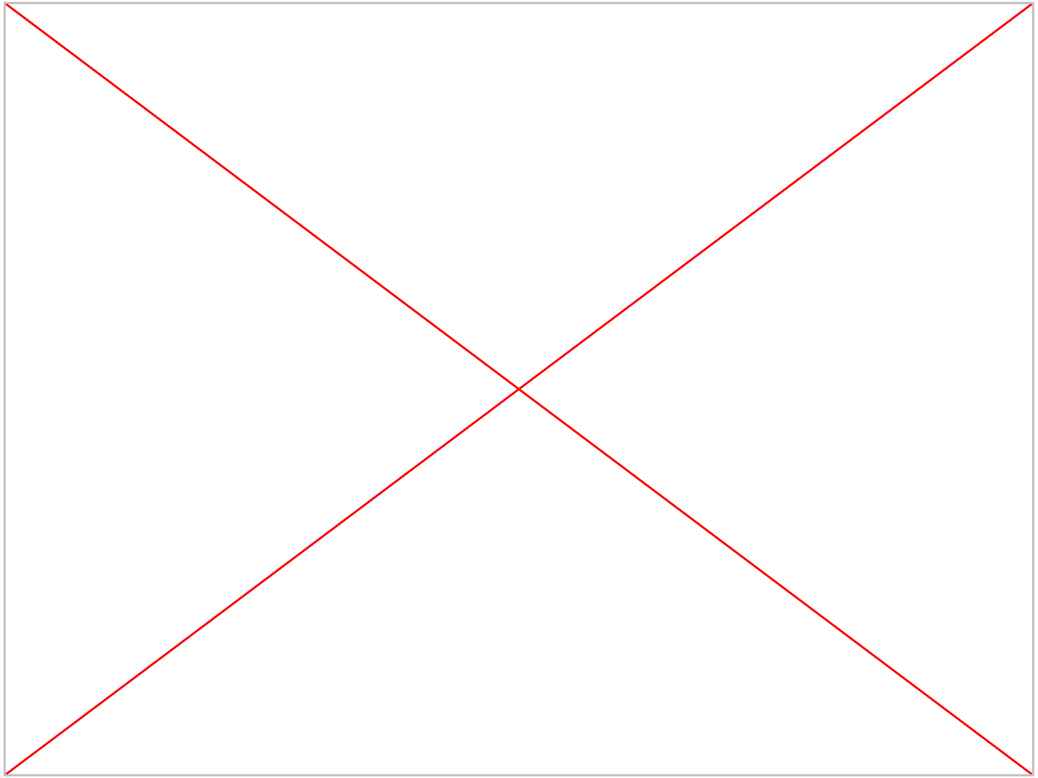
чудово, зміни освітлення - нехай все буде як є, але відбувається без освітлення і тільки свічки

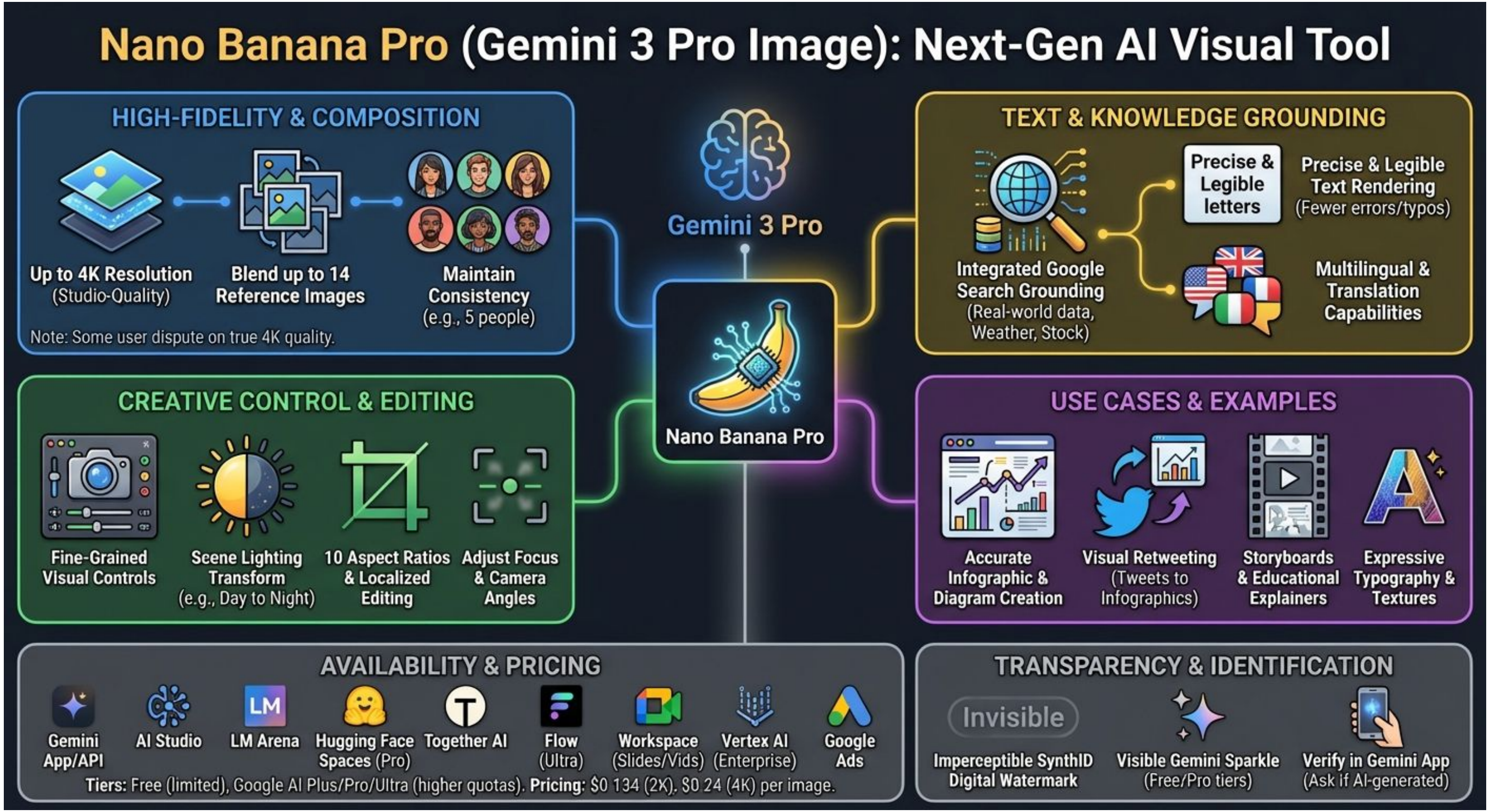
зміни сорочку у лисого чоловіка - зараз воно червона, зроби її кольоровою з цукерками


LAUNDRY
LAUNDRY
LAUNDRY MAT
DIRTY MONEY
CLEAN CASH
НАБУ ВИКРИЛО СХЕМУ
СХЕМУ В ЕНЕРГОАТОМІ

зроби маленьку сцену на основі цієї картинки. хай персонажи говорять українською

зроби маленьку сцену на основі цієї картинки:
Карлсон улітає, лисий чувак теж тікає.
лишається лише ЧеГевара, якого заарештовують, але він за дивні гроші теж виходить на свободу
хай персонажи говорять українською





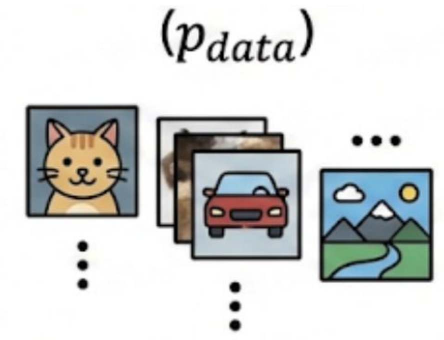
https://news.smol.ai/

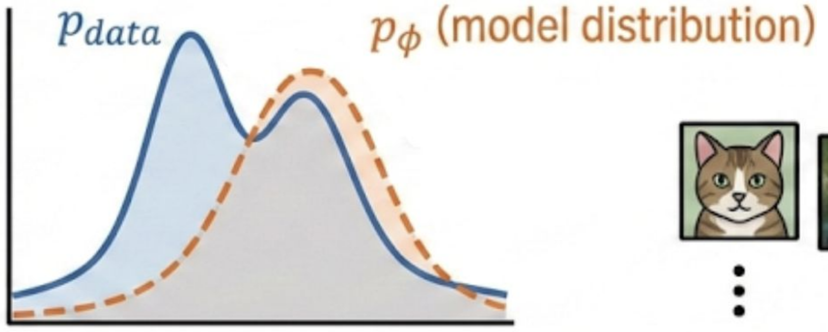
Content of today's lecture

- Generative modeling
    - appetiser
    - **approaches and metrics**
- Diffusion models
    - intro
    - tale of marvelous Gaussians
    - more details
- Stable Diffusion

Generative modeling

Training data



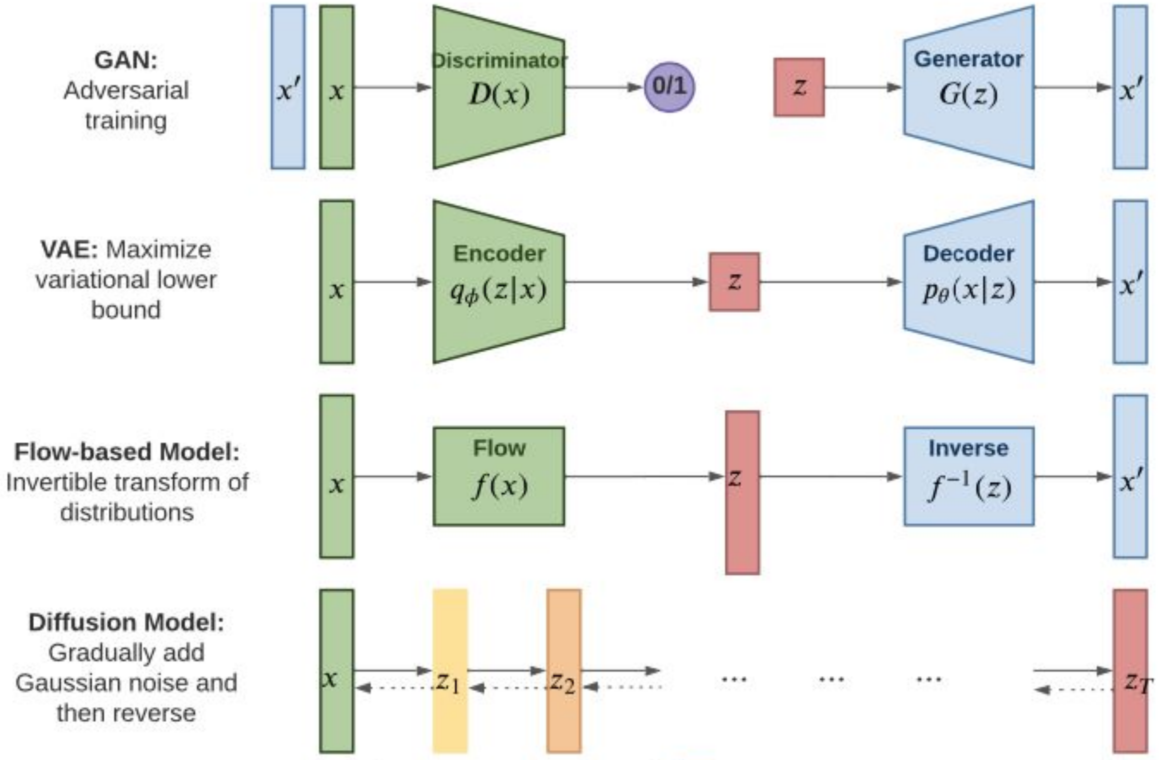


Generated samples

lean

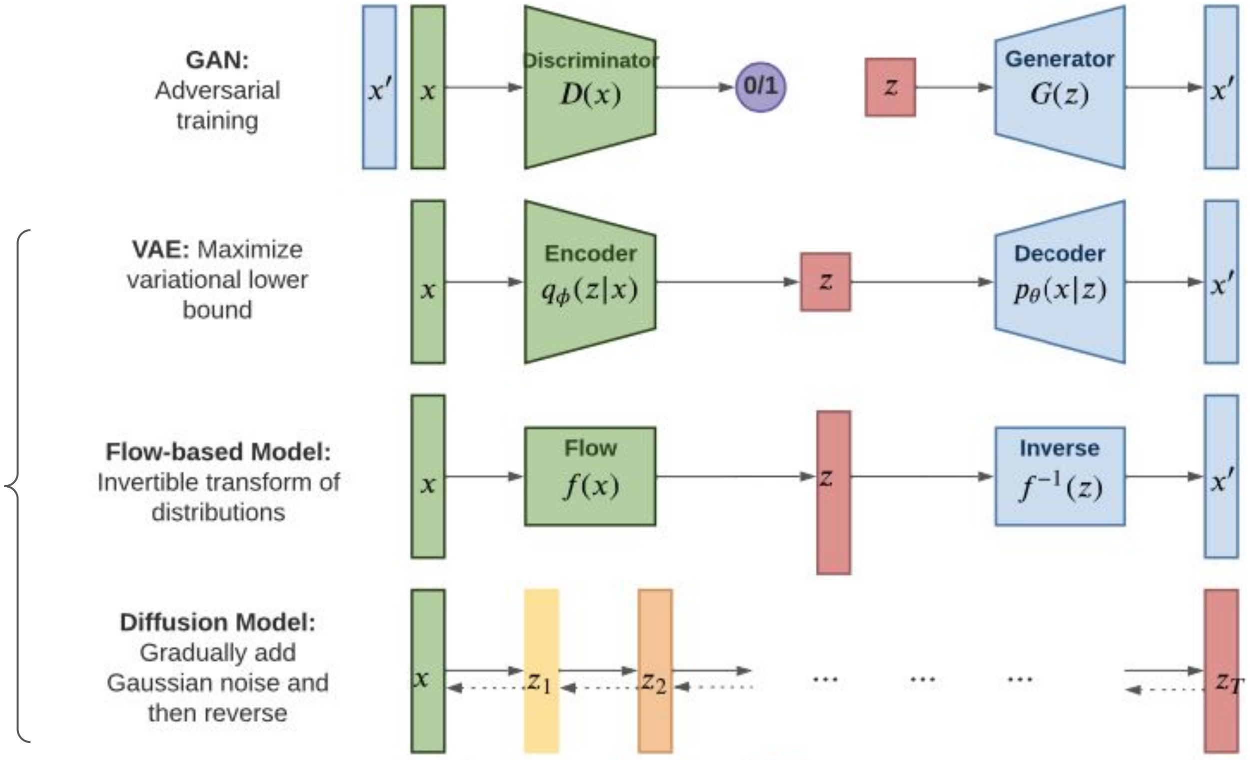
sample

Deep Generative model



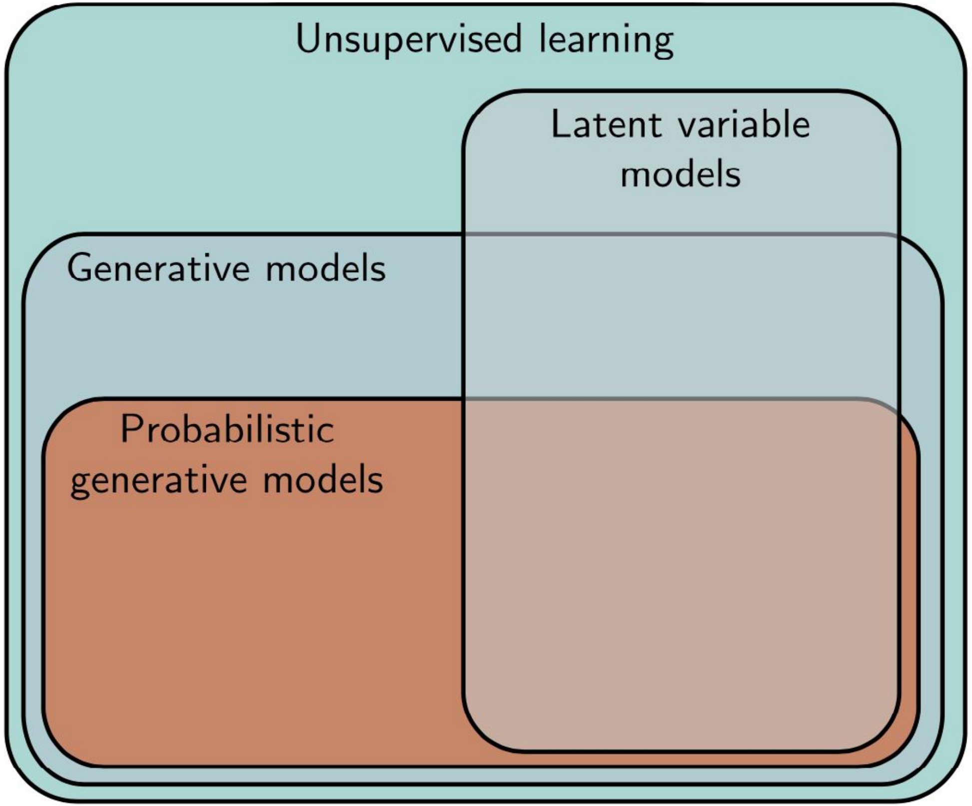
Kevin P. Murphy
"Probabilistic Machine Learning: Advanced Topics", 2023

originally based on
What are Diffusion Models? | Lil'Log



[Kevin P. Murphy](#)
[“Probabilistic Machine Learning: Advanced Topics”, 2023](#)

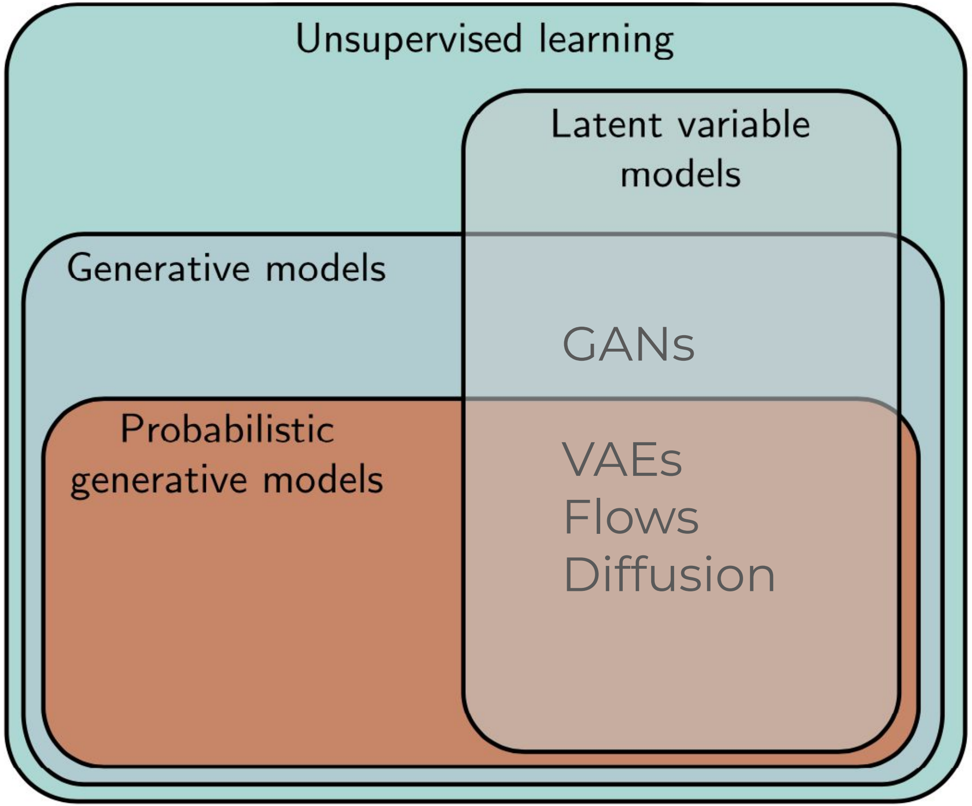
originally based on
[What are Diffusion Models? | Lil'Log](#)



Latent variable models:

$$\mathbf{X} \sim \mathbf{Z}$$

* image from Understanding Deep Learning, book by Simon J.D. Prince, 2023

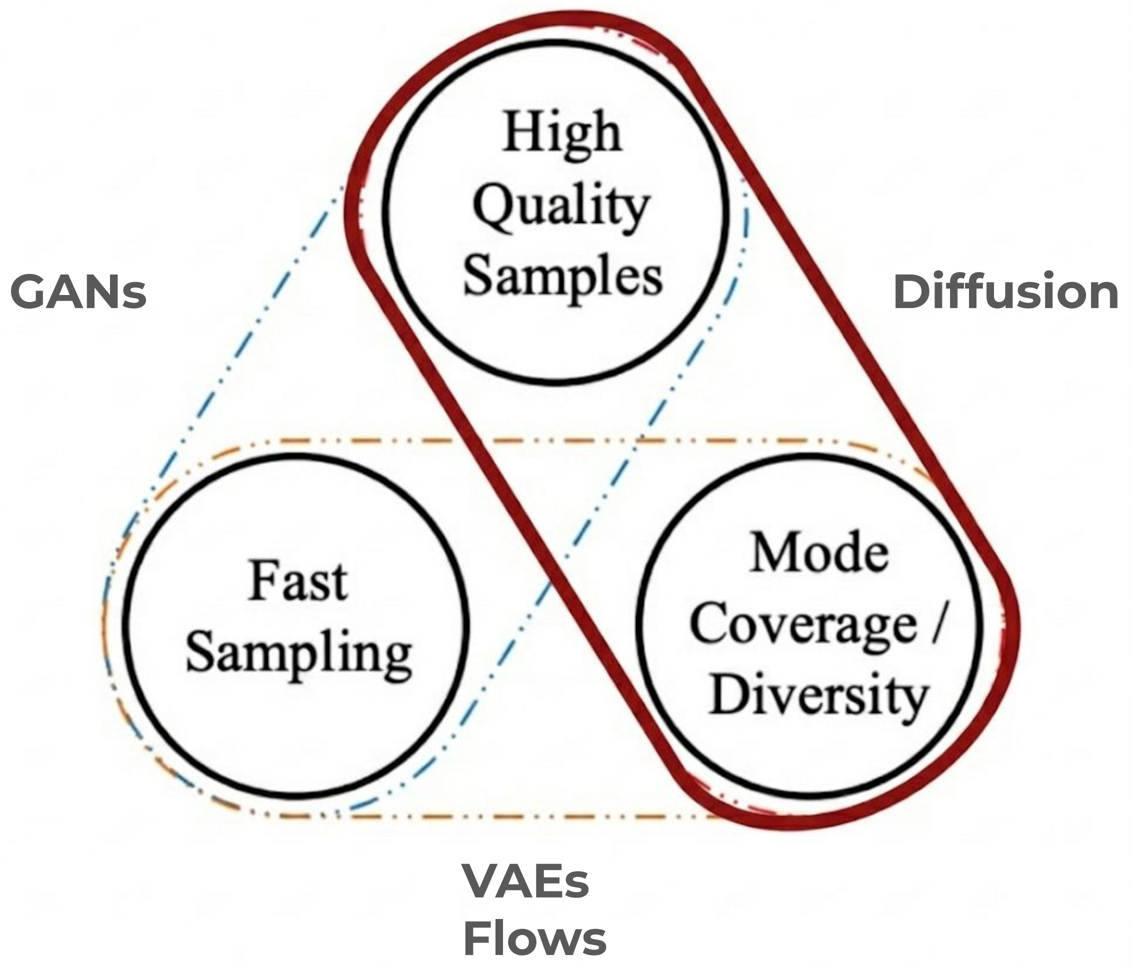


Latent variable models:

$$\mathbf{x} \sim \mathbf{z}$$

* image from Understanding Deep Learning, book by Simon J.D. Prince, 2023

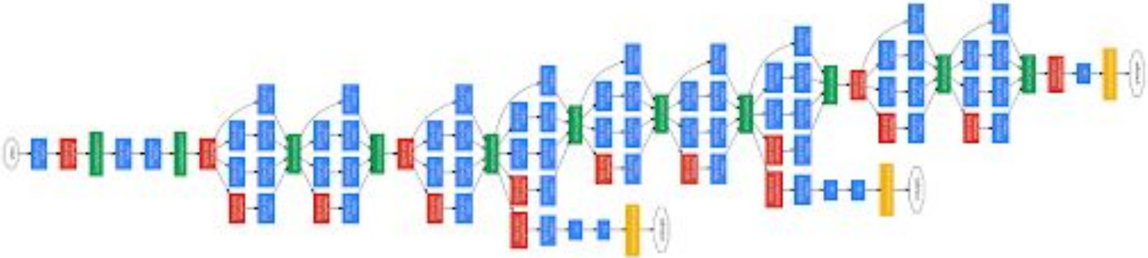
- **High-quality Samples**: The samples should be indistinguishable from the real data with which the model was trained.
- **Fast sampling**: Generating samples from the model should be computationally inexpensive.
- **Mode Coverage / Diversity**: Samples should represent the entire training distribution. It is insufficient to cover only a subset of the training examples.
- **Well-behaved latent space**: Every latent variable z corresponds to a plausible data example x. Smooth changes in z correspond to smooth changes in x.
- **Disentangled latent space**: Manipulating each dimension of z should correspond to changing an interpretable property of the data.

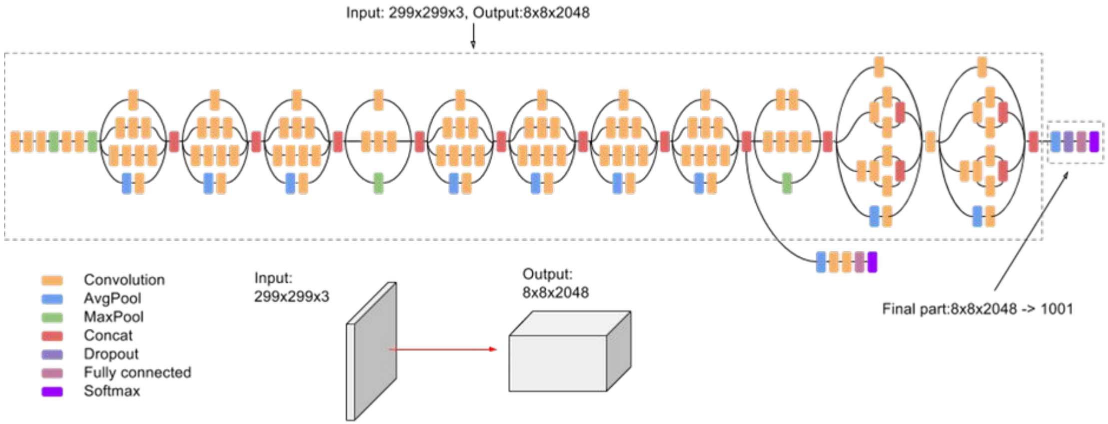

High Quality Samples
Fast Sampling
Mode Coverage / Diversity
GANs
Diffusion
VAEs
Flows

source

Metrics

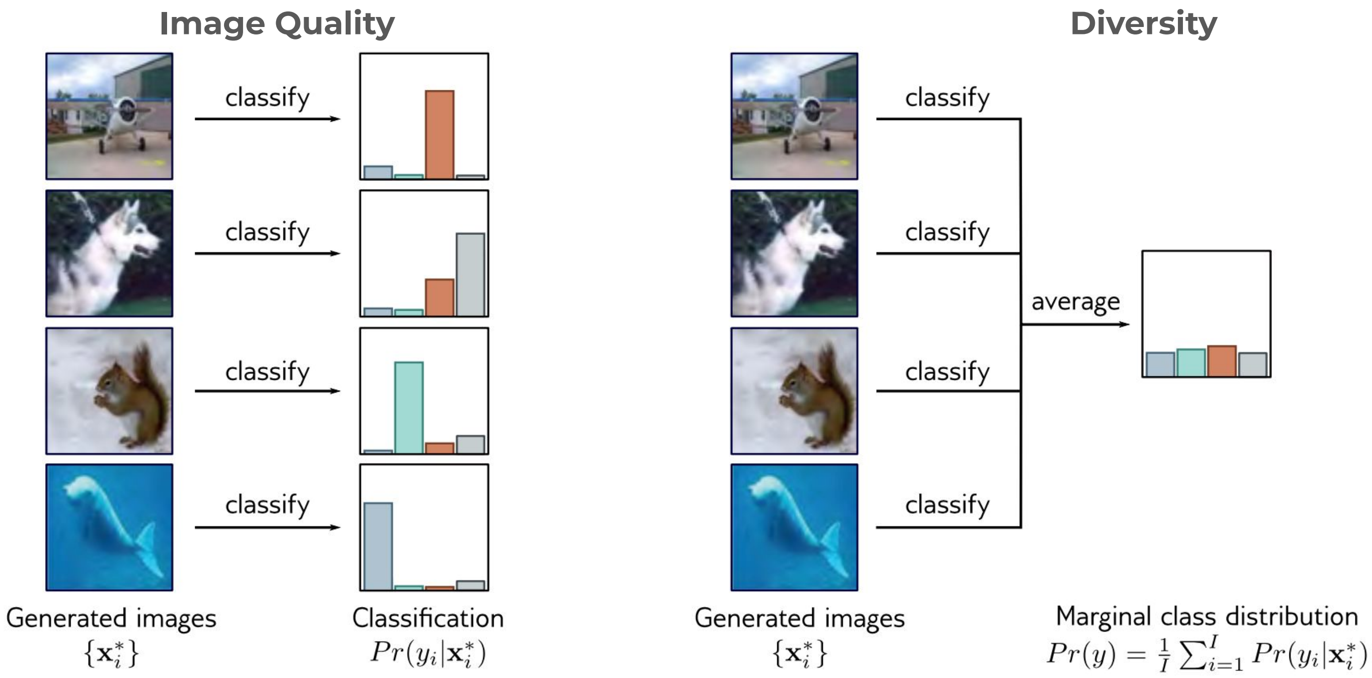
Metrics [Inception score]





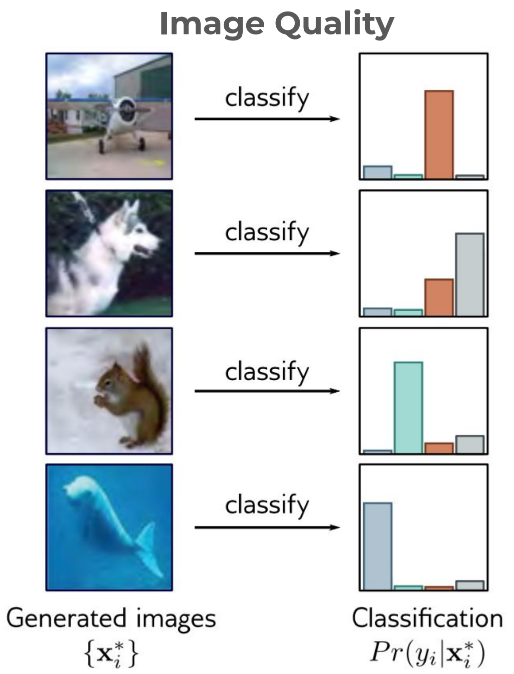
[Improved Techniques for Training GANs]

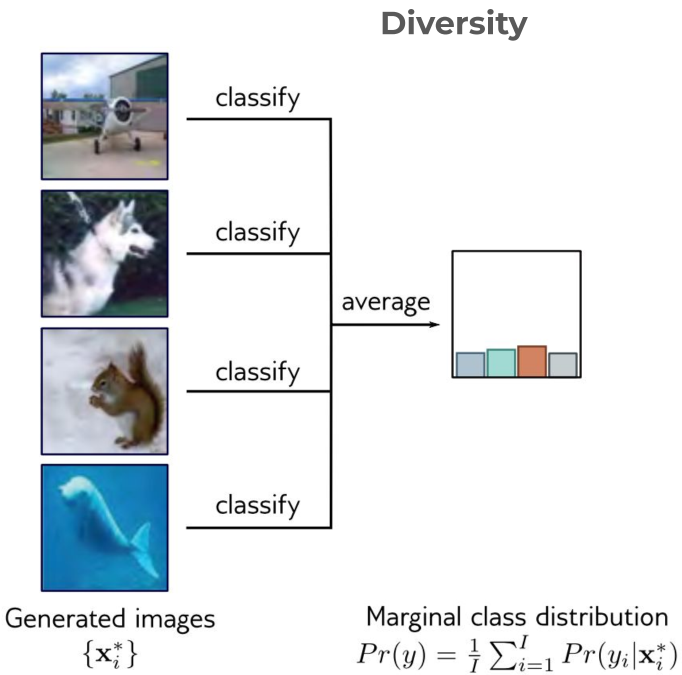
Metrics [Inception score]



- **Image Quality** : each generated image x∗ should look like one and only one of the 1000 possible classes y in the ImageNet database. The probability distribution Pr(y|x∗) should be highly peaked at the correct class.
- **Diversity** : the entire set of generated images should be assigned to the classes with equal probability, so Pr(y) should be flat when averaged over all generated examples.

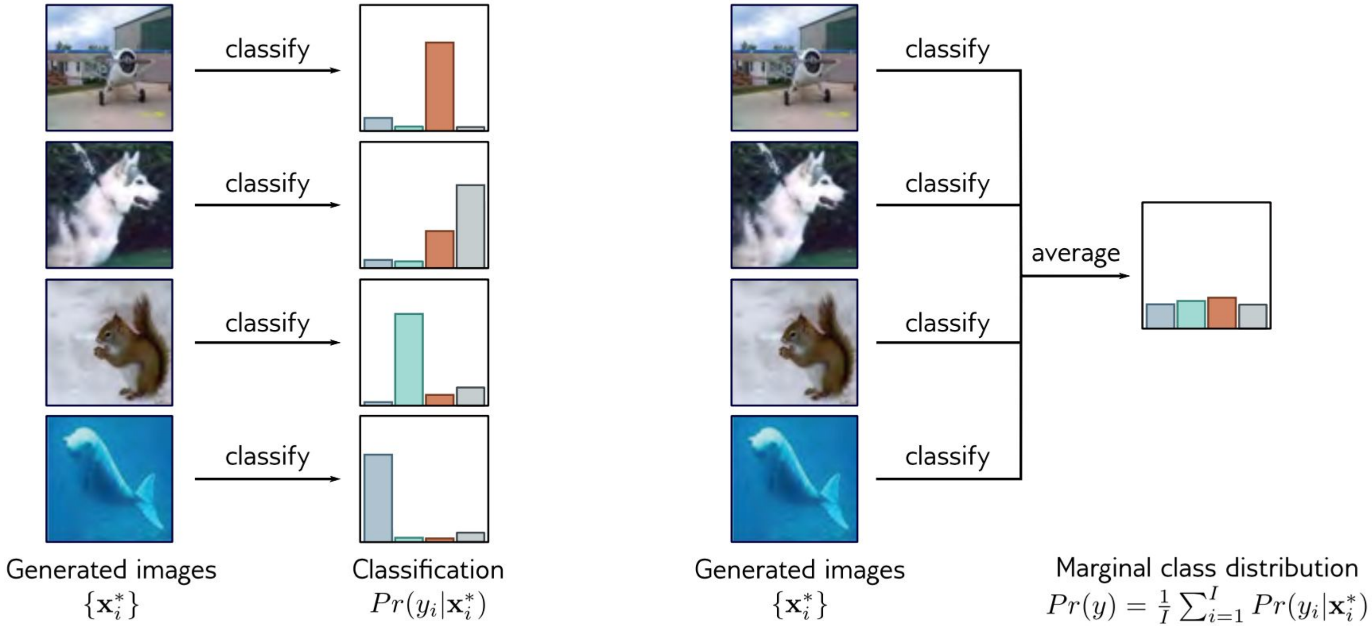
Metrics [Inception score]





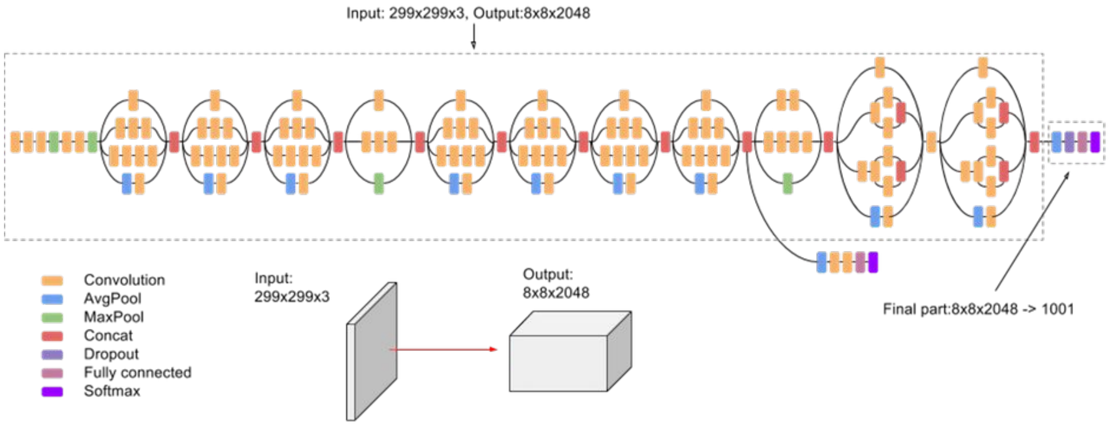
$$IS = \exp\left[\frac{1}{I}\sum_{i=1}^{I} D_{KL}\Big[Pr(y_i|\mathbf{x}_i^*)||Pr(y)\Big]\right]$$

Metrics [Inception score]



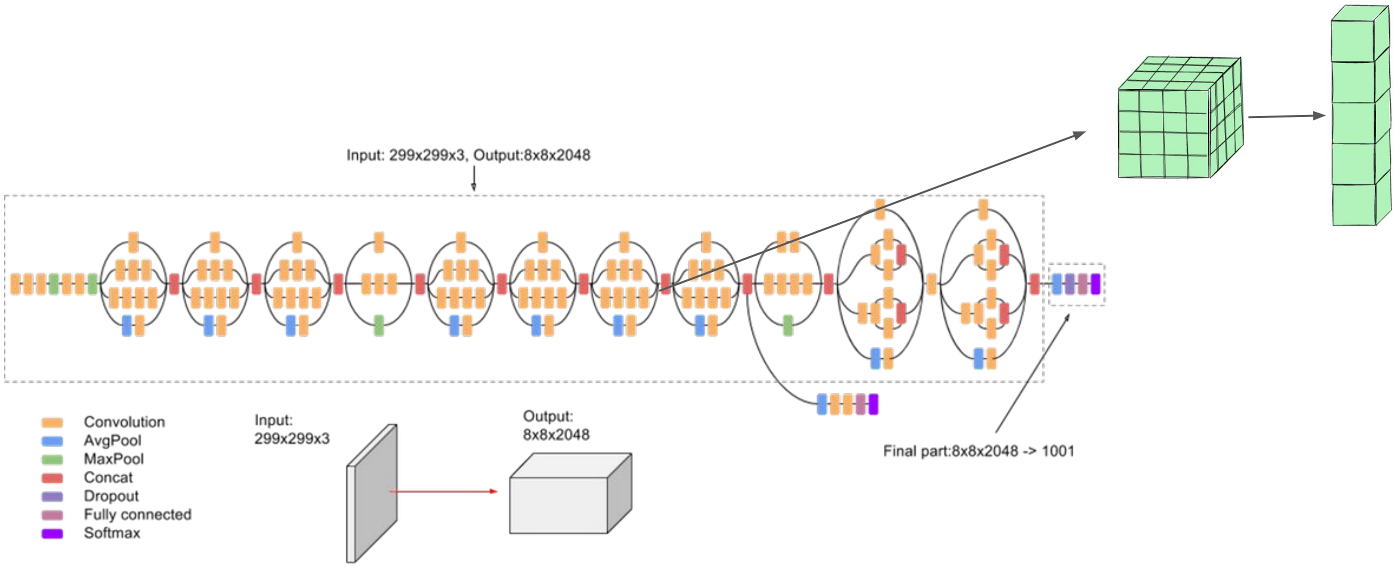
- **Higher is Better**
- IS metric is only sensible for generative models of the ImageNet;
- it does not reward diversity within an object class; it returns a high value if the model only generates one realistic example of each class.

Metrics [Fréchet Inception Score]

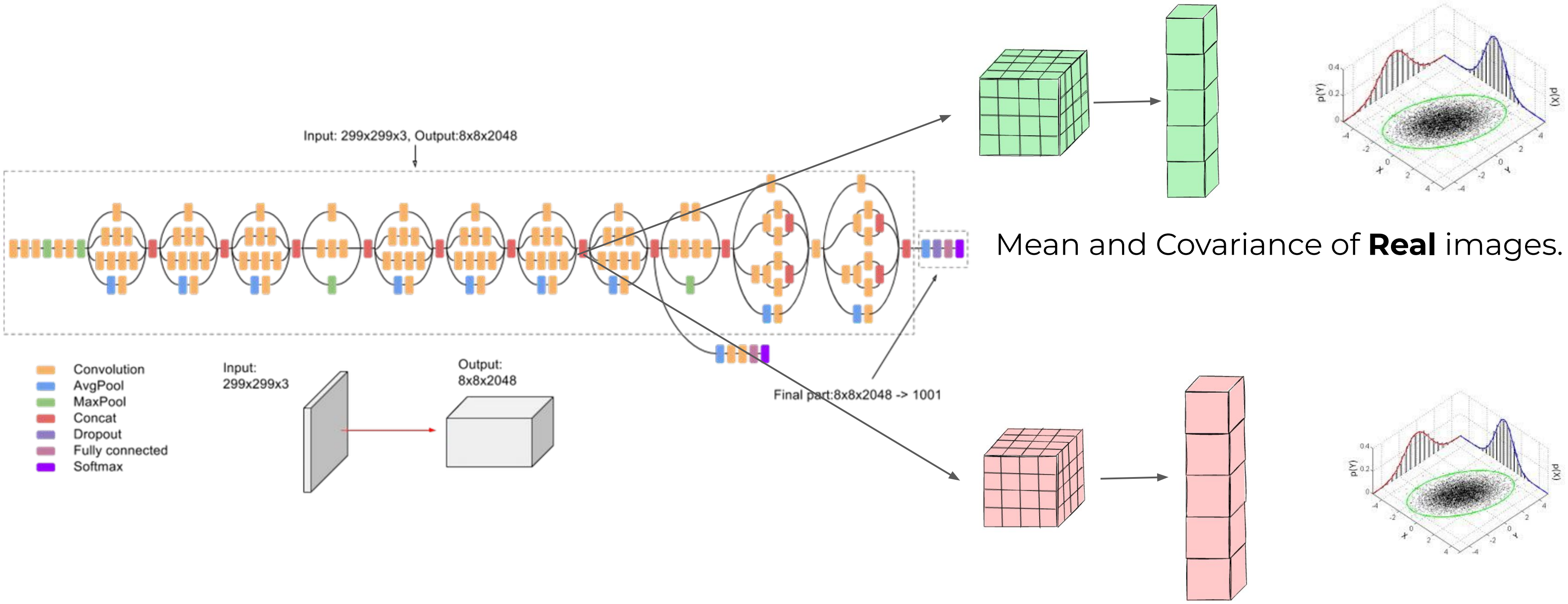


[GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium]

Metrics [Fréchet Inception Score]



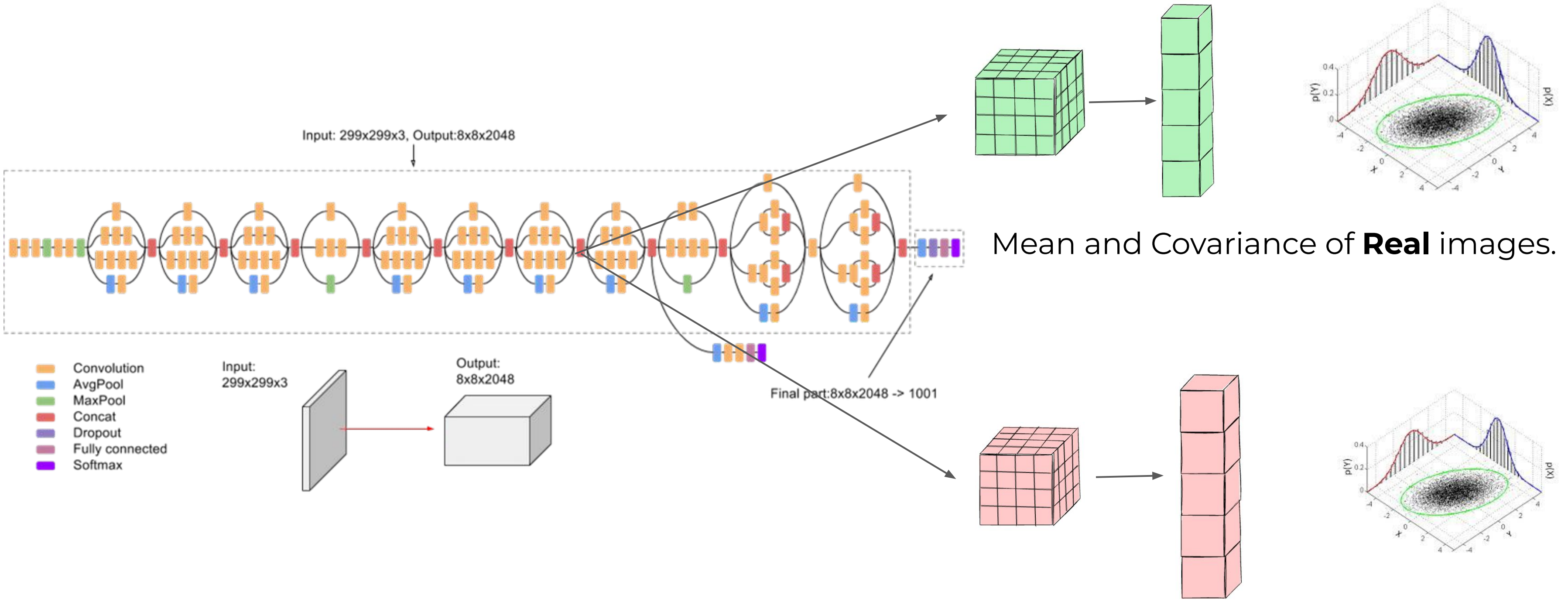
Metrics [Fréchet Inception Score]



Mean and Covariance of **Real** images.

Mean and Covariance of **Generated** images.

Metrics [Fréchet Inception Score]

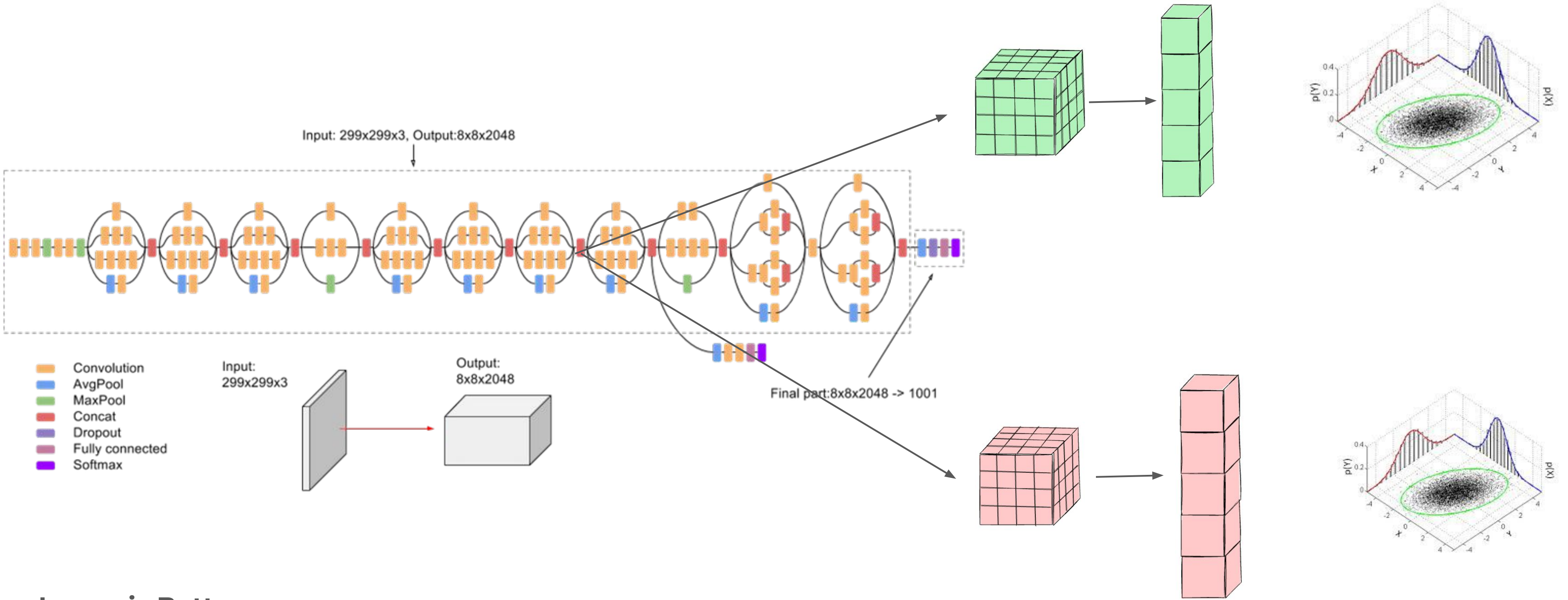


Mean and Covariance of **Real** images.

Mean and Covariance of **Generated** images.

$$D^2_{Fr/W}\left[\mathrm{Norm}[\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1] \middle\| \mathrm{Norm}[\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2]\right] = |\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2|^2 + \mathrm{tr}\left[\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2 - 2\left(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2\right)^{1/2}\right]$$

Metrics [Fréchet Inception Score]



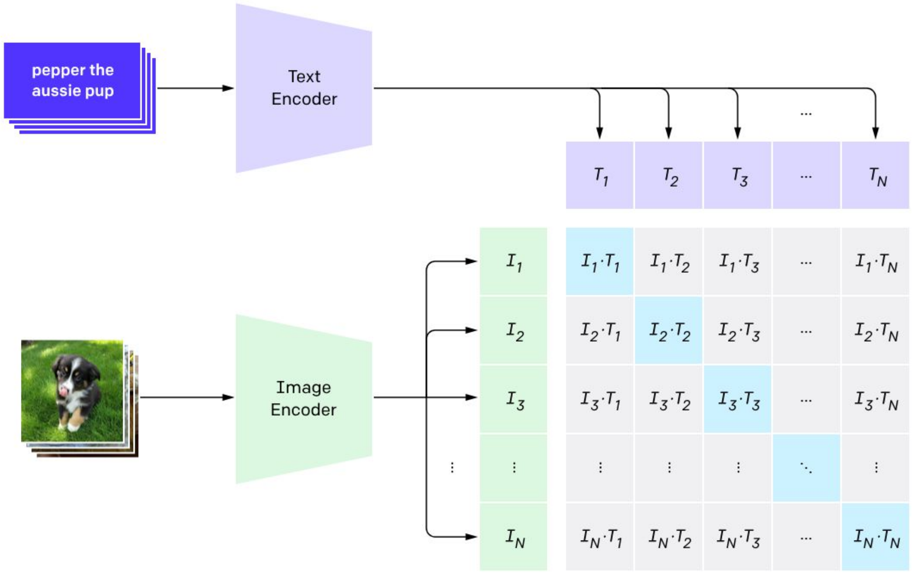
- **Lower is Better**
- **Captures Both Quality and Diversity**: By considering both means and covariances, FID assesses not just how close individual generated images are to real images (quality) but also how diverse the generated images are (diversity).
- **Sensitive to Mode Collapse**: FID can detect when a model generates limited varieties of images (mode collapse), a common issue with GANs.
- **Alignment with Human Perception**: Empirical studies have shown that FID correlates well with human judgment of image quality.

Metrics [Fréchet Inception Score vs. Inception score]

Feature	Inception Score (IS)	Fréchet Inception Distance (FID)
Goal	Measure Clarity & Diversity	Measure Realism (Distance to Ground Truth)
Reference Data	None (Uses pre-trained ImageNet knowledge)	**Requires Real Data** (Compares Real vs. Fake)
Mechanism	Class Probabilities (Softmax Output)	Feature Embeddings (Internal Layer)
Direction	**Higher** is Better	**Lower** is Better
Pros	Good for checking if objects are recognizable.	Captures texture/realism better; sensitive to mode collapse.
Cons	Can't measure similarity to a specific dataset.	Computationally expensive; requires large sample size.

Metrics [CLIP score]

$$\text{CLIPScore(I, C)} = max(100 * cos(E_I, E_C), 0)$$



Concept: Measures the semantic similarity between the **generated image** and the **text prompt**.

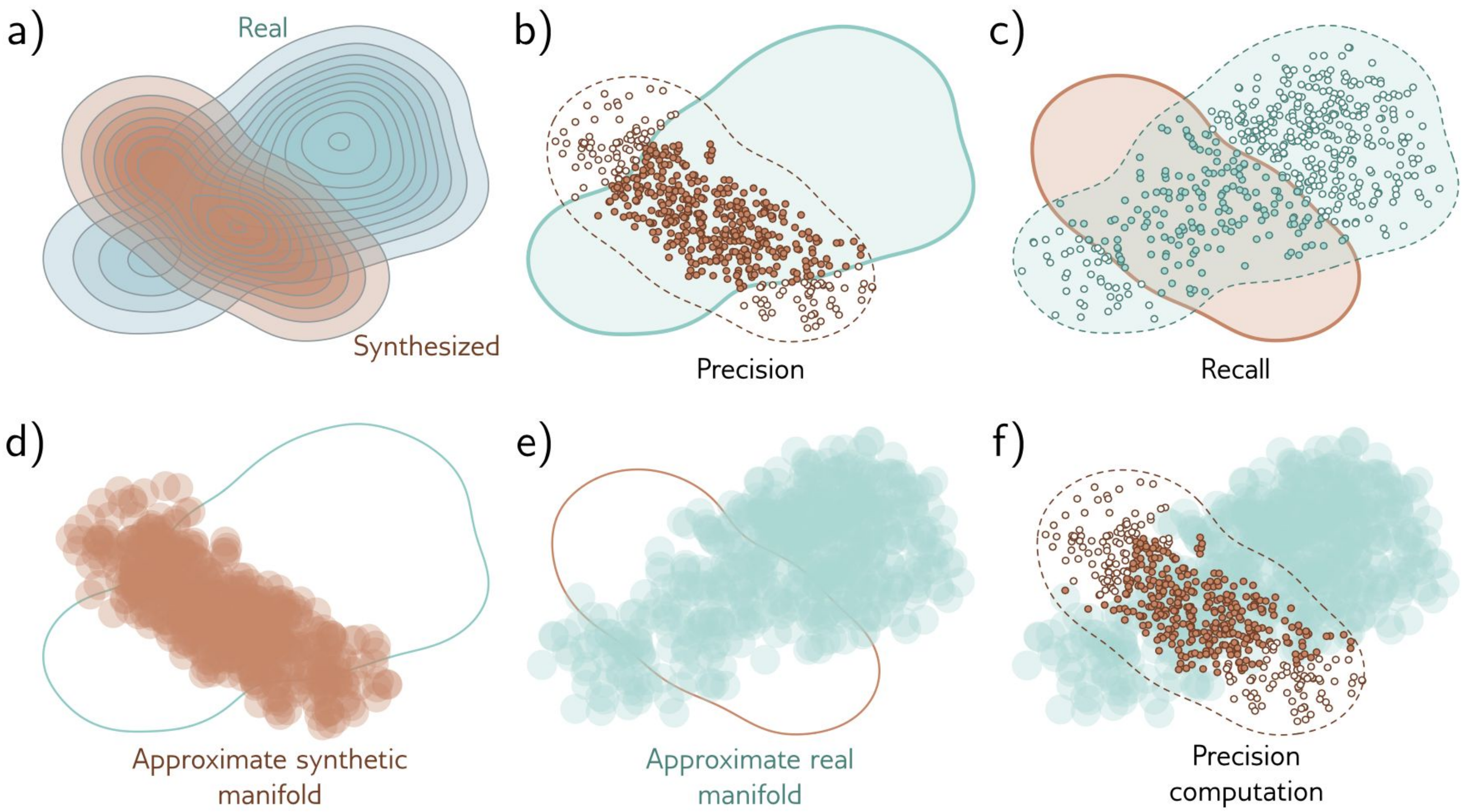
Mechanism: Uses OpenAI's CLIP model to embed both the image and text into the same latent space, then calculates the **Cosine Similarity** between them.

Interpretation:

- **Higher is Better.**
- Range: [-1, 1]
- A high FID but low CLIP score means the image looks realistic but has nothing to do with the prompt.

[CLIP Score — PyTorch-Metrics 1.5.2 documentation]

Metrics [Manifold precision/recall]



ImgNet 256×256	pre-training token	perc.	self-sup.	params	Gflops	FID↓	IS↑
Latent-space Diffusion							
DiT-XL/2 [46]	SD-VAE	VGG	-	675+49M	119	2.27	278.2
SiT-XL/2 [40]	SD-VAE	VGG	-	675+49M	119	2.06	277.5
REPA [74], SiT-XL/2	SD-VAE	VGG	DINOv2	675+49M	119	1.42	305.7
LightningDiT-XL/2 [73]	VA-VAE	VGG	DINOv2	675+49M	119	1.35	295.3
DDT-XL/2 [71]	SD-VAE	VGG	DINOv2	675+49M	119	1.26	310.6
RAE [78], DiT^{DH}-XL/2	RAE	VGG	DINOv2	839+415M	146	**1.13**	262.6
***Pixel**-space (non-diffusion)*							
JetFormer [65]	-	-	-	2.8B	-	6.64	-
FractalMAR-H [36]	-	-	-	848M	-	6.15	348.9
***Pixel**-space Diffusion*							
ADM-G [12]	-	-	-	559M	1120	7.72	172.7
RIN [28]	-	-	-	320M	334	3.95	216.0
SiD [25] , UViT/2	-	-	-	2B	555	2.44	256.3
VDM++, UViT/2	-	-	-	2B	555	2.12	267.7
SiD2 [26], UViT/2	-	-	-	N/A	137	1.73	-
SiD2 [26], UViT/1	-	-	-	N/A	653	**1.38**	-
PixelFlow [6], XL/4	-	-	-	677M	2909	1.98	282.1
PixNerd [70], XL/16	-	-	DINOv2	700M	134	2.15	297
JiT-B/16	-	-	-	131M	25	3.66	275.1
JiT-L/16	-	-	-	459M	88	2.36	298.5
JiT-H/16	-	-	-	953M	182	1.86	303.4
JiT-G/16	-	-	-	2B	383	1.82	292.6

Table 7. **Reference results on ImageNet 256×256.** FID [21] and

[2511.13720] Back to Basics: Let Denoising Generative Models Denoise

Method	FID ↓	CLIP-score ↑	CLIP-aes. ↑
Stable Diffusion	6.09	0.26	6.32
VQGAN [19](seg.)*	26.28	0.17	5.14
LDM [72](seg.)*	25.35	0.18	5.15
PITI [89](seg.)	19.74	0.20	5.77
ControlNet-lite	17.92	0.26	6.30
ControlNet	15.27	0.26	6.31

Table 3: Evaluation for image generation conditioned by semantic segmentation. We report FID, CLIP text-image score, and CLIP aesthetic scores for our method and other baselines. We also report the performance of Stable Diffusion without segmentation conditions. Methods marked with "*" are trained from scratch.

[2302.05543] Adding Conditional Control to Text-to-Image Diffusion Models

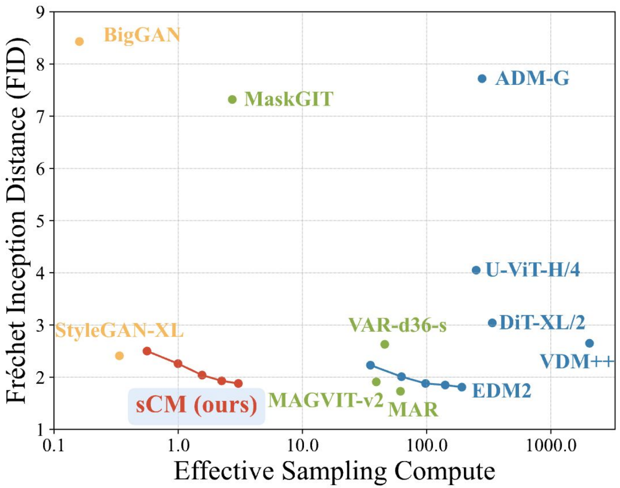


Figure 1: Sample quality vs. effective sampling compute (billion parameters × number of function evaluations during sampling). We compare the sample quality of different models on ImageNet 512×512, measured by FID (↓). Our 2-step sCM achieves sample quality comparable to the best previous generative models while using less than 10% of the effective sampling compute.

[2410.11081] Simplifying, Stabilizing and Scaling Continuous-Time Consistency Models

```
> pip install torchmetrics[image]

import torch
from torchmetrics.image.inception import InceptionScore
from torchmetrics.image.fid import FrechetInceptionDistance

BATCH_SIZE = 32
NUM_BATCHES = 2

def get_dummy_images (batch_size):
    return torch.randint(0, 255, (batch_size, 3, 299, 299),
dtype=torch.uint8)

# Inception Score (IS)
metric_is = InceptionScore(feature= "logits_unbiased" )

print("Calculating Inception Score..." )
for _ in range(NUM_BATCHES):
    fake_imgs = get_dummy_images(BATCH_SIZE)
    metric_is.update(fake_imgs)

is_mean, is_std = metric_is.compute()
print(f"Inception Score: {is_mean.item():.4f} +/-
{is_std.item():.4f}")
```

```
# Fréchet Inception Distance (FID)
metric_fid =
FrechetInceptionDistance(feature= 2048)

print("\nCalculating FID..." )
for _ in range(NUM_BATCHES):
    # 1. Feed REAL images (ground truth)
    real_imgs = get_dummy_images(BATCH_SIZE)
    metric_fid.update(real_imgs, real= True)

    # 2. Feed GENERATED images (fake)
    fake_imgs = get_dummy_images(BATCH_SIZE)
    metric_fid.update(fake_imgs, real= False)

fid_score = metric_fid.compute()
print(f"FID Score: {fid_score.item():.4f}")

# Cleanup
metric_is.reset()
metric_fid.reset()
```

```
> Calculating Inception Score...
> Inception Score: 1.0453 +/- 0.0222

> Calculating FID...
> FID Score: 7.8714
```

Content of today's lecture

- Generative modeling
  - appetiser
  - approaches

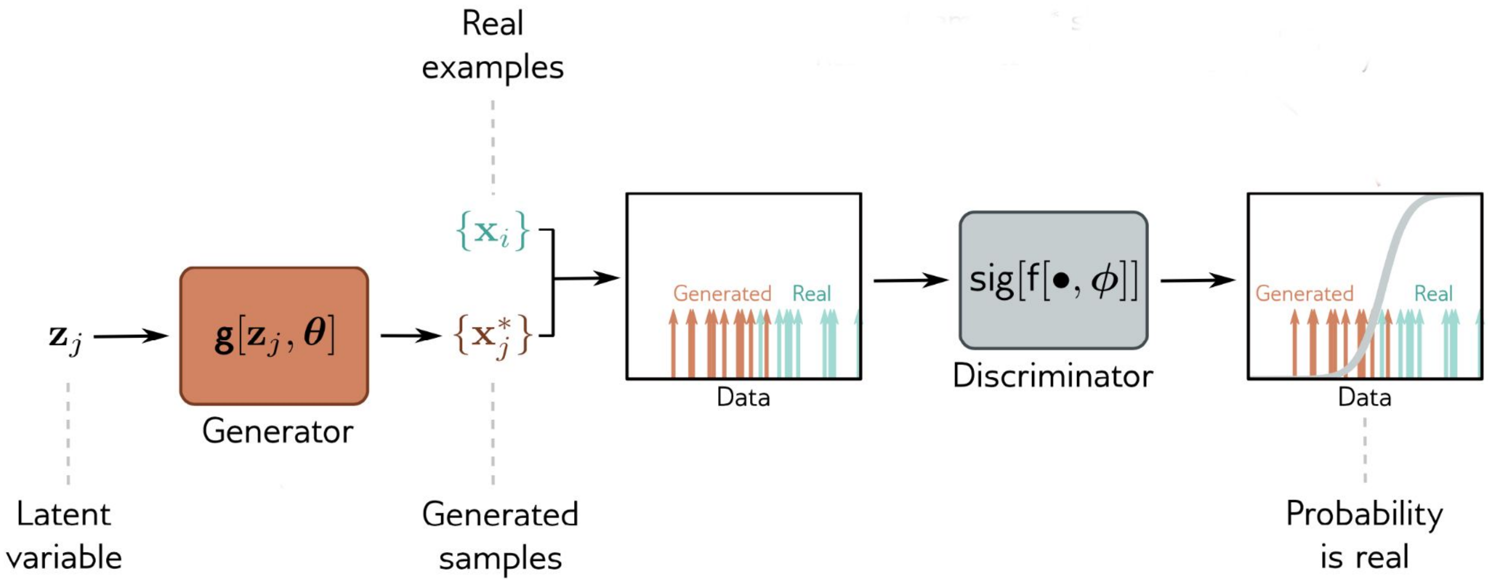
- GAN

- Diffusion models

  - intro
  - tale of marvelous Gaussians
  - more details

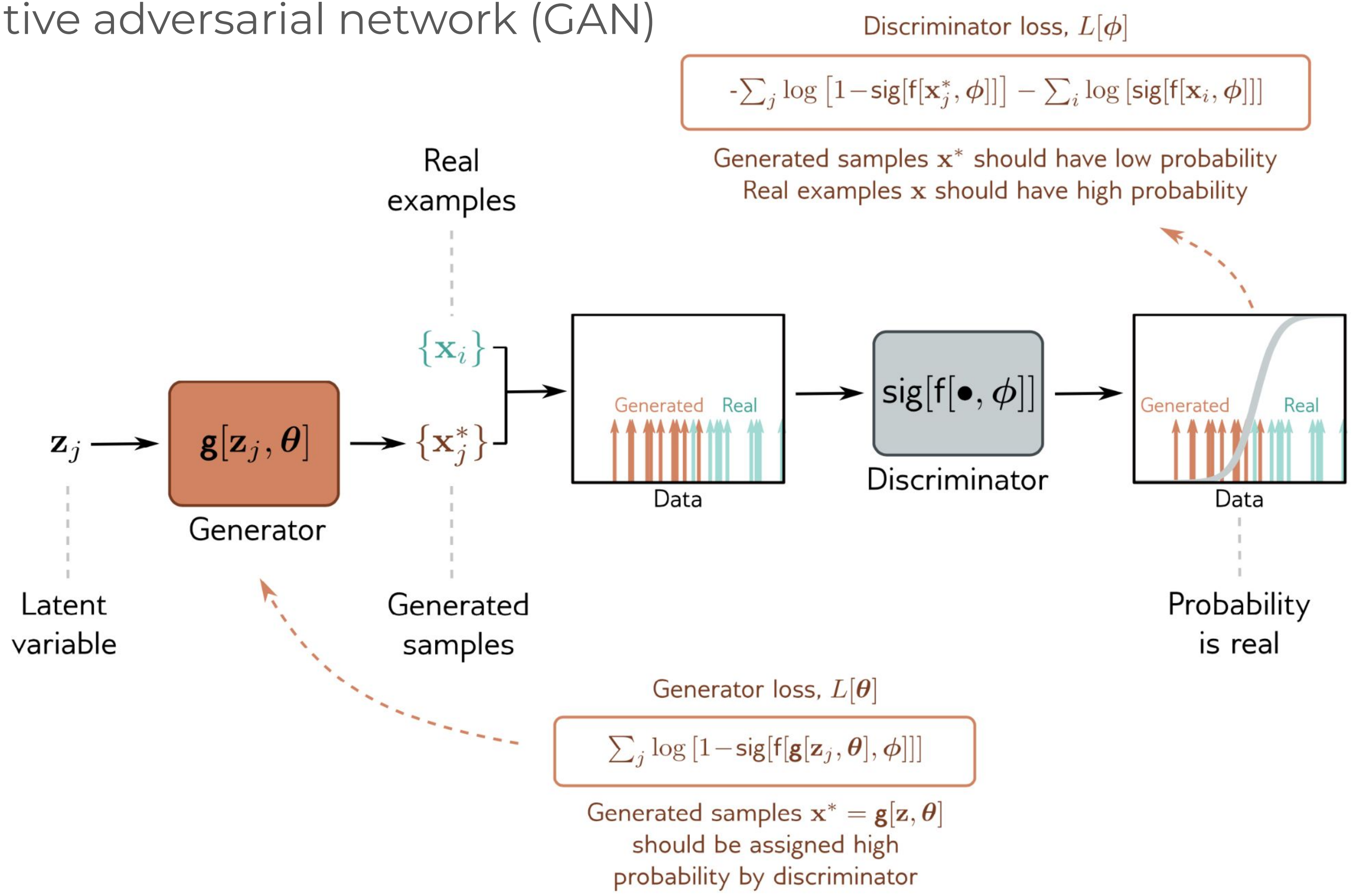
- Stable Diffusion

Generative adversarial network (GAN)

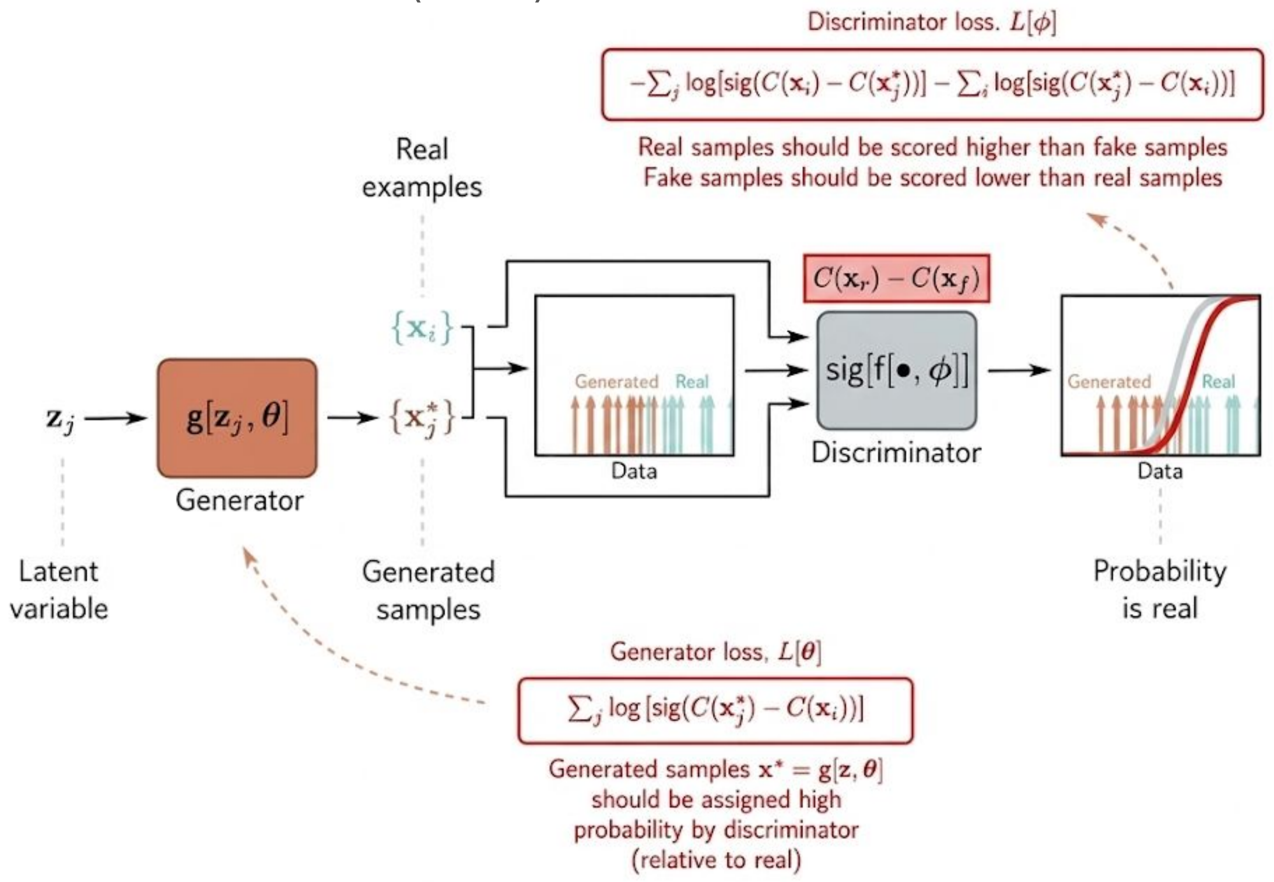


[1406.2661] Generative Adversarial Networks
[2501.05441] The GAN is dead; long live the GAN! A Modern GAN Baseline

Generative adversarial network (GAN)

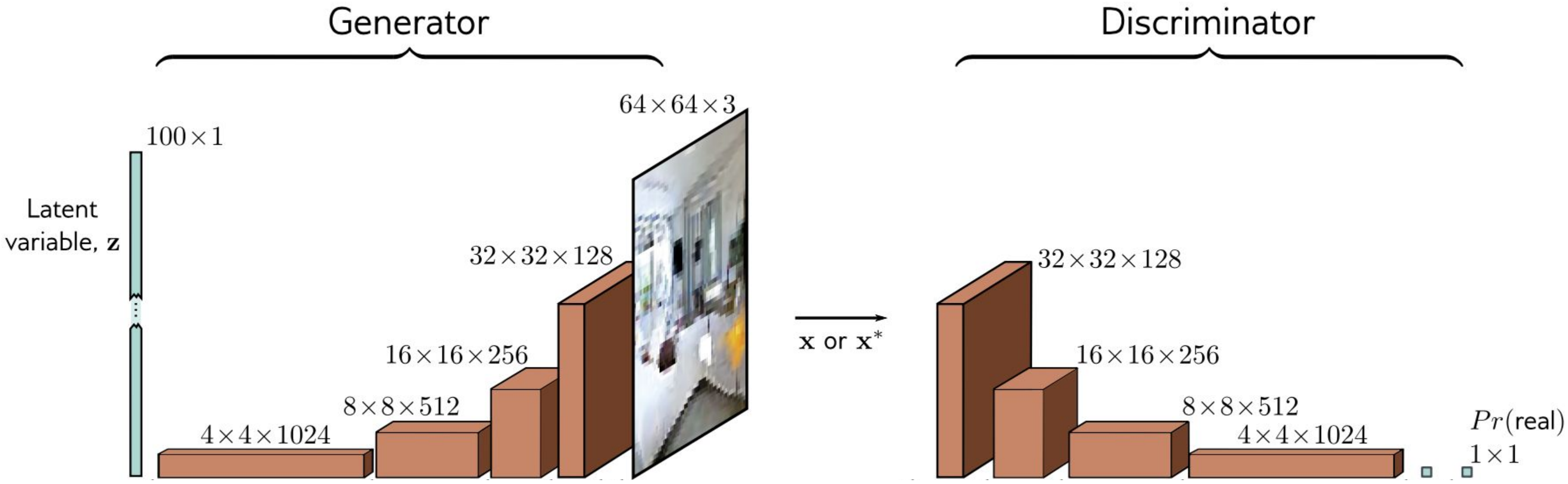


Generative adversarial network (GAN)



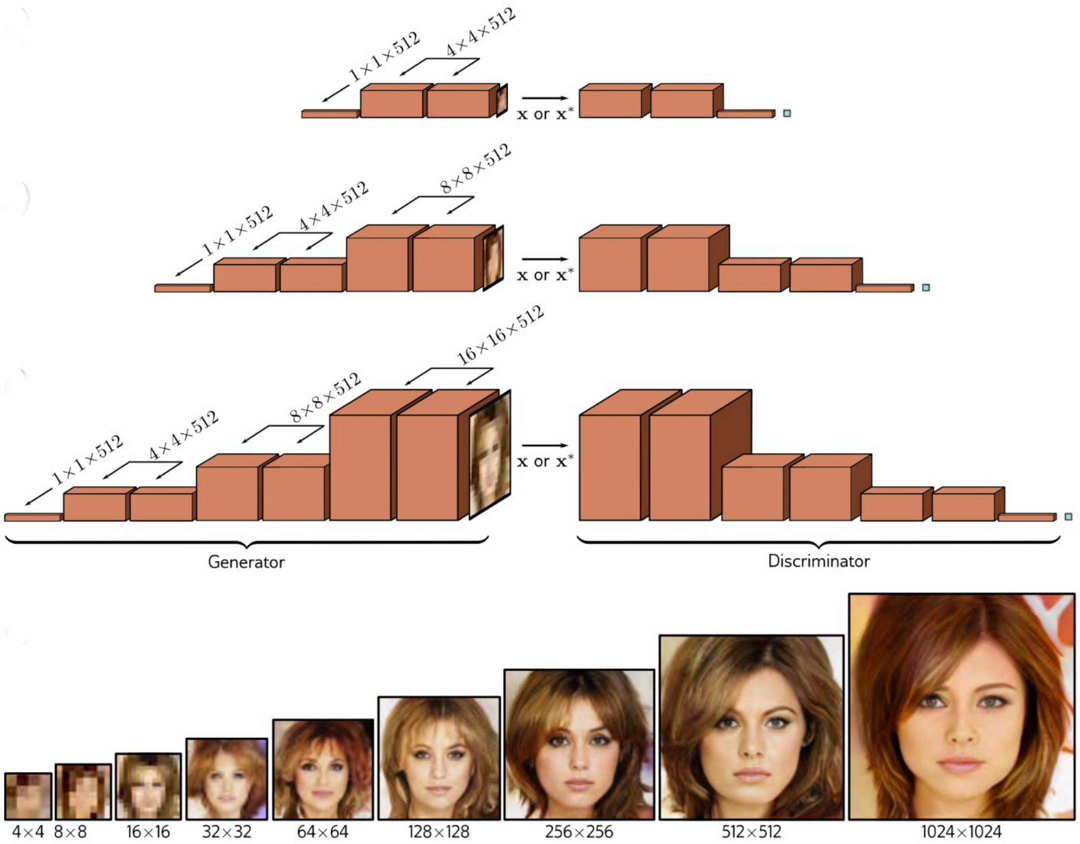
[2501.05441] The GAN is dead; long live the GAN! A Modern GAN Baseline

Generative adversarial network (GAN)



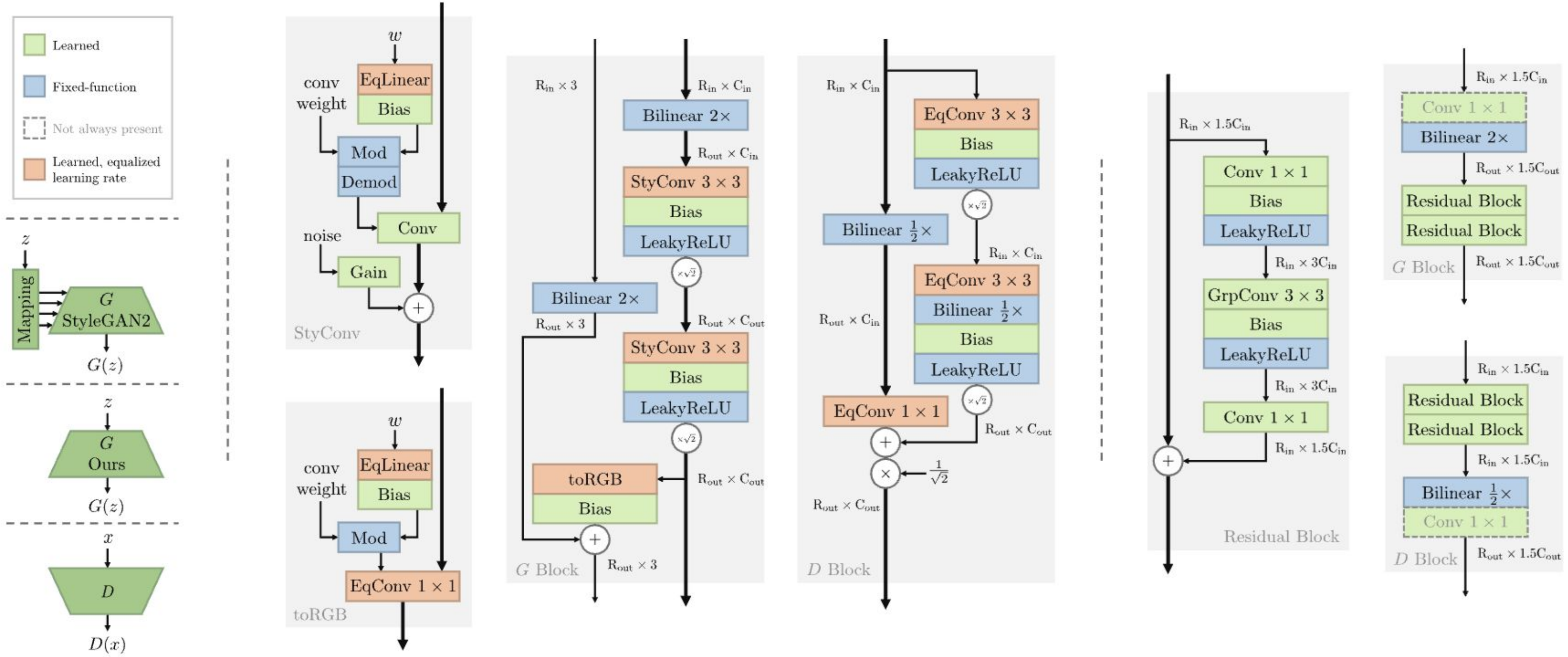
[1511.06434] Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks

Generative adversarial network (GAN)



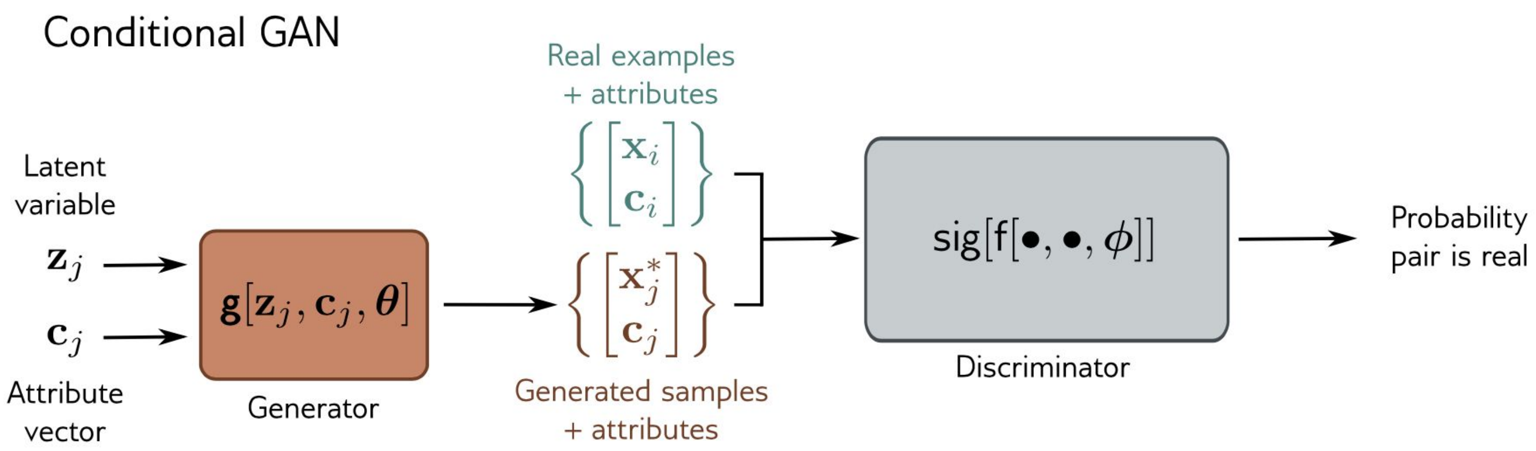
[1710.10196] Progressive Growing of GANs for Improved Quality, Stability, and Variation

Generative adversarial network (GAN)



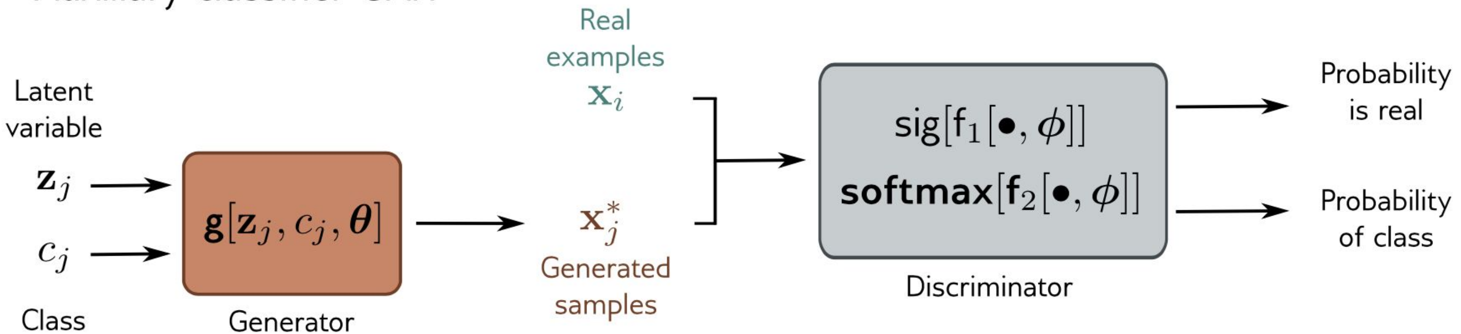
(a) Overall view (b) StyleGAN2 architecture blocks [31] (Config A) (c) Ours (Config E)

[2501.05441] The GAN is dead; long live the GAN! A Modern GAN Baseline



Conditional GAN
Real examples
+ attributes
$\left\{\begin{bmatrix}\mathbf{x}_i \\ \mathbf{c}_i\end{bmatrix}\right\}$
Latent
variable
$\mathbf{z}_j$
$\mathbf{c}_j$
Attribute
vector
$\mathbf{g}[\mathbf{z}_j, \mathbf{c}_j, \boldsymbol{\theta}]$
Generator
$\left\{\begin{bmatrix}\mathbf{x}_j^* \\ \mathbf{c}_j\end{bmatrix}\right\}$
Generated samples
+ attributes
$\text{sig}[\text{f}[\bullet, \bullet, \phi]]$
Discriminator
Probability
pair is real

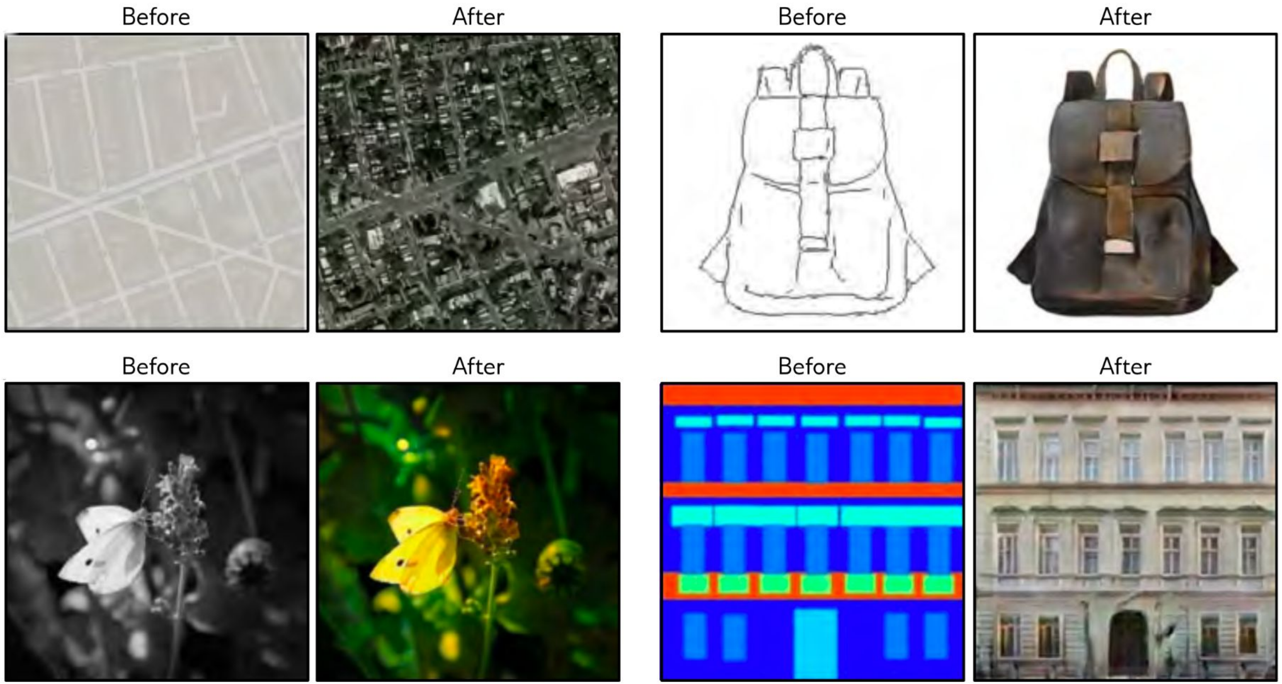
Auxiliary classifier GAN



Conditional generation

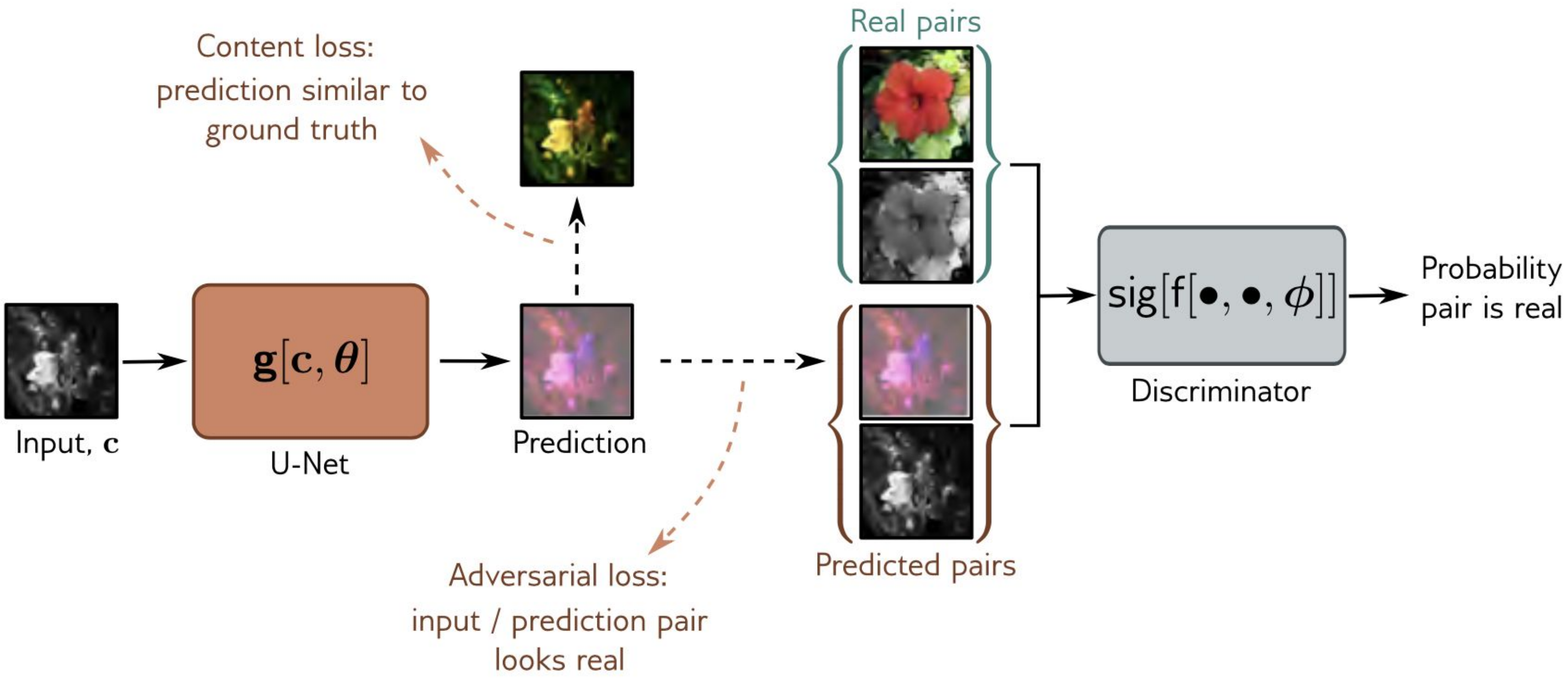
[1610.09585] Conditional image synthesis with auxiliary classifier GANs

Image translation



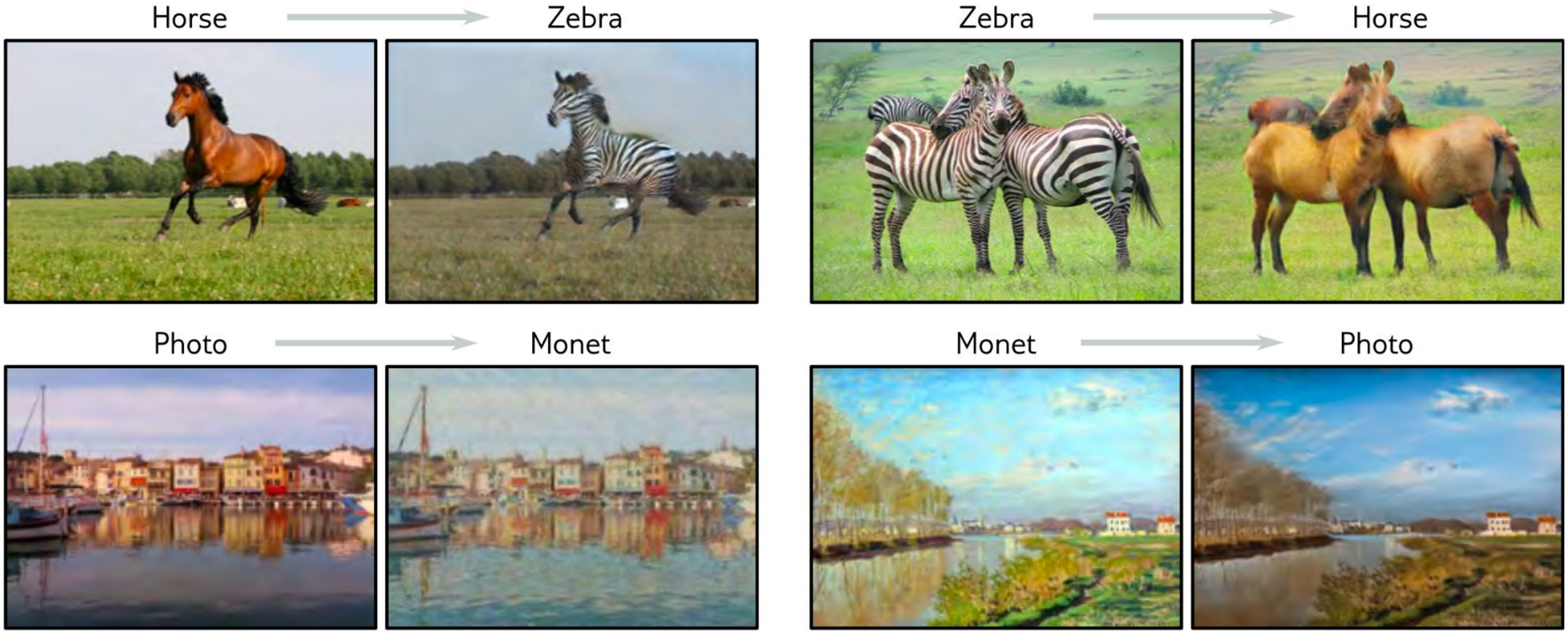
[Pix2Pix]

Image translation



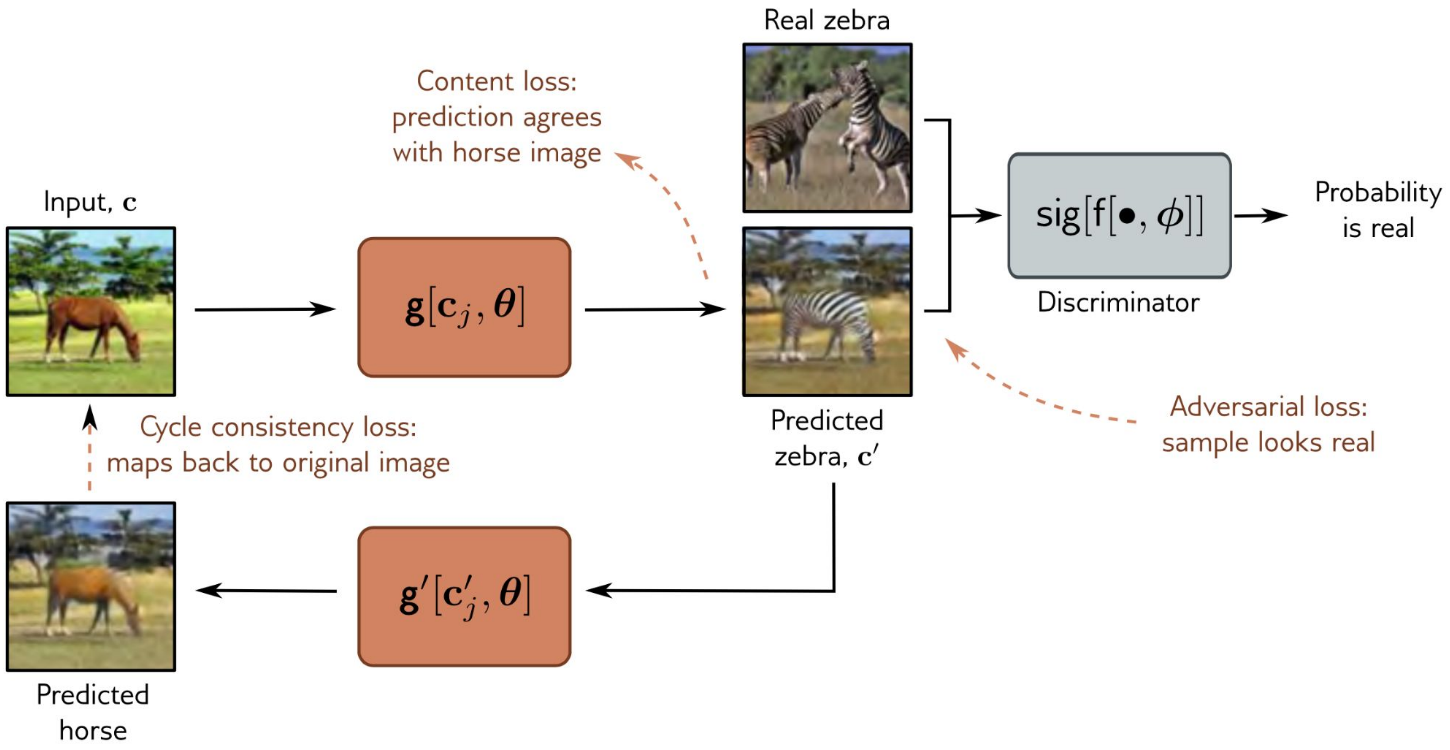
Pix2Pix

Image translation



CycleGAN

Image translation



CycleGAN

Super-resolution

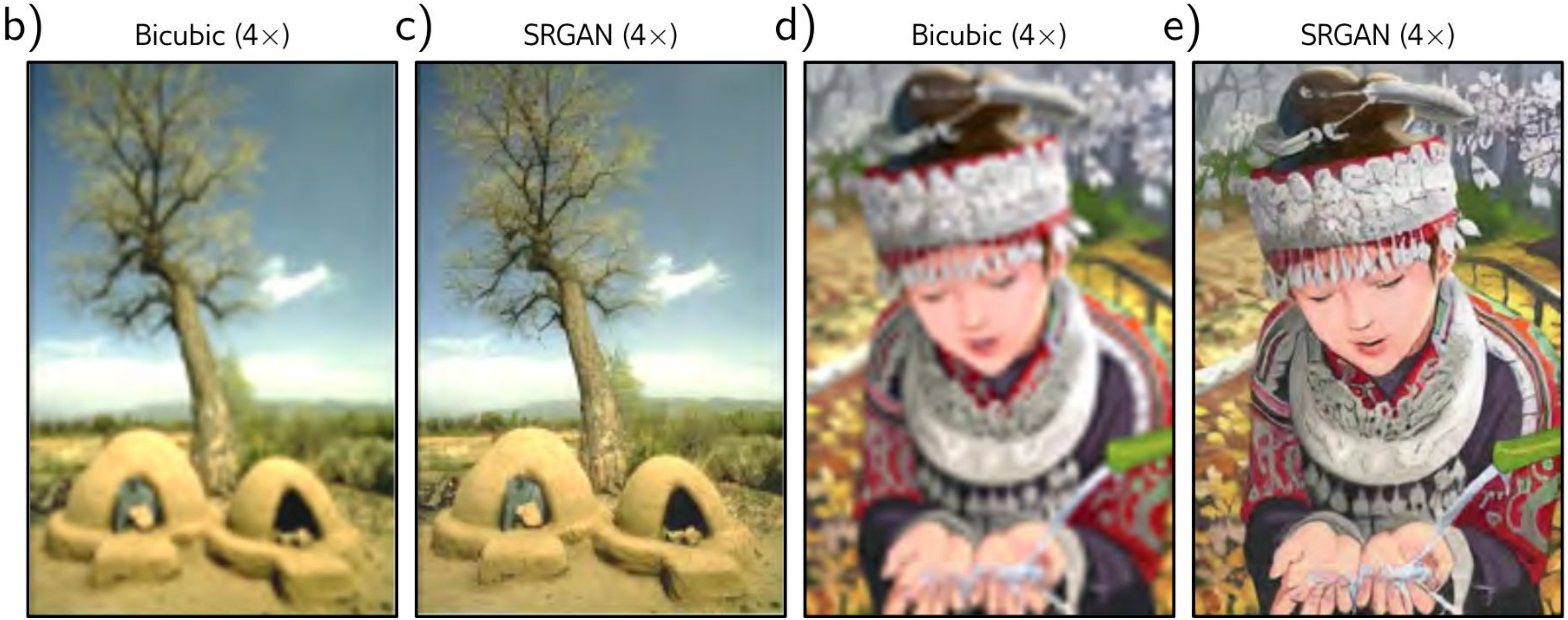


Photo-realistic single image super- resolution using a generative adversarial network

Super-resolution

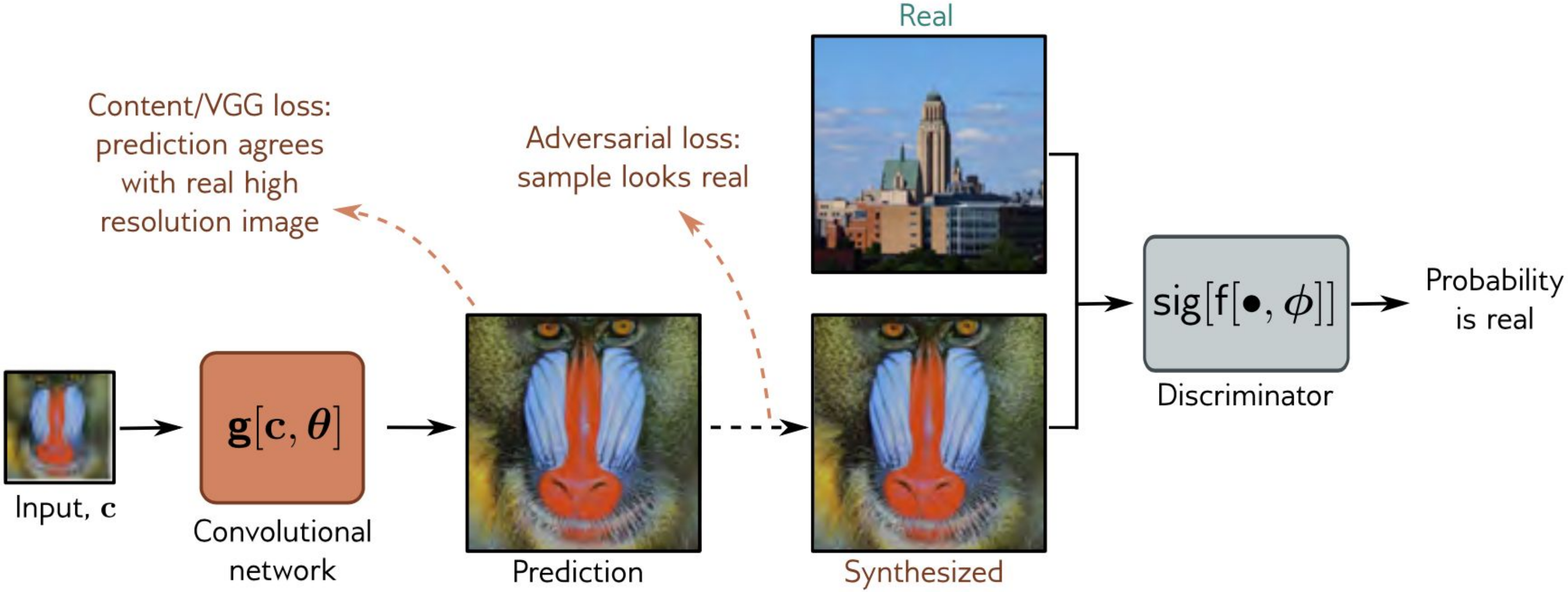


Photo-realistic single image super- resolution using a generative adversarial network

Stopping GAN Violence: Generative Unadversarial Networks

Samuel Albanie*
Institute of Deep Statistical Harmony
Shelfanger, UK

Sébastien Ehrhardt*
French Foreign Legion
Location Redacted

João F. Henriques*
Centre for Discrete Peace, Love and Understanding
Coimbra, Portugal

2015) or Soumith Chintala (Chintala, 2012-present). We note that we can further improve training efficiency by trivially rewriting our motivator objective as follows[9]:

$$\theta_M^* = \min_{\theta_M} \oint_{S(G)} \log(R) + \log(1 - \zeta) \tag{3}$$

Equation 3 describes the flow of reward and personal well-being on the generator network surface. ζ is a constant which improves the appearance of the equation. In all our experiments, we fixed the value of ζ to zero.

[1703.02528] Stopping GAN Violence: Generative Unadversarial Networks

Content of today's lecture

- Generative modeling
  - appetiser
  - approaches and metrics
- GAN
- **Diffusion models**
  - intro
  - tale of marvelous Gaussians
  - more details
- Stable Diffusion

“The sculpture is already complete within the marble block, before I start my work. It is already there, I just have to chisel away the superfluous material.”
© Michelangelo

“Creating noise from data is easy;
creating data from noise is generative modeling”
© 2011.13456

Large-language models are the foundation of generative AI today. We're using a technique called diffusion to explore a new kind of language model that gives users greater control, creativity, and speed in text generation.

Deep unsupervised learning using nonequilibrium thermodynamics

Authors	Jascha Sohl-Dickstein, Eric A Weiss, Niru Maheswaranathan, Surya Ganguli
Publication date	2015/3/12
Journal	International Conference on Machine Learning
Description	A central problem in machine learning involves modeling complex data-sets using highly flexible families of probability distributions in which learning, sampling, inference, and evaluation are still analytically or computationally tractable. Here, we develop an approach that simultaneously achieves both flexibility and tractability. The essential idea, inspired by non-equilibrium statistical physics, is to systematically and slowly destroy structure in a data distribution through an iterative forward diffusion process. We then learn a reverse diffusion process that restores structure in data, yielding a highly flexible and tractable generative model of the data. This approach allows us to rapidly learn, sample from, and evaluate probabilities in deep generative models with thousands of layers or time steps, as well as to compute conditional and posterior probabilities under the learned model. We additionally release an open source reference implementation of the algorithm.
Total citations	Cited by 10582

2015

[1503.03585] Deep Unsupervised Learning using Nonequilibrium Thermodynamics
#foundational, #images,
Jascha Sohl-Dickstein [Stanford University]

[2006.11239] Denoising Diffusion Probabilistic Models #foundational, #images, #connecting score based method (1907.05600)
Jonathan Ho [UC Berkeley]

[2011.13456] Score-Based Generative Modeling through Stochastic Differential Equations blog
#foundational, #extending to Stochastic stuff
Yang Song, Jascha Sohl-Dickstein [Stanford University + Google Brain]

[2105.05233] Diffusion Models Beat GANs on Image Synthesis
#images, [openAI], Yang Song, Jascha Sohl-Dickstein [Stanford University + Google Brain]

[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models
#StableDiffision

[2302.05543] Adding Conditional Control to Text-to-Image Diffusion Models
#ControlNet

[2403.03206] Scaling Rectified Flow Transformers for High-Resolution Image Synthesis
#StableDiffision latest

[2505.17638] Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training
#NeurIPS 2025 Best Paper Awards

2025

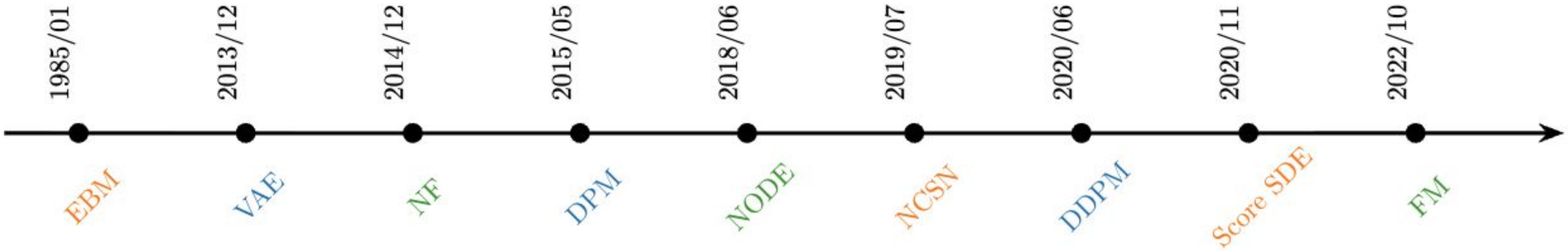


Figure 1: Timeline of diffusion model perspectives. Each group shares the same color.

- In Chapter 2, Variational Autoencoder (VAE) (Kingma and Welling, 2013) → Diffusion Probabilistic Models (DPM) (Sohl-Dickstein *et al.*, 2015) → DDPM (Ho *et al.*, 2020).
- In Chapters 3 and 4, Energy-Based Model (EBM) (Ackley *et al.*, 1985) → Noise Conditional Score Network (NCSN) (Song and Ermon, 2019) → Score SDE (Song *et al.*, 2020c).
- In Chapter 5, Normalizing Flow (NF) (Rezende and Mohamed, 2015) → Neural ODE (NODE) (Chen *et al.*, 2018) → Flow Matching (FM) (Lipman *et al.*, 2022).

[2510.21890] The Principles of Diffusion Models

D🧨ffusers

license Apache-2.0 | release v0.35.2 | downloads/month 8M | Contributor Covenant 2.1 | Follow @diffuserslib

🤗 Diffusers is the go-to library for state-of-the-art pretrained diffusion models for generating images, audio, and even 3D structures of molecules. Whether you're looking for a simple inference solution or training your own diffusion models, 🤗 Diffusers is a modular toolbox that supports both. Our library is designed with a focus on usability over performance, simple over easy, and customizability over abstractions.

🤗 Diffusers offers three core components:

- State-of-the-art diffusion pipelines that can be run in inference with just a few lines of code.
- Interchangeable noise schedulers for different diffusion speeds and output quality.
- Pretrained models that can be used as building blocks, and combined with schedulers, for creating your own end-to-end diffusion systems.

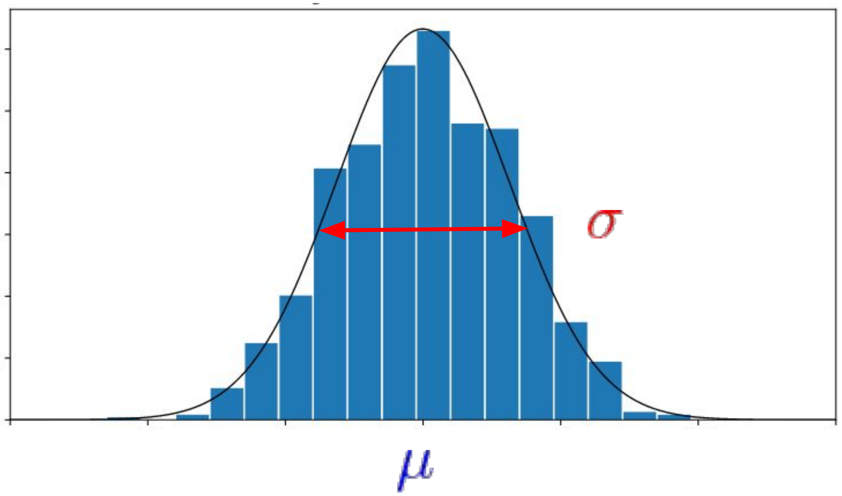
https://github.com/huggingface/diffusers

Content of today’s lecture

- Generative modeling
    - appetiser
    - approaches and metrics
- GAN
- Diffusion models
    - intro
    - **tale of marvelous Gaussians**
    - more details
- Stable Diffusion

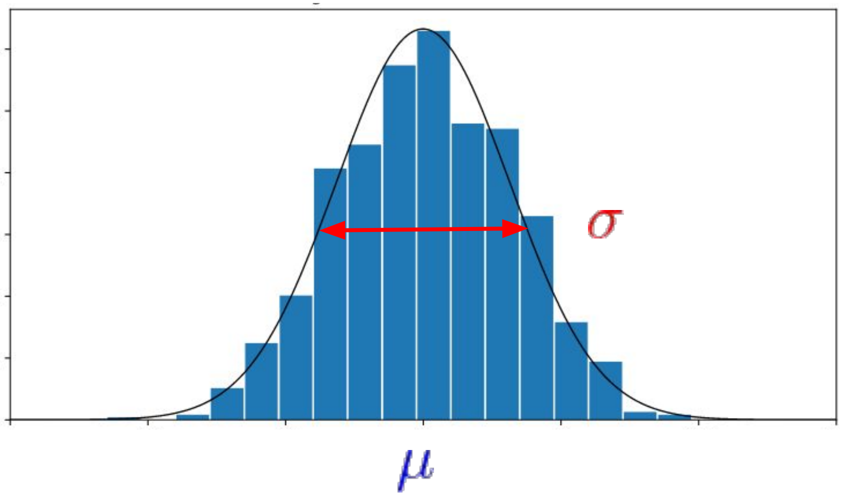
Gaussian distribution

$$\mathcal{N}(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$



Gaussian distribution

$$\mathcal{N}(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$



The Unreasonable Effectiveness of Mathematics in the Natural Sciences

Richard Courant Lecture in Mathematical Sciences delivered at New York University, May 11, 1959

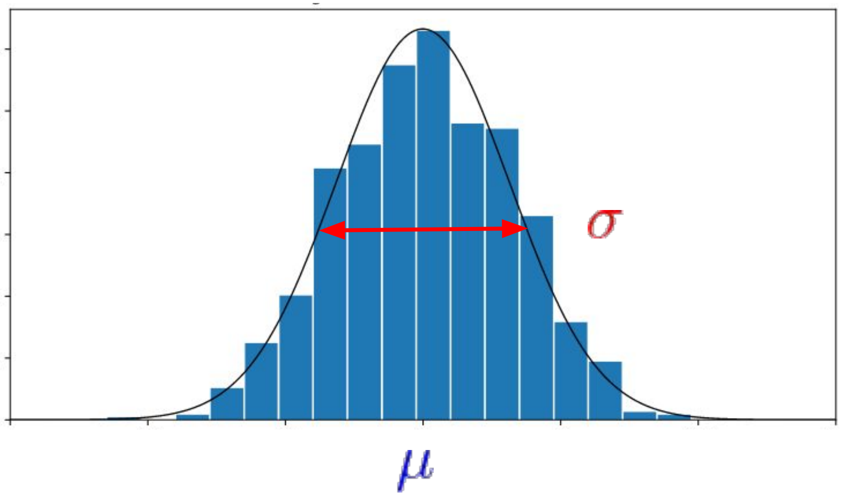
EUGENE P. WIGNER
Princeton University

"and it is probable that there is some secret here which remains to be discovered." (*C. S. Peirce*)

There is a story about two friends, who were classmates in high school, talking about their jobs. One of them became a statistician and was working on population trends. He showed a reprint to his former classmate. The reprint started, as usual, with the Gaussian distribution and the statistician explained to his former classmate the meaning of the symbols for the actual population, for the average population, and so on. His classmate was a bit incredulous and was not quite sure whether the statistician was pulling his leg. "How can you know that?" was his query. "And what is this symbol here?" "Oh," said the statistician, "this is π." "What is that?" "The ratio of the circumference of the circle to its diameter." "Well, now you are pushing your joke too far," said the classmate, "surely the population has nothing to do with the circumference of the circle."

Gaussian distribution

$$\mathcal{N}(x;\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$



The Unreasonable Effectiveness of Mathematics in the Natural Sciences

Richard Courant Lecture in Mathematical Sciences delivered at New York University, May 11, 1959

EUGENE P. WIGNER

Princeton University

"and it is probable that there is some secret here which remains to be discovered." (C. S. Peirce)

Історія про двох друзів, колишніх однокласників. Один став математиком і займався зміною населення. Він показав другові статтю, що починалася з розподілу Гаусса. Однокласник поставився до цього недовірливо: «А що це за символ?» «Це π», – відповів математик. «Що то таке?»«Відношення довжини кола до його діаметра». «Ти вже заходиш занадто далеко зі своїм жартом, – сказав однокласник, – населення точно не має нічого спільного з довжиною кола».

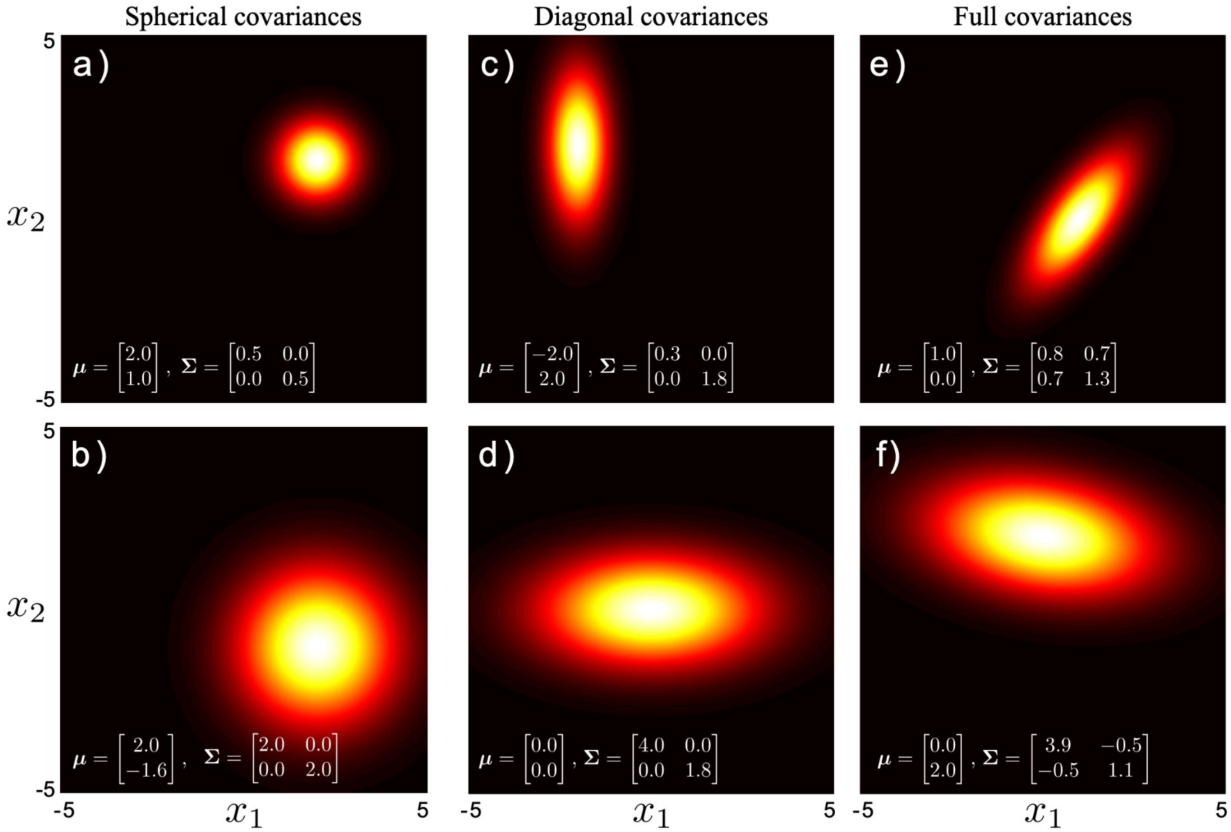
Multivariate Gaussian distribution

$$\mathcal{N}(\mathbf{x}; \mu, \Sigma) = \frac{1}{|\Sigma|^{1/2} (2\pi)^{D/2}} \exp\left[-\frac{1}{2} (\mathbf{x} - \mu)^T \cdot \Sigma^{-1} \cdot (\mathbf{x} - \mu)\right]$$

μ is *[D x 1]* vector that describes the position of the distribution

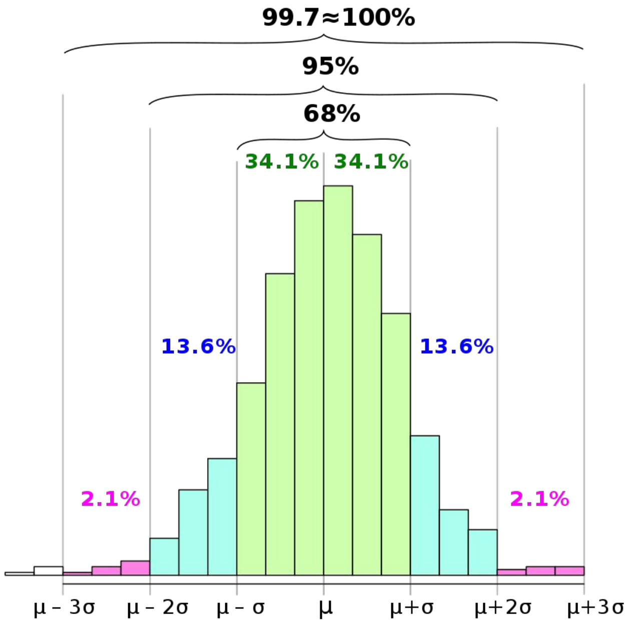
Σ is *[D x D]* positive definite matrix
(implying that $\boldsymbol{z}^{\boldsymbol{T}} \cdot \boldsymbol{\Sigma} \cdot \boldsymbol{z}$ is positive for any real vector ***z***)
and describes the shape of the distribution

$$\boldsymbol{\Sigma}_{spher} = \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix} \quad \boldsymbol{\Sigma}_{diag} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \quad \boldsymbol{\Sigma}_{full} = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12}^2 \\ \sigma_{21}^2 & \sigma_{22}^2 \end{bmatrix}$$



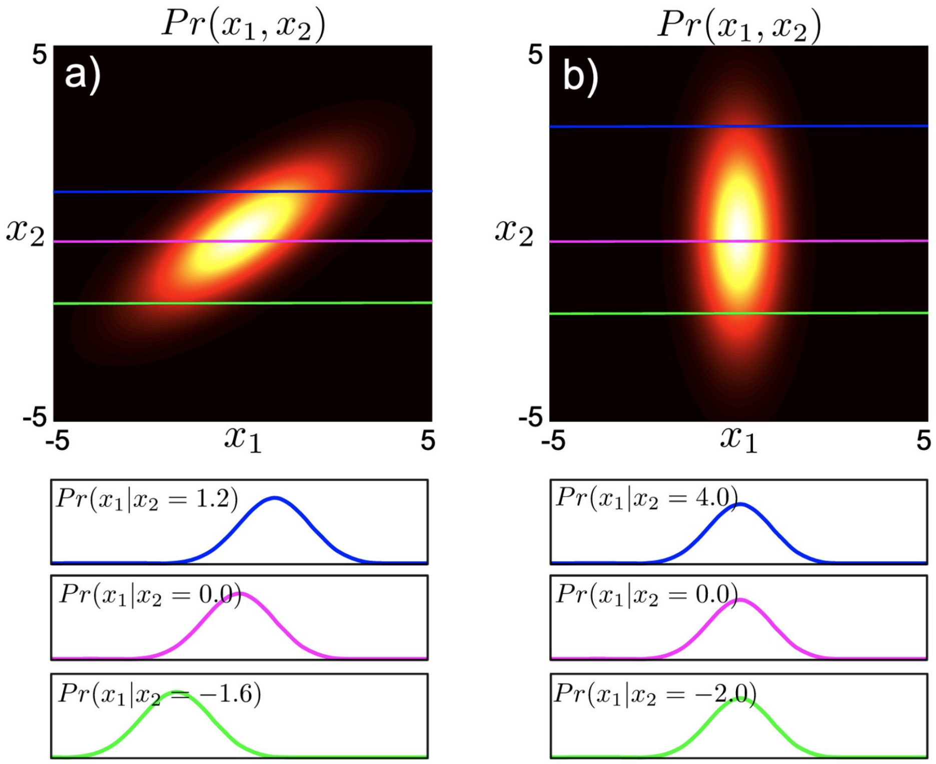
[Simon J.D. Prince, “Computer Vision: Models, Learning, and Inference”]

Multivariate Gaussian distribution



n = 1	99.73 %
n = 10	97.33 %
n = 100	76.31 %
n = 300	44.44 %
n = 1000	6.70 %

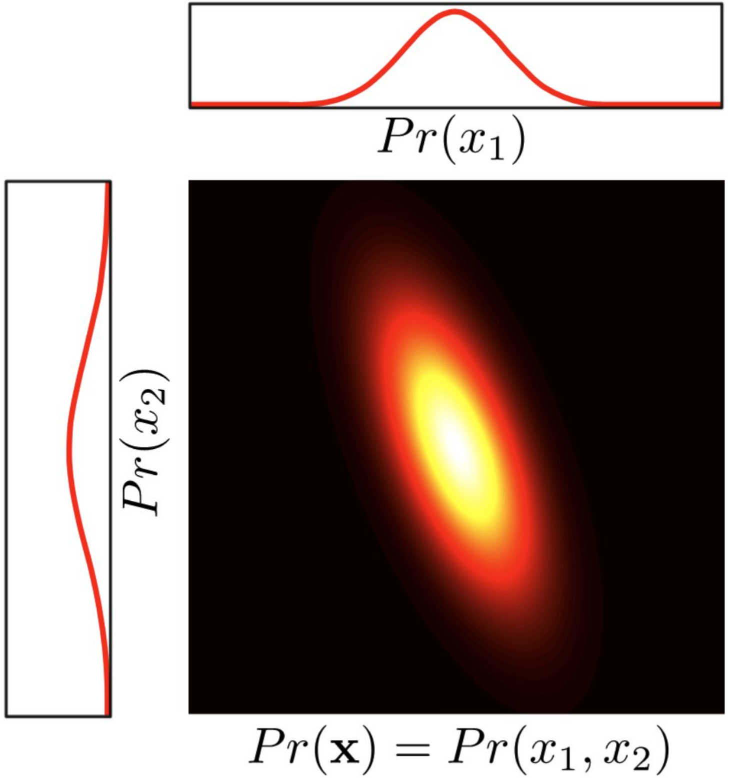
Conditional distributions (of multivariate Gaussian)



If we take any multivariate normal distribution, fix a subset of the variables, and look at the distribution of the remaining variables, this distribution will also take the form of a normal. The mean of this new normal depends on the values that we fixed the subset to, but the covariance is always the same.

Simon J.D. Prince, "Computer Vision: Models, Learning, and Inference"

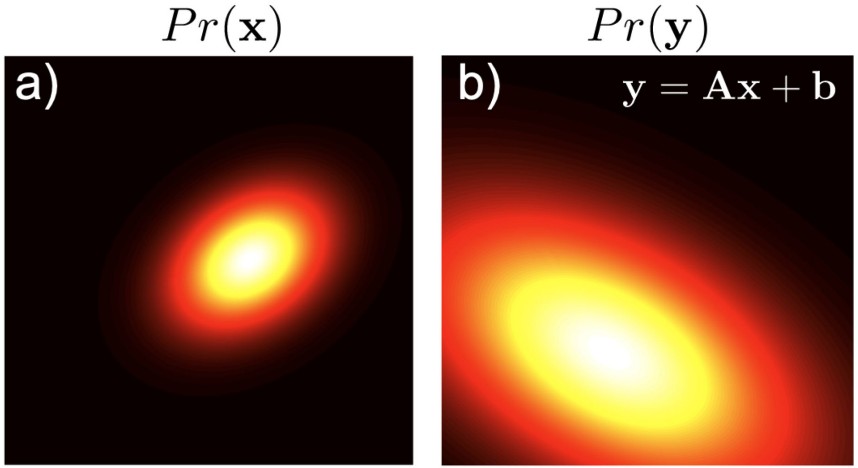
Marginal distributions (of multivariate Gaussian)



The marginal distribution of any subset of variables in a normal distribution is also normally distributed. In other words, if we sum over the distribution in any direction, the remaining quantity is also normally distributed. To find the mean and the covariance of the new distribution, we can simply extract the relevant entries from the original mean and covariance matrix.

Simon J.D. Prince, "Computer Vision: Models, Learning, and Inference"

Linear transformations of variables



The form of the multivariate normal is preserved under linear transformations

y = Ax + b

If the original distribution was

$$P(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu, \mathbf{\Sigma})$$

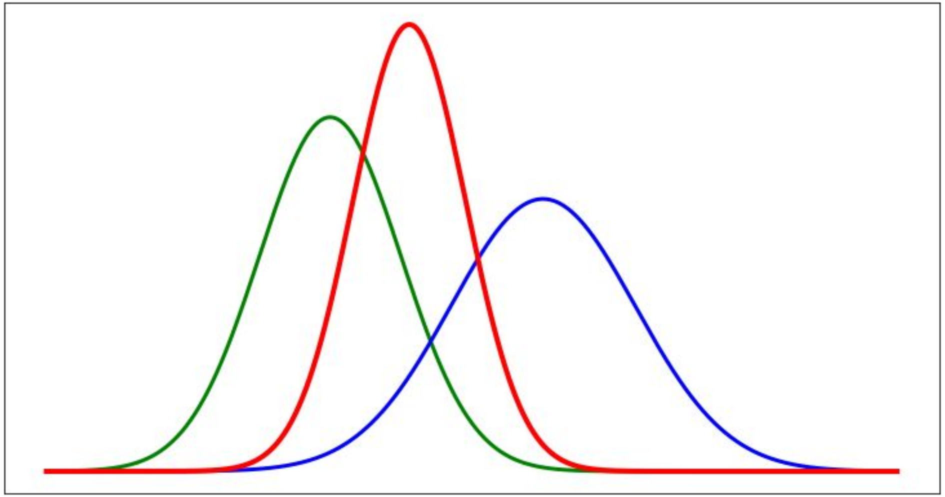
then the transformed variable **y** is distributed as:

$$Pr(\mathbf{y}) = \mathcal{N}(\mathbf{y}; \mathbf{A}\mu + \mathbf{b}, \mathbf{A}\mathbf{\Sigma}\mathbf{A}^T)$$

This relationship provides a simple method to draw samples from a normal distribution with mean μ and covariance Σ. We first draw a sample x from a standard normal distribution (μ = 0 and Σ = I) and then apply the transform $y = \Sigma^{1/2}x + \mu$ [“*Reparameterization Trick*”]

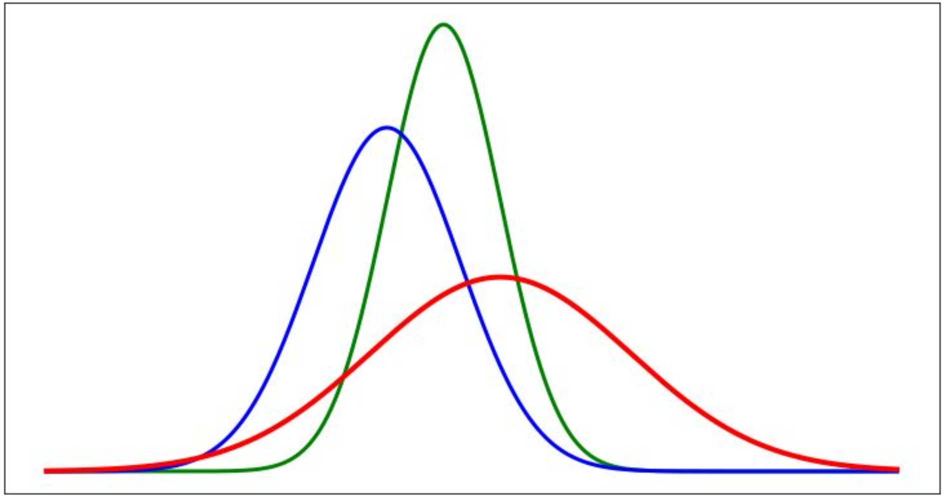
Simon J.D. Prince, “Computer Vision: Models, Learning, and Inference”

Product of two Gaussians



The product of any two normals **N1** and **N2** is proportional to a third normal **N3** distribution, with a mean between the two original means and a variance that is smaller than either of the original distributions.

Sum of two Gaussians



The sum of any two normals **N1** and **N2** is proportional to a third normal **N3** distribution, with a mean as a sum of two original means and a variance that is also sum of original variances

Content of today's lecture

- Generative modeling
  - appetiser
  - approaches
- Diffusion models
  - intro
  - tale of marvelous Gaussians
  - more details
- Stable Diffusion

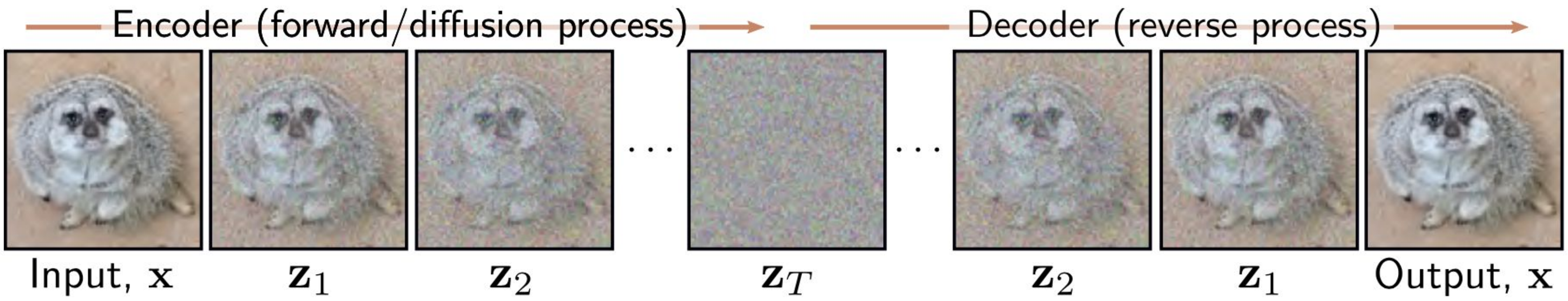
Structure → Uniform distribution

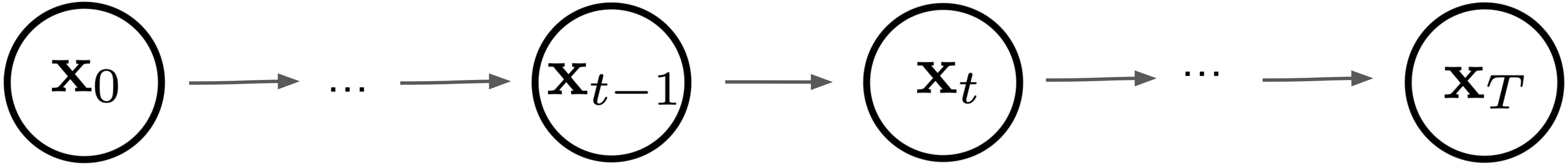
https://en.wikipedia.org/wiki/Diffusion

Structure ← Uniform distribution

https://en.wikipedia.org/wiki/Diffusion

- Destroy all structure in data distribution using diffusion process
- Learn reversal of diffusion process
- *Reverse diffusion process* is the *model* of the data


Encoder (forward/diffusion process)
Decoder (reverse process)
Input, $\mathbf{x}$
$\mathbf{z}_1$
$\mathbf{z}_2$
$\mathbf{z}_T$
$\mathbf{z}_2$
$\mathbf{z}_1$
Output, $\mathbf{x}$

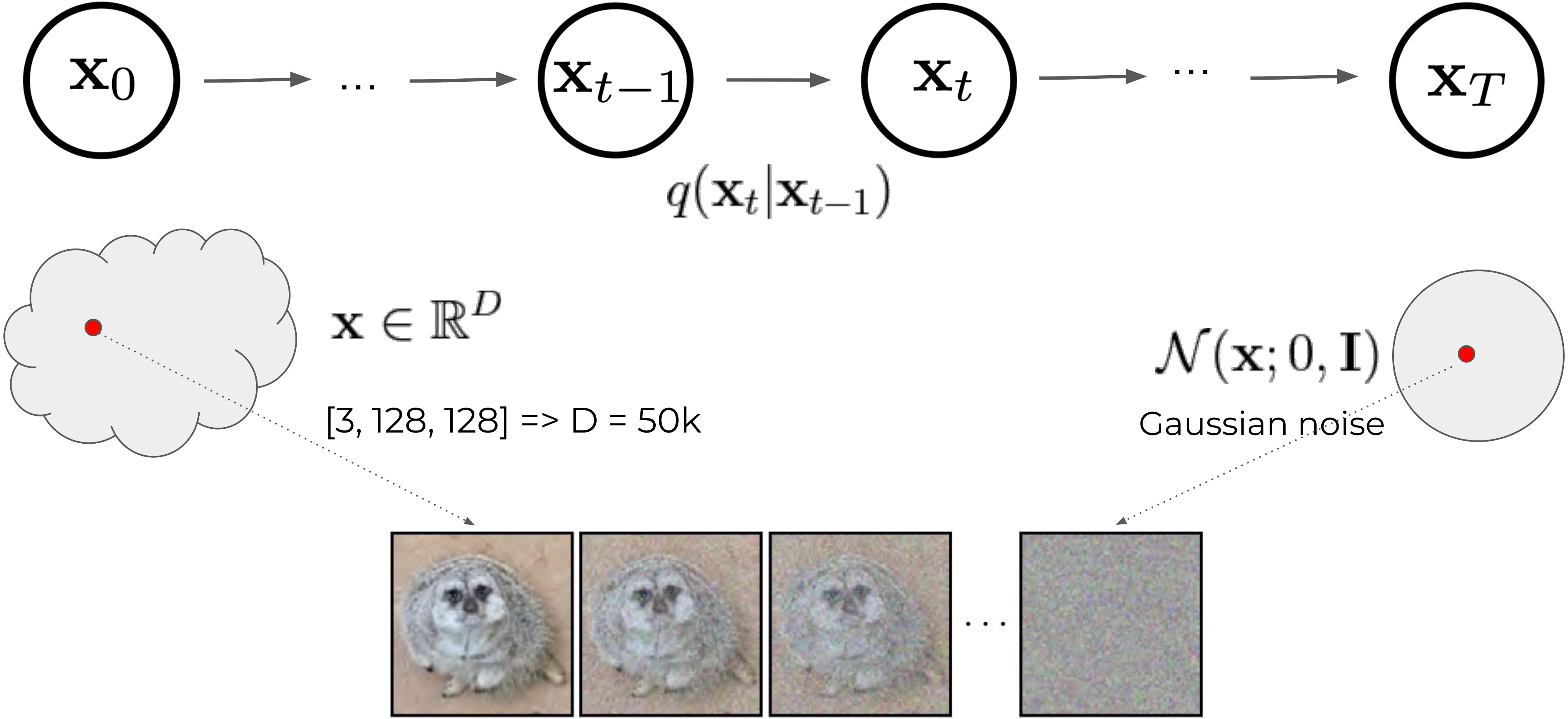


$$q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

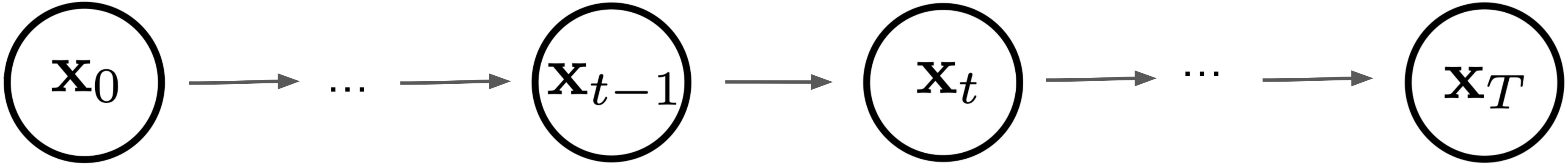
$$\mathbf{x}_1 = \sqrt{1-\beta_1} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_1} \cdot \epsilon_1$$

...

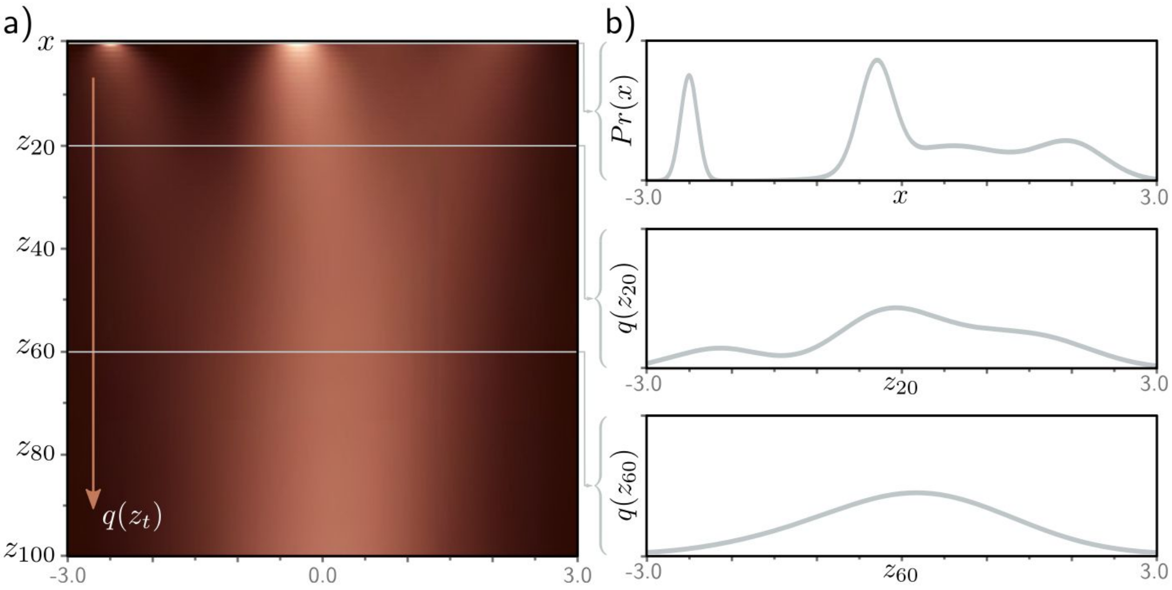
$$\mathbf{x}_t = \sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon_t$$

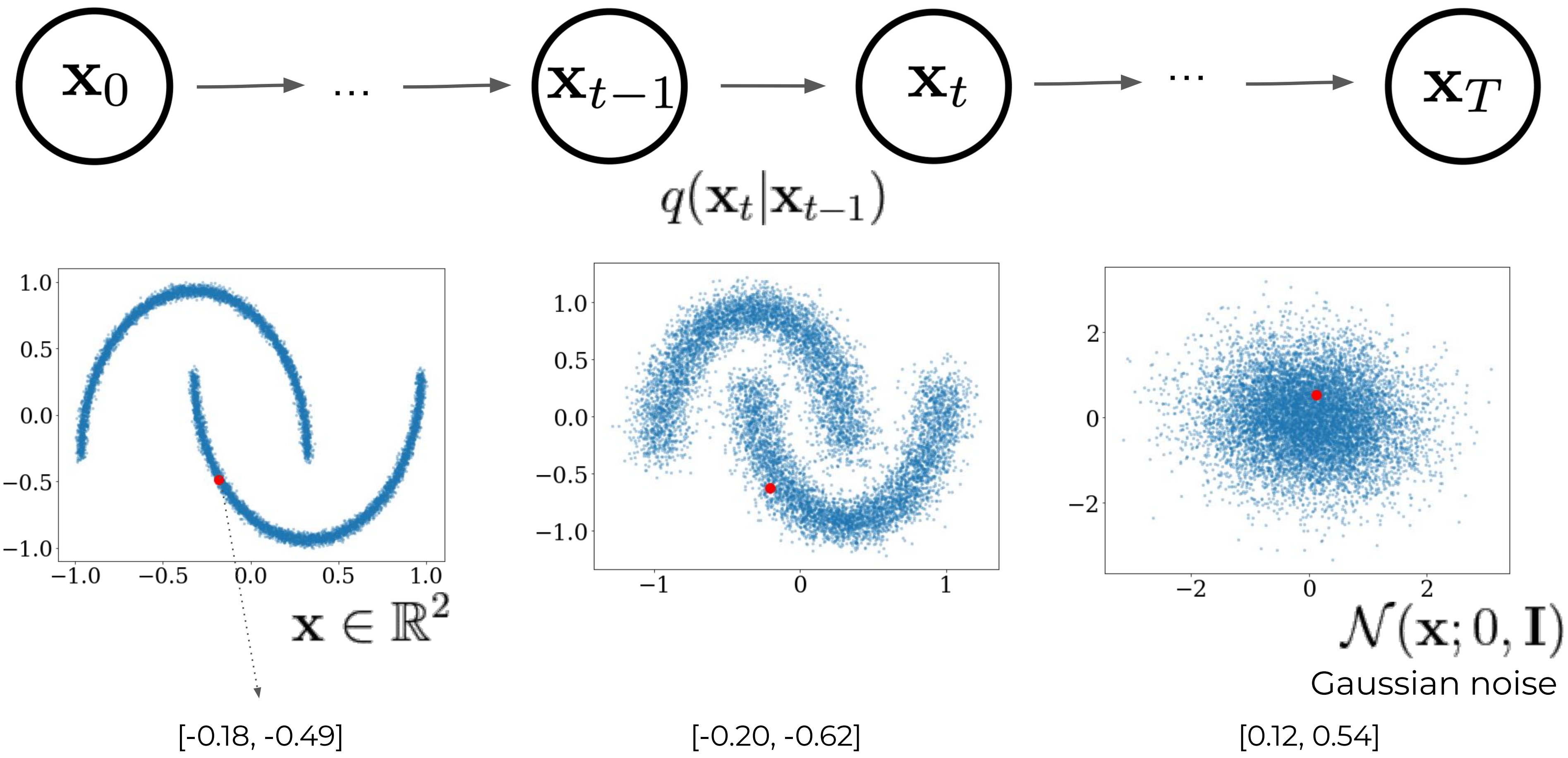


$\mathbf{x}_0$
$\cdots$
$\mathbf{x}_{t-1}$
$\mathbf{x}_t$
$\cdots$
$\mathbf{x}_T$
$q(\mathbf{x}_t|\mathbf{x}_{t-1})$
$\mathbf{x} \in \mathbb{R}^D$
[3, 128, 128] => D = 50k
$\mathcal{N}(\mathbf{x}; 0, \mathbf{I})$
Gaussian noise

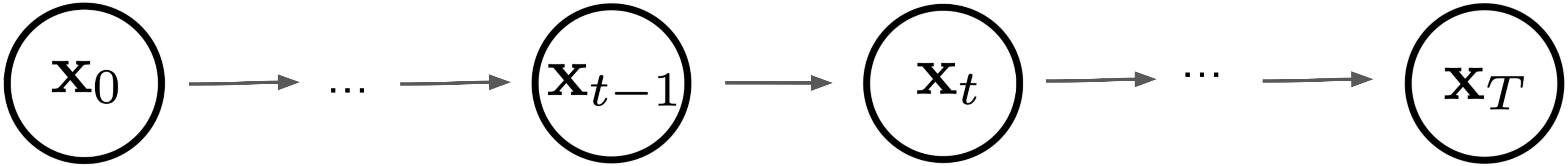


$$q(\mathbf{x}_t|\mathbf{x}_{t-1})$$





$\mathbf{x}_0$
$\mathbf{x}_{t-1}$
$\mathbf{x}_t$
$\mathbf{x}_T$
$q(\mathbf{x}_t|\mathbf{x}_{t-1})$
$\mathbf{x} \in \mathbb{R}^2$
$\mathcal{N}(\mathbf{x}; 0, \mathbf{I})$
Gaussian noise
[-0.18, -0.49]
[-0.20, -0.62]
[0.12, 0.54]



$$q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

$$\mathbf{x}_1 = \sqrt{1-\beta_1} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_1} \cdot \epsilon_1$$

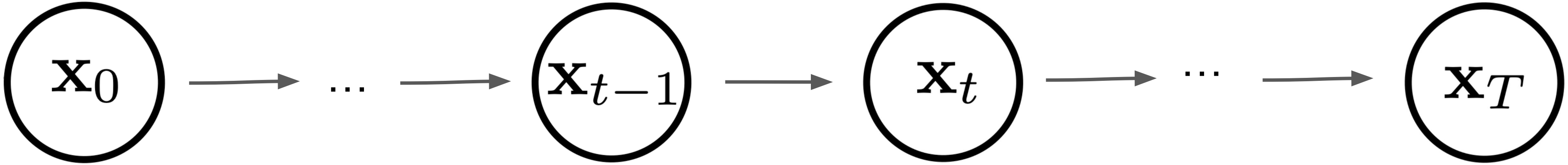
$$\ldots$$

$$\mathbf{x}_t = \sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon_t$$

$$\mathbf{x}_0 \rightarrow \cdots \rightarrow \mathbf{x}_{t-1} \xrightarrow[q(\mathbf{x}_t|\mathbf{x}_{t-1})]{} \mathbf{x}_t \rightarrow \cdots \rightarrow \mathbf{x}_T$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

$$0 < \beta_1 < \beta_2 < ... < \beta_T < 1$$



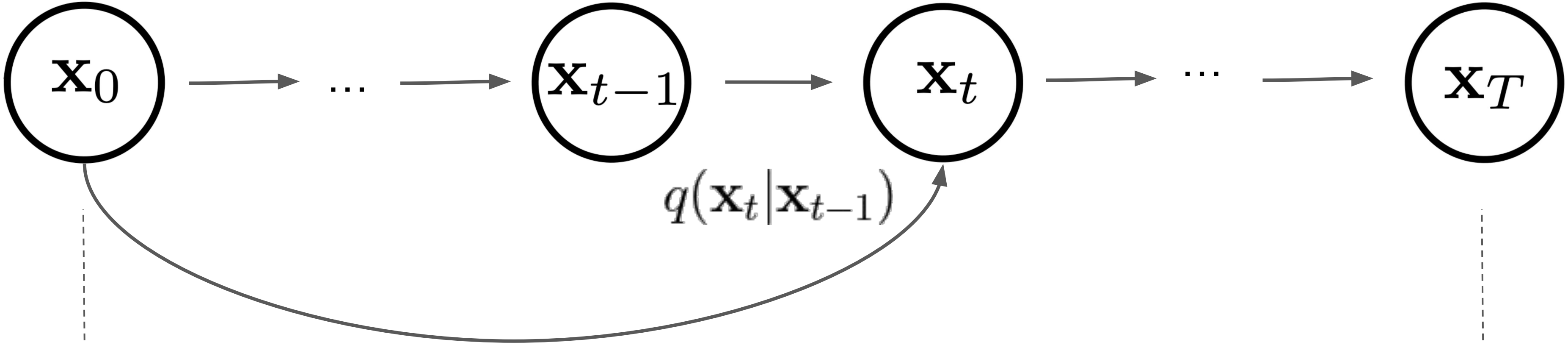
$q(\mathbf{x}_t|\mathbf{x}_{t-1})$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

$$0 < \beta_1 < \beta_2 < ... < \beta_T < 1$$

$\mathcal{N}(\mathbf{x}; \mathbf{x}_0, \mathbf{0})$ limiting cases $\mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$

Gaussian noise

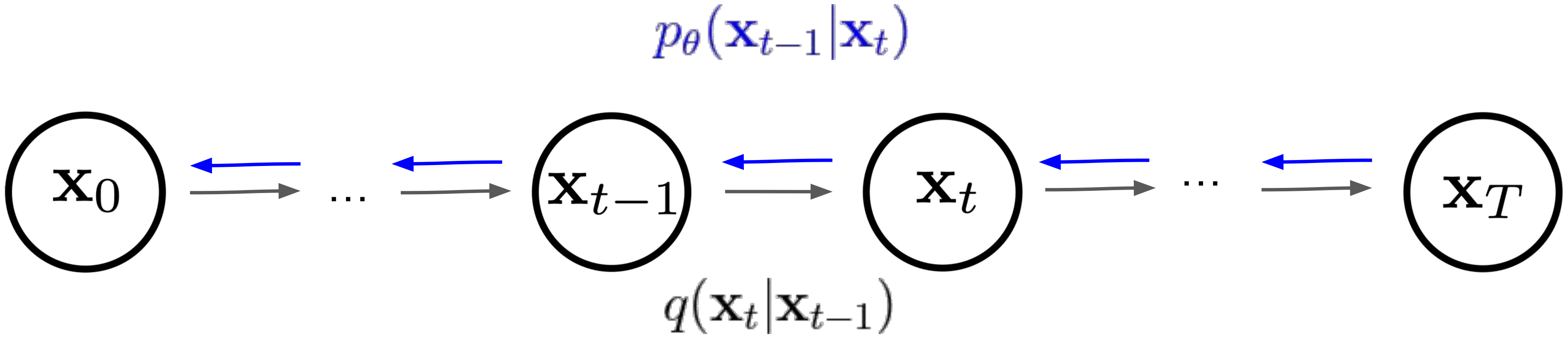


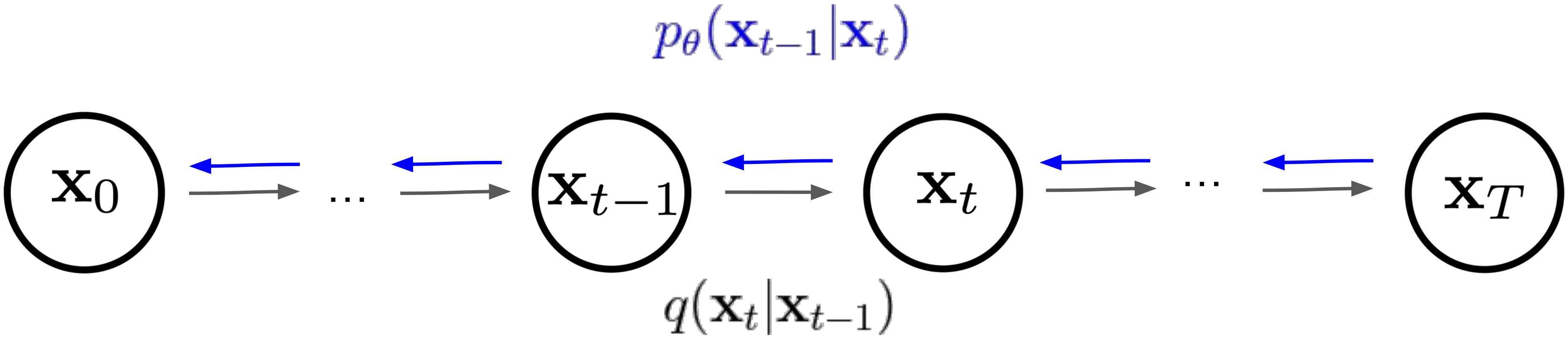
$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$

$$\bar{\alpha}_t := \prod_{s=1}^{T} \alpha_s \quad \alpha_t := 1 - \beta_t$$

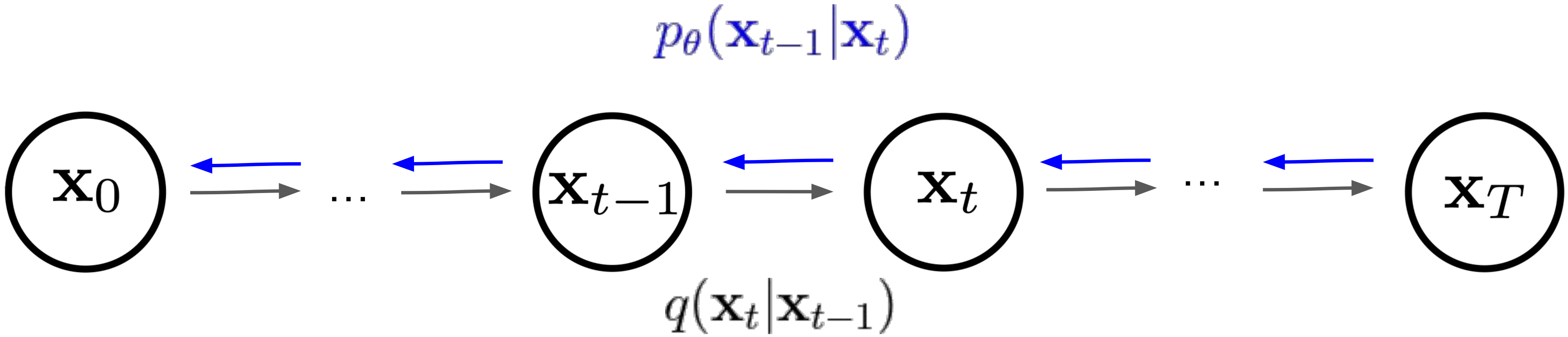
$\mathcal{N}(\mathbf{x}; \mathbf{x}_0, \mathbf{0})$ ------ limiting cases ------ $\mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$

Gaussian noise

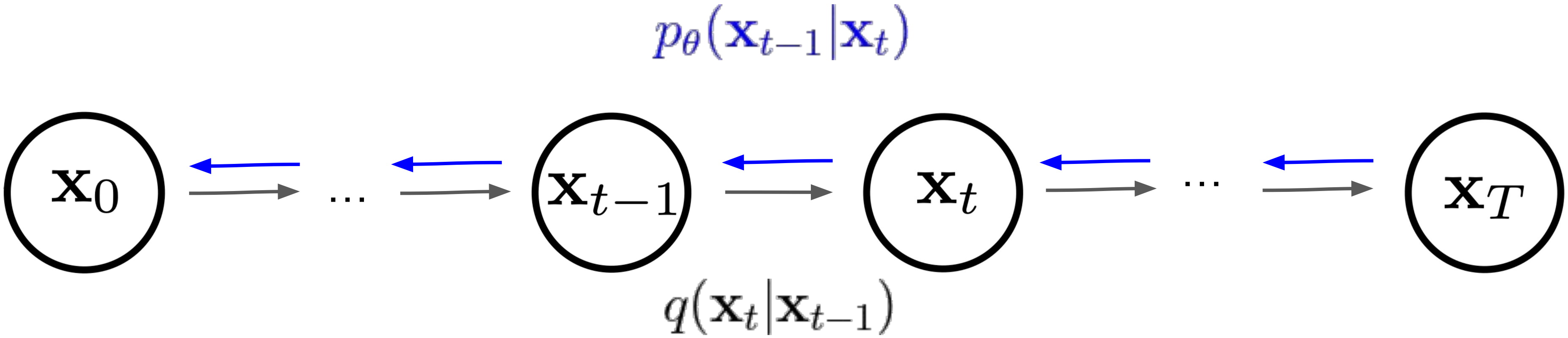

$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$
$\mathbf{x}_0$
$\cdots$
$\mathbf{x}_{t-1}$
$\mathbf{x}_t$
$\cdots$
$\mathbf{x}_T$
$q(\mathbf{x}_t|\mathbf{x}_{t-1})$



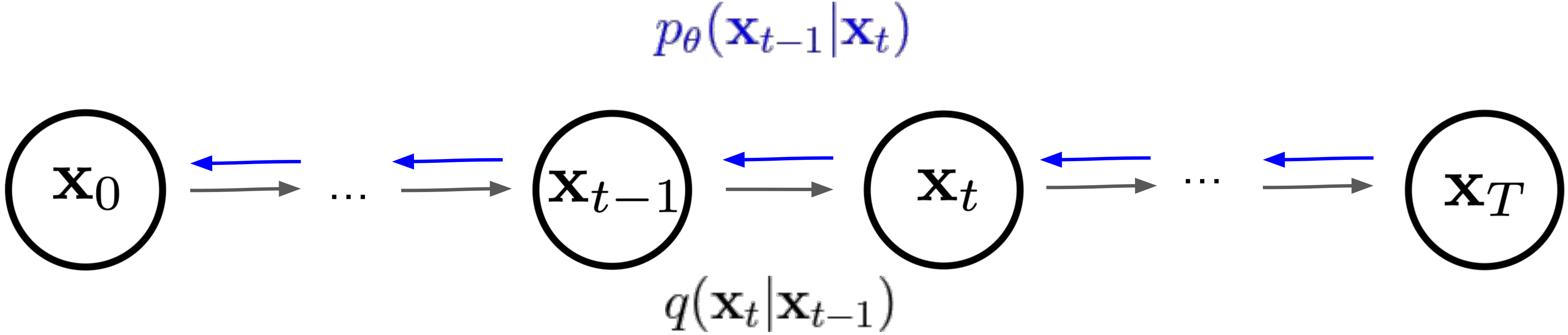
$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$$



$$
\begin{aligned}
p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)) \\
&= \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2 \mathbf{I})
\end{aligned}
$$



$$
\begin{aligned}
p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)) \\
&= \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\mu_\theta(\mathbf{x}_t, t)}_{\text{Trainable network}}, \sigma_t^2 \mathbf{I})
\end{aligned}
$$



$$
\begin{aligned}
p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)) \\
&= \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2 \mathbf{I})
\end{aligned}
$$

$$
\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\alpha_t}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \cdot \underbrace{\epsilon(\mathbf{x}_t, t)}_{\text{Trainable network}}\right)
$$

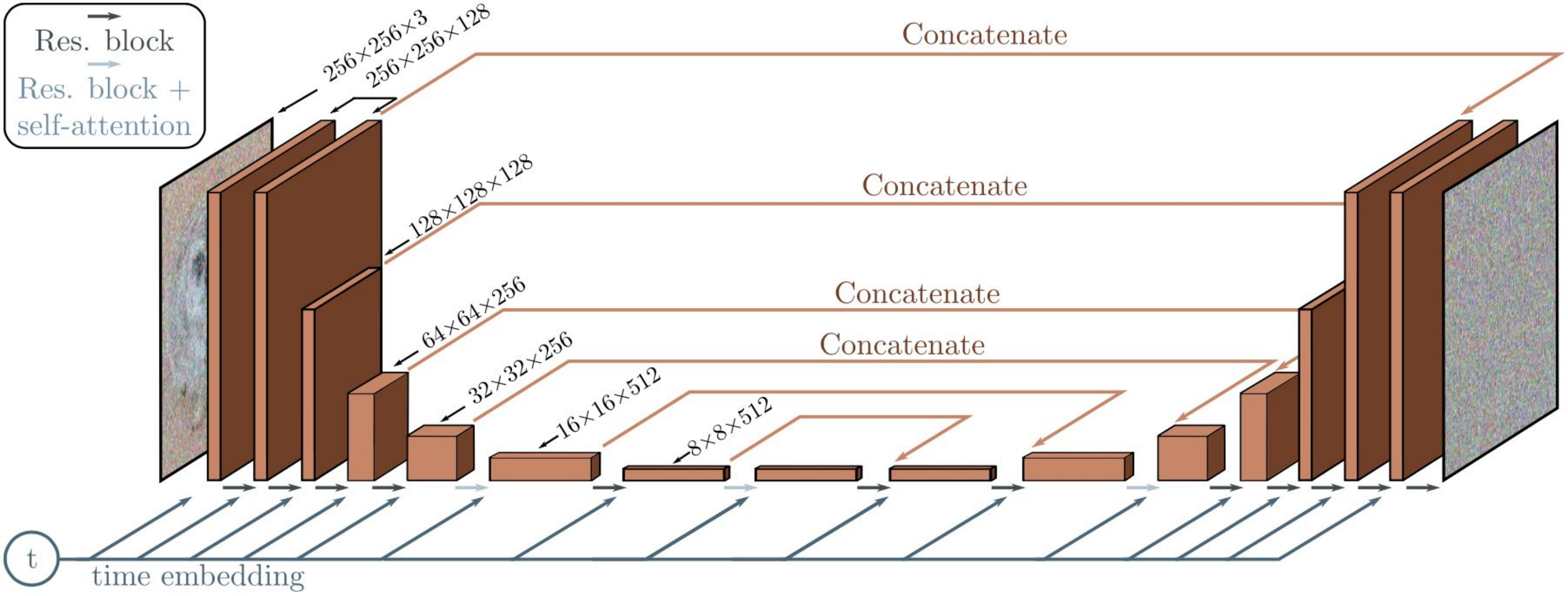
Algorithm 1 Training

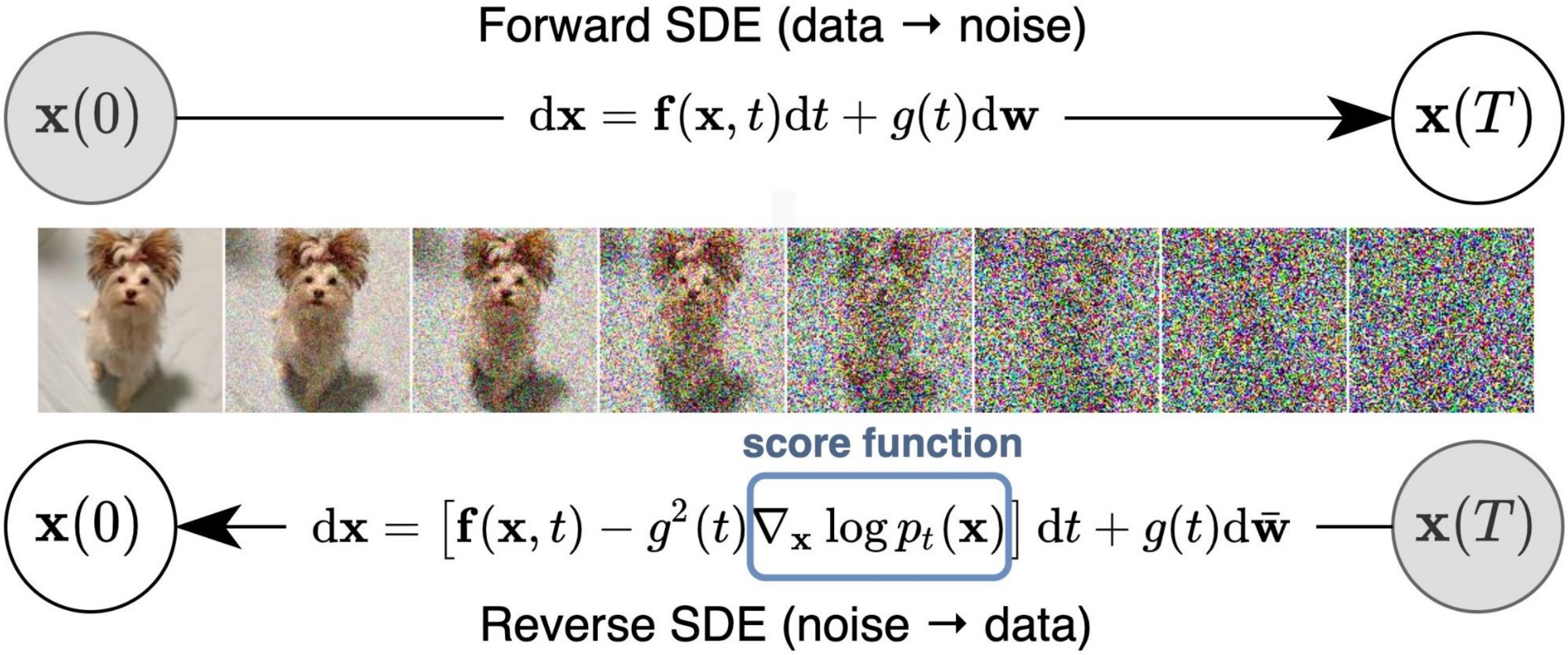
repeat
  $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
  $t \sim \mathrm{Uniform}(\{1, \dots, T\})$
  $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  Take gradient descent step on
    $\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\boxed{\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}}, t\right) \right\|^2$
until converged

Algorithm 2 Sampling

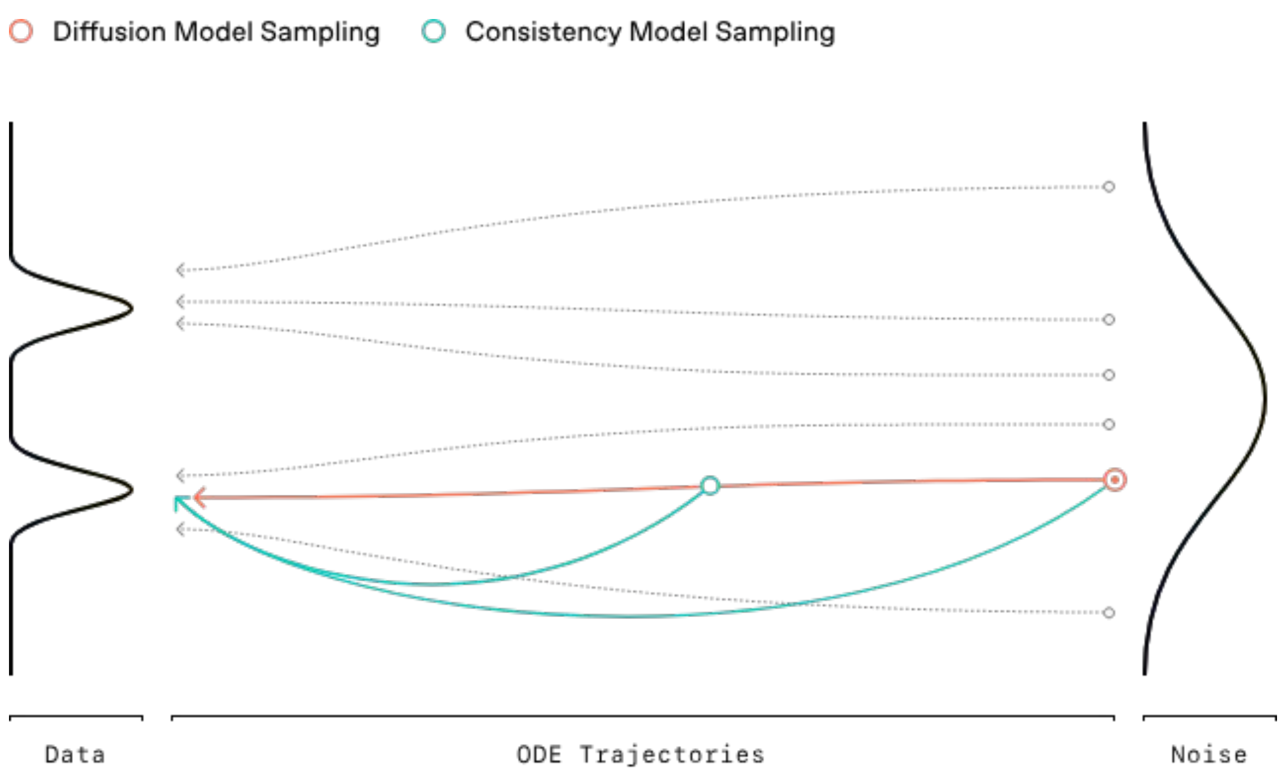
$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \dots, 1$ **do**
  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
  $\mathbf{x}_{t-1} = \boxed{\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\mathbf{z}_\theta(\mathbf{x}_t, t)\right)} + \sigma_t \mathbf{z}$
end for
return $\mathbf{x}_0$

$$\epsilon(\mathbf{x}_t, t)$$





[2011.13456] Score-Based Generative Modeling through Stochastic Differential Equations



[blog](#)

[[2410.11081] Simplifying, Stabilizing and Scaling Continuous-Time Consistency Models](#)

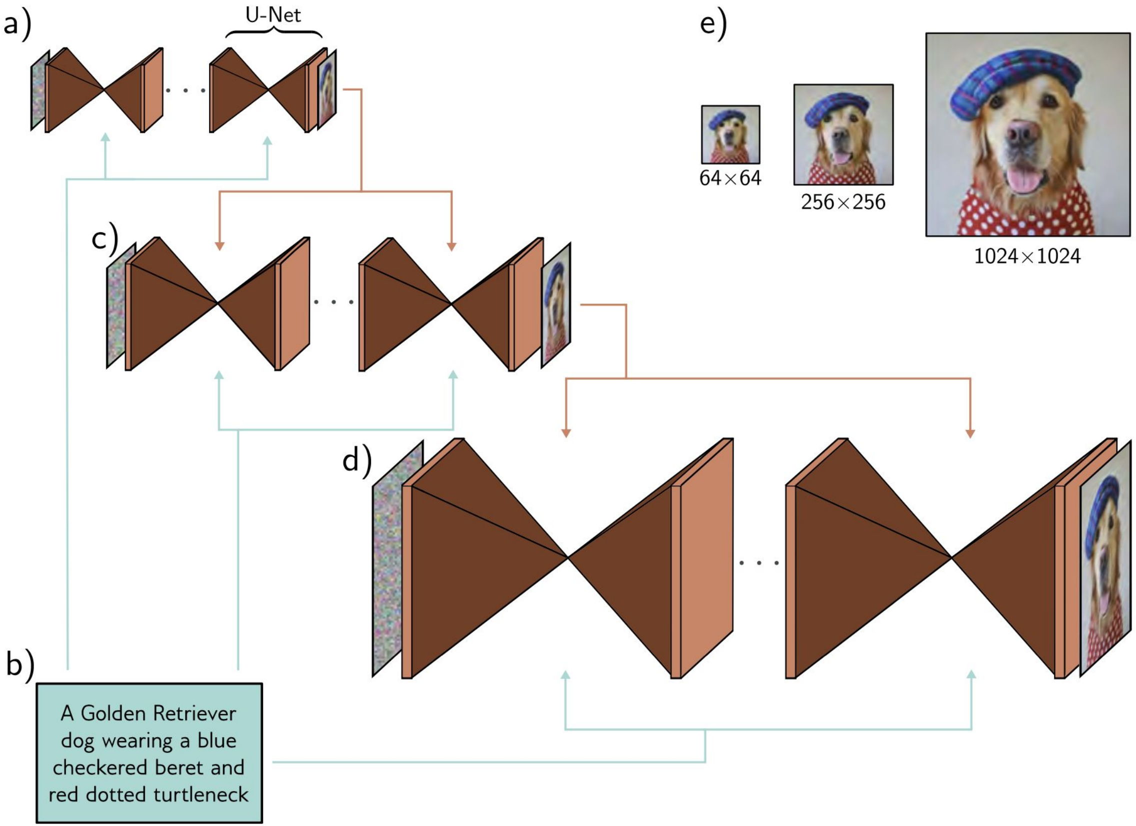
Conditional generation

To explicit incorporate class information into the diffusion process, Dhariwal & Nichol (2021) trained a classifier $f_\phi(y|\mathbf{x}_t, t)$ on noisy image and use gradients $\nabla_\mathbf{x} \log f_\phi(y|\mathbf{x}_t, t)$ to guide the diffusion sampling process toward the target class label .

Algorithm 2 Classifier guided DDIM sampling, given a diffusion model $\epsilon_\theta(x_t)$, classifier $f_\phi(y|x_t)$, and gradient scale s.

Input: class label y, gradient scale s
$x_T \leftarrow$ sample from $\mathcal{N}(0, \mathbf{I})$
for all t from T to 1 **do**
  $\hat{\epsilon} \leftarrow \epsilon_\theta(x_t) - \sqrt{1-\bar{\alpha}_t}\, \nabla_{x_t} \log f_\phi(y|x_t)$
  $x_{t-1} \leftarrow \sqrt{\bar{\alpha}_{t-1}} \left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\hat{\epsilon}}{\sqrt{\bar{\alpha}_t}} \right) + \sqrt{1-\bar{\alpha}_{t-1}}\hat{\epsilon}$
end for
return x_0

Conditional generation



[2205.11487] Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding

Content of today's lecture

- Generative modeling

    - appetiser
    - approaches

- GAN

- Diffusion models

    - intro
    - tale of marvelous Gaussians
    - more details

- Stable Diffusion

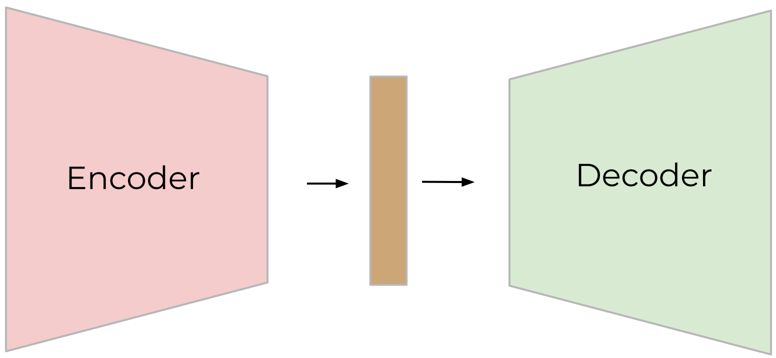
Stable Diffusion

[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models

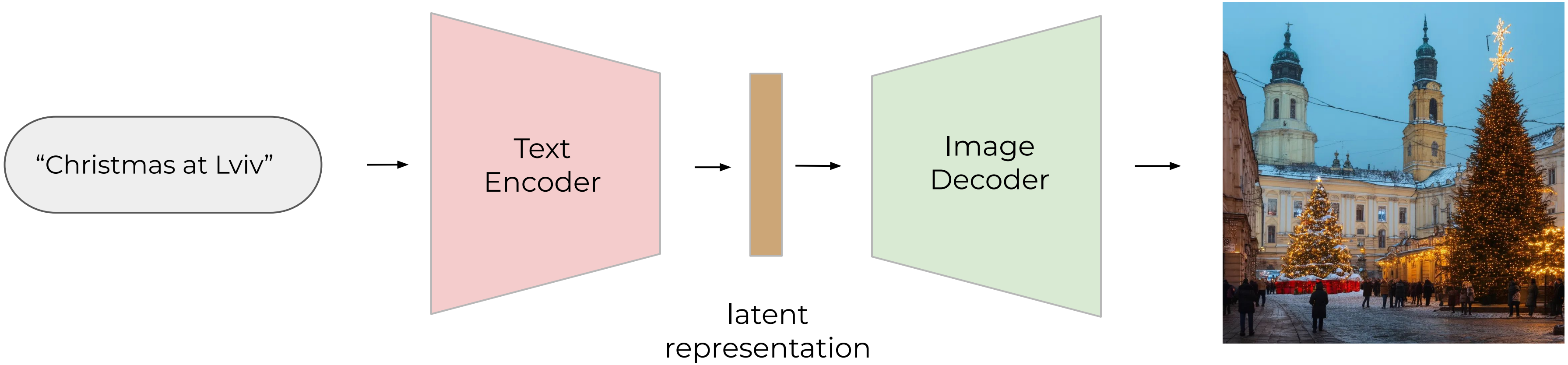
https://stability.ai/news/introducing-stable-diffusion-3-5

https://huggingface.co/stabilityai

Stable Diffusion

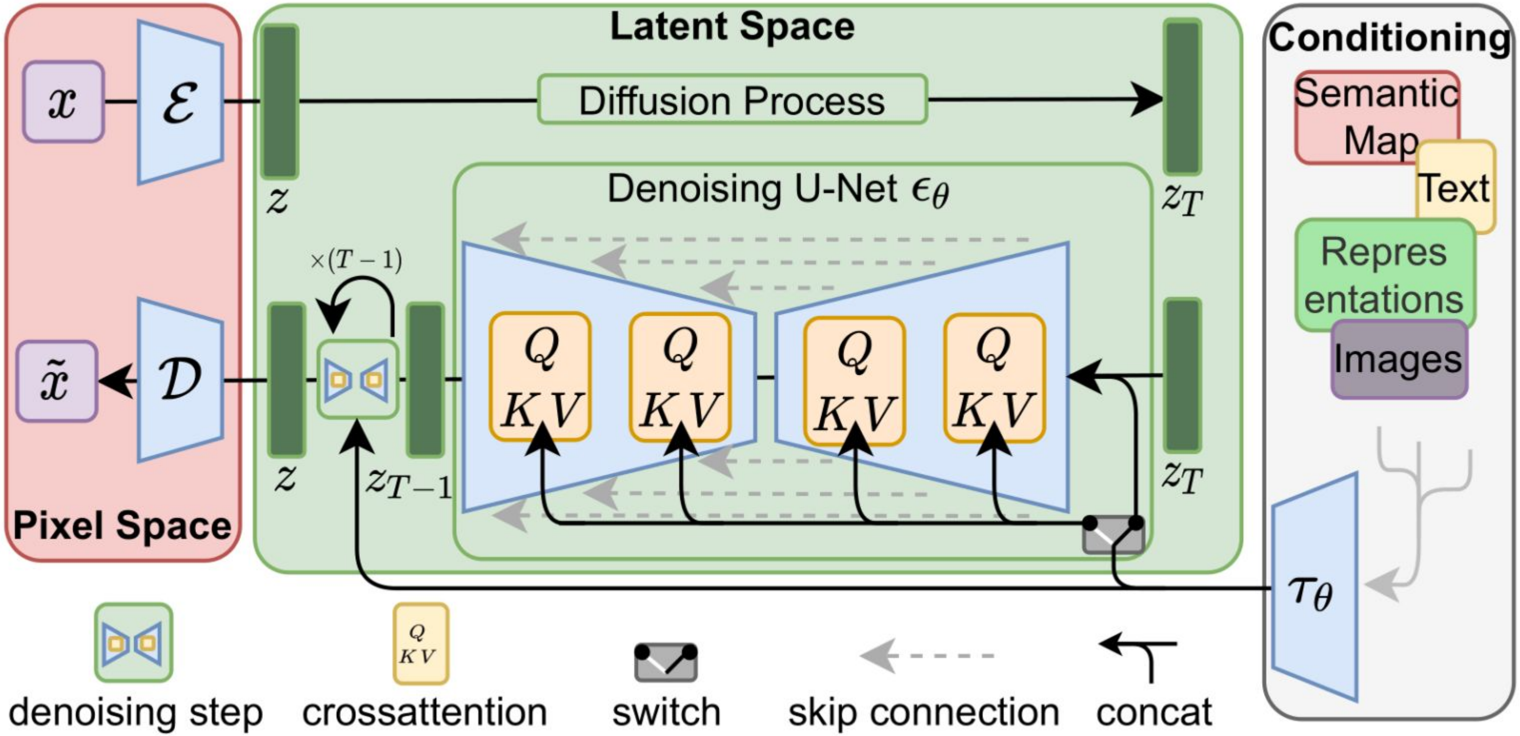


Stable Diffusion



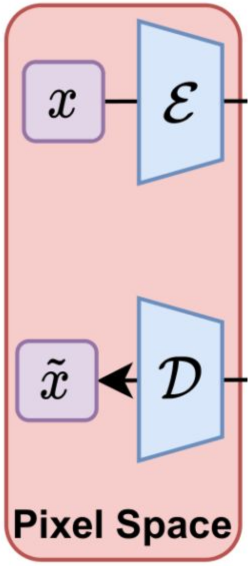
[Stable Diffusion]

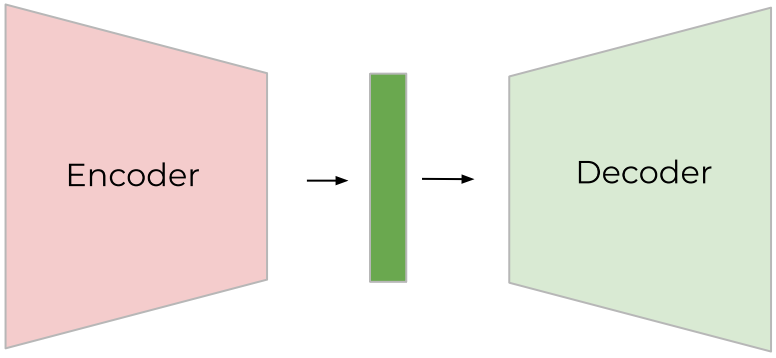
Stable Diffusion



[[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models]

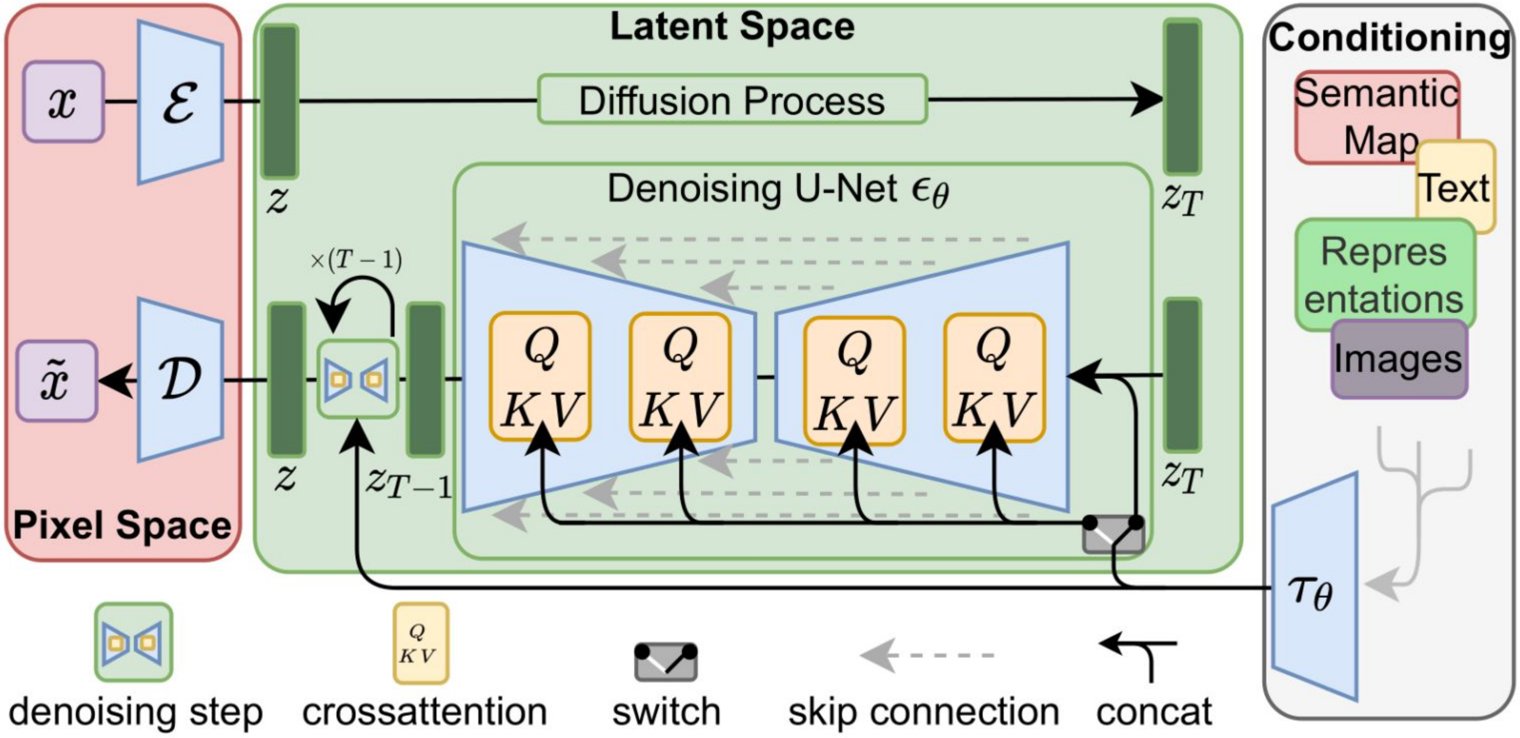
Stable Diffusion





[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models

Stable Diffusion



[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models

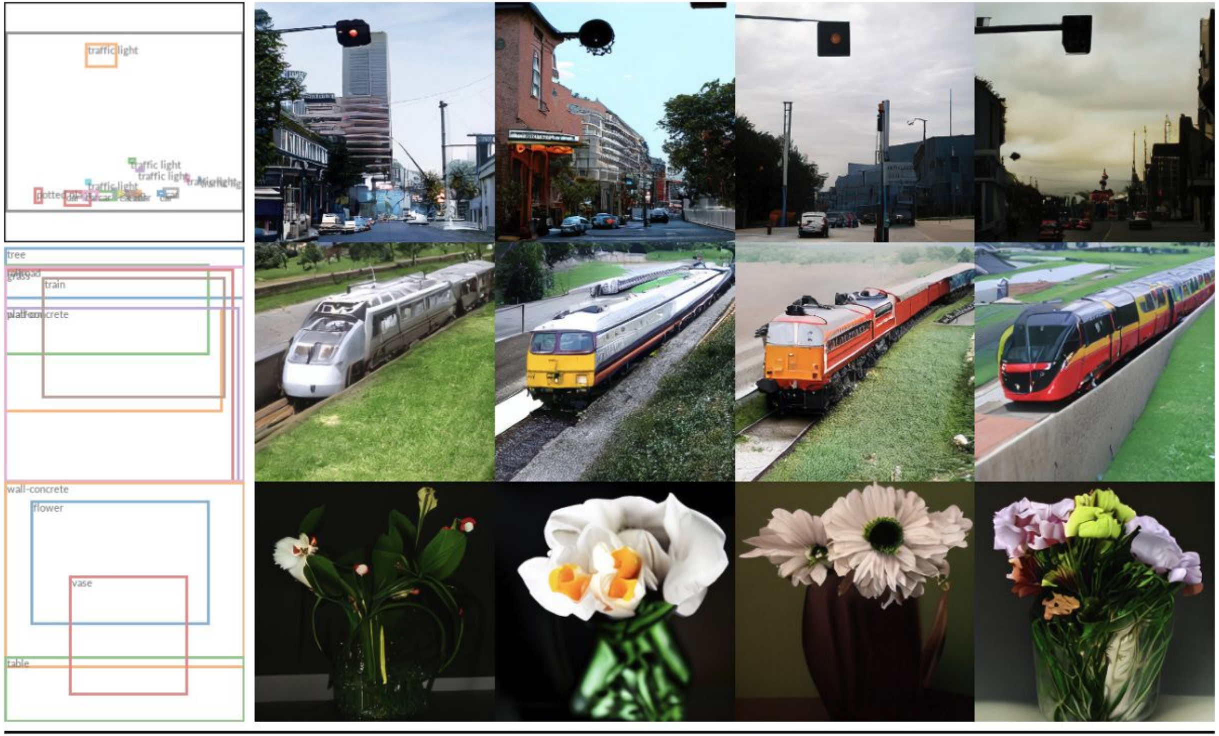
Stable Diffusion

[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models

Stable Diffusion

[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models

Stable Diffusion



[2112.10752] High-Resolution Image Synthesis with Latent Diffusion Models

MIDJOURNEY VERSIONS COMPARED

V1
Feb 2022

V2
Apr 2022

V3
Jul 2022

V4
Nov 2022

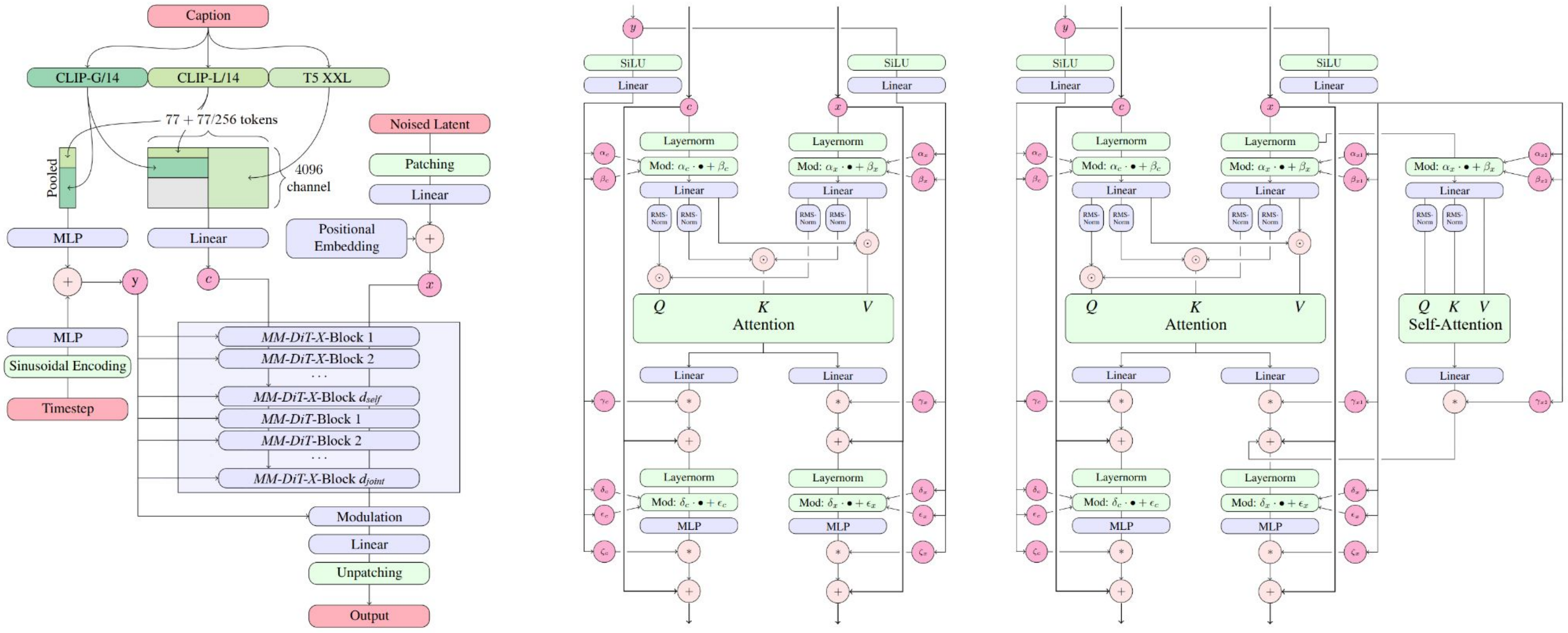
V5
Mar 2023

V5.1
May 2023

V5.1 --style raw
May 2023

V5.2
Jun 2023

V5.2 --style raw
Jun 2023



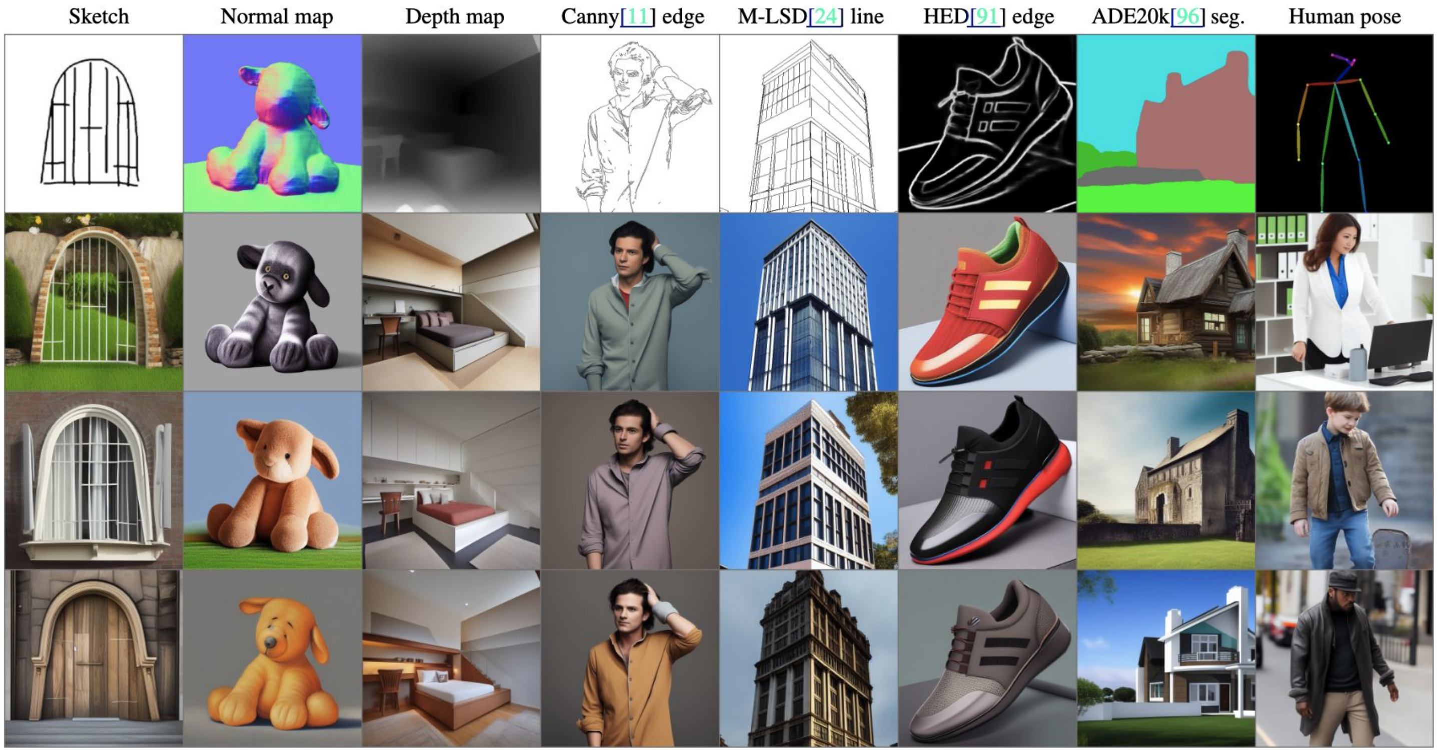
(a) Overview of all components.

(b) One *MM-DiT* block

(c) One *MM-DiT-X* block

[2403.03206] Scaling Rectified Flow Transformers for High-Resolution Image Synthesis

ControlNet



[2302.05543] Adding Conditional Control to Text-to-Image Diffusion Models

ControlNet

Input Canny edge

Default

“masterpiece of fairy tale, giant deer, golden antlers”

“..., quaint city Galic”

Input human pose

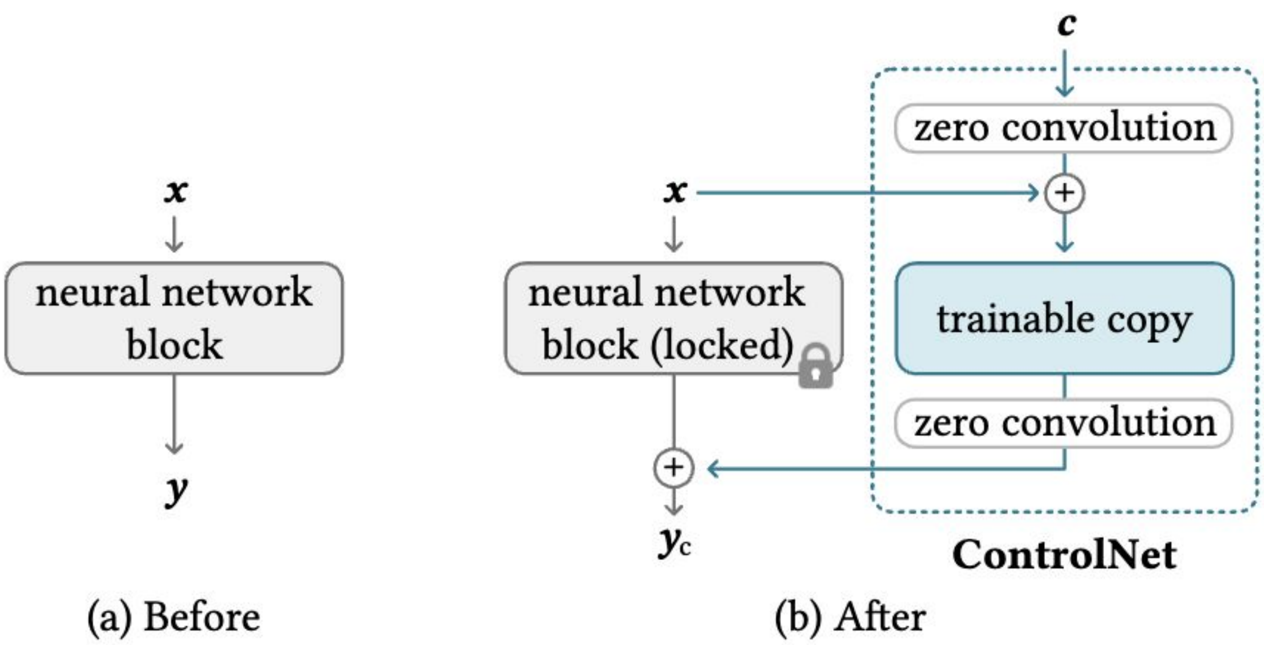
Default

“chef in kitchen”

“Lincoln statue”

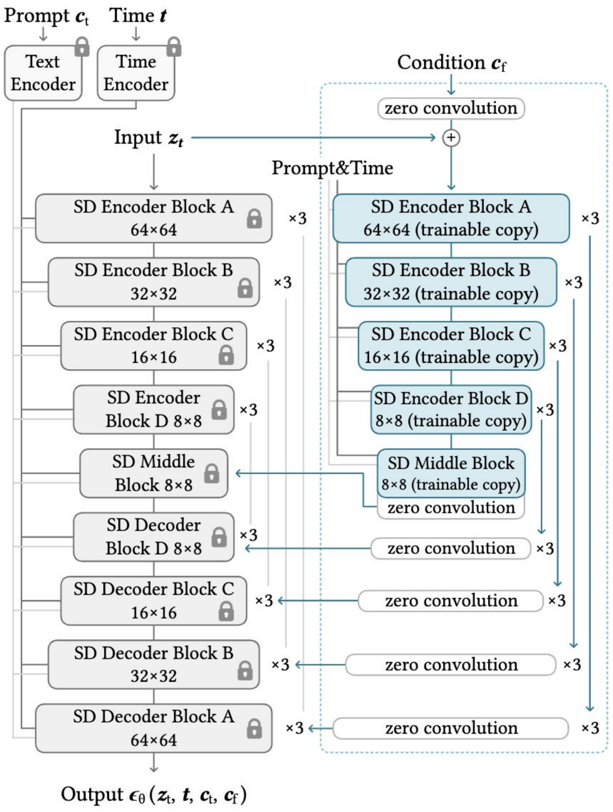
[2302.05543] Adding Conditional Control to Text-to-Image Diffusion Models

ControlNet



[2302.05543] Adding Conditional Control to Text-to-Image Diffusion Models

ControlNet



(a) Stable Diffusion (b) ControlNet

[[2302.05543] Adding Conditional Control to Text-to-Image Diffusion Models]

Summary

- We are witnessing a revolution in computer-assisted creativity.

- Generation is not limited to images, already there are video generation, 3D generation, and it will become better and better.

Other materials

- Great course
  Diffusion and Flow Models

- [2510.21890] The Principles of Diffusion Models